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References

Aalen et al.: Can we believe the DAGs? A comment on the relationship between causal DAGs and mechanisms **Aalen-Roysland-Gran-etal-2016**

Odd O. Aalen, Kjetil Røysland, Jon Michael Gran, Roger Kouyos, et al. "Can we believe the DAGs? A comment on the relationship between causal DAGs and mechanisms". In: *Statistical Methods in Medical Research* 25.5 (July 2016), pp. 2294–2314. ISSN: 1477-0334. DOI: 10.1177/0962280213520436.

Abstract: Directed acyclic graphs (DAGs) play a large role in the modern approach to causal inference. DAGs describe the relationship between measurements taken at various discrete times including the effect of interventions. The causal mechanisms, on the other hand, would naturally be assumed to be a continuous process operating over time in a cause-effect fashion. How does such immediate causation, that is causation occurring over very short time intervals, relate to DAGs constructed from discrete observations? We introduce a time-continuous model and simulate discrete observations in order to judge the relationship between the DAG and the immediate causal model. We find that there is no clear relationship; indeed the Bayesian network described by the DAG may not relate to the causal model. Typically, discrete observations of a process will obscure the conditional dependencies that are represented in the underlying mechanistic model of the process. It is therefore doubtful whether DAGs are always suited to describe causal relationships unless time is explicitly considered in the model. We relate the issues to mechanistic modeling by using the concept of local (in)dependence. An example using data from the Swiss HIV Cohort Study is presented.

Aalen et al.: Causality, mediation and time: A dynamic viewpoint

Aalen-Roysland-Gran-etal-2012

Odd O. Aalen, Kjetil Røysland, Jon Michael Gran, and Bruno Ledergerber. "Causality, mediation and time: A dynamic viewpoint". In: *Journal of the Royal Statistical Society. Series A (Statistics in Society)* 175.4 (2012), pp. 831–861. ISSN: 09641998, 1467985X. DOI: 10.1111/j.1467-985X.2011.01030.x.

Abstract: Time dynamics are often ignored in causal modelling. Clearly, causality must operate in time and we show how this corresponds to a mechanistic, or system, understanding of causality. The established counterfactual definitions of direct and indirect effects depend on an ability to manipulate the mediator which may not hold in practice, and we argue that a mechanistic view may be better. Graphical representations based on local independence graphs and dynamic path analysis are used to facilitate communication as well as providing an overview of the dynamic relations 'at a glance'. The relationship between causality as understood in a mechanistic and in an interventionist sense is discussed. An example using data from the Swiss HIV Cohort Study is presented.

Antonakis et al.: On ignoring the random effects assumption in multilevel models: Review, critique, and recommendations

Antonakis-Bastardoz-Ronkko-2019

John Antonakis, Nicolas Bastardoz, and Mikko Rönkkö. "On ignoring the random effects assumption in multilevel models: Review, critique, and recommendations". In: *Organizational Research Methods* 24.2 (Oct. 2019), pp. 443–483. ISSN: 1552-7425. DOI: 10.1177/1094428119877457.

Abstract: Entities such as individuals, teams, or organizations can vary systematically from one another. Researchers typically model such data using multilevel models, assuming that the random effects are uncorrelated with the regressors. Violating this testable assumption, which is often ignored, creates an endogeneity problem thus preventing causal interpretations. Focusing on two-

level models, we explain how researchers can avoid this problem by including cluster means of the Level 1 explanatory variables as controls; we explain this point conceptually and with a large-scale simulation. We further show why the common practice of centering the predictor variables is mostly unnecessary. Moreover, to examine the state of the science, we reviewed 204 randomly drawn articles from macro and micro organizational science and applied psychology journals, finding that only 106 articles—with a slightly higher proportion from macro-oriented fields—properly deal with the random effects assumption. Alarmingly, most models also failed on the usual exogeneity requirement of the regressors, leaving only 25 mostly macro-level articles that potentially reported trustworthy multilevel estimates. We offer a set of practical recommendations for researchers to model multilevel data appropriately.

Appelbaum et al.: Journal article reporting standards for quantitative research in psychology: The APA Publications and Communications Board task force report
Appelbaum-Cooper-Kline-etal-2018

Mark Appelbaum et al. "Journal article reporting standards for quantitative research in psychology: The APA Publications and Communications Board task force report". In: *American Psychologist* 73.7 (Oct. 2018), pp. 947–947. ISSN: 0003-066X. DOI: 10.1037/amp0000389.

Abstract: Following a review of extant reporting standards for scientific publication, and reviewing 10 years of experience since publication of the first set of reporting standards by the American Psychological Association (APA; APA Publications and Communications Board Working Group on Journal Article Reporting Standards, 2008), the APA Working Group on Quantitative Research Reporting Standards recommended some modifications to the original standards. Examples of modifications include division of hypotheses, analyses, and conclusions into 3 groupings (primary, secondary, and exploratory) and some changes to the section on meta-analysis. Several new modules are included that report standards for observational studies, clinical trials, longitudinal studies, replication studies, and N-of-1 studies. In addition, standards for analytic methods with unique characteristics and output (structural equation modeling and Bayesian analysis) are included. These proposals were accepted by the Publications and Communications Board of APA and supersede the standards included in the 6th edition of the Publication Manual of the American Psychological Association (APA, 2010).

Asparouhov et al.: Dynamic structural equation models

Asparouhov-Hamaker-Muthen-2018

Tihomir Asparouhov, Ellen L. Hamaker, and Bengt Muthén. "Dynamic structural equation models". In: *Structural Equation Modeling: A Multidisciplinary Journal* 25.3 (2018), pp. 359–388. DOI: 10.1080/10705511.2017.1406803.

Abstract: This article presents dynamic structural equation modeling (DSEM), which can be used to study the evolution of observed and latent variables as well as the structural equation models over time. DSEM is suitable for analyzing intensive longitudinal data where observations from multiple individuals are collected at many points in time. The modeling framework encompasses previously published DSEM models and is a comprehensive attempt to combine time-series modeling with structural equation modeling. DSEM is estimated with Bayesian methods using the Markov chain Monte Carlo Gibbs sampler and the Metropolis-Hastings sampler. We provide a detailed description of the estimation algorithm as implemented in the Mplus software package. DSEM can be used for longitudinal analysis of any duration and with any number of observations across time. Simulation studies are used to illustrate the framework and study the performance of the estimation method. Methods for evaluating model fit are also discussed.

Asparouhov et al.: Latent variable centering of predictors and mediators in multilevel and time-series models **Asparouhov-Muthen-2018**

Tihomir Asparouhov and Bengt Muthén. "Latent variable centering of predictors and mediators in multilevel and time-series models". In: *Structural Equation Modeling: A Multidisciplinary Journal* 26.1 (Sept. 2018), pp. 119–142. ISSN: 1532-8007. DOI: 10.1080/10705511.2018.1511375.

Bainter et al.: Advantages of integrative data analysis for developmental research Bainter-Curran-2015

Sierra A. Bainter and Patrick J. Curran. “Advantages of integrative data analysis for developmental research”. In: *Journal of Cognition and Development* 16.1 (Jan. 2015), pp. 1–10. ISSN: 1532-7647. DOI: 10.1080/15248372.2013.871721.

Abstract: Amid recent progress in cognitive development research, high-quality data resources are accumulating, and data sharing and secondary data analysis are becoming increasingly valuable tools. Integrative data analysis (IDA) is an exciting analytical framework that can enhance secondary data analysis in powerful ways. IDA pools item-level data across multiple studies to make inferences possible both within and across studies and can be used to test questions not possible in individual contributing studies. Some of the potential benefits of IDA include the ability to study longer developmental periods, examine how the measurement of key constructs changes over time, increase subject heterogeneity, and improve statistical power and capability to study rare behaviors. Our goal in this article is to provide a brief overview of the benefits and challenges of IDA in developmental research and to identify additional resources that provide more detailed discussions of this topic.

Barker et al.: Habitual alcohol seeking: Modeling the transition from casual drinking to addiction Barker-Taylor-2014

Jacqueline M Barker and Jane R. Taylor. “Habitual alcohol seeking: Modeling the transition from casual drinking to addiction”. In: *Neuroscience & Biobehavioral Reviews* 47 (Nov. 2014), pp. 281–294. ISSN: 0149-7634. DOI: 10.1016/j.neubiorev.2014.08.012.

Abstract: The transition from goal-directed actions to habitual ethanol seeking models the development of addictive behavior that characterizes alcohol use disorders. The progression to habitual ethanol-seeking behavior occurs more rapidly than for natural rewards, suggesting that ethanol

may act on habit circuit to drive the loss of behavioral flexibility. This review will highlight recent research that has focused on the formation and expression of habitual ethanol seeking, and the commonalities and distinctions between ethanol and natural reward-seeking habits, with the goal of highlighting important, understudied research areas that we believe will lead toward the development of novel treatment and prevention strategies for uncontrolled drinking.

Barlow et al.: Toward a unified treatment for emotional disorders - republished article
Barlow-Allen-Choate-2016

David H. Barlow, Laura B. Allen, and Molly L. Choate. "Toward a unified treatment for emotional disorders - republished article". In: *Behavior Therapy* 47.6 (Nov. 2016), pp. 838–853. ISSN: 0005-7894. DOI: 10.1016/j.beth.2016.11.005.

Abstract: Over 40 years of development of cognitive behavioral approaches to treating anxiety and related emotional disorders have left us with highly efficacious treatments that are increasingly widely accepted. Nevertheless, these manualized protocols have become numerous and somewhat complex, restricting effective training and dissemination. Deepening understanding of the nature of emotional disorders reveals that commonalities in etiology and latent structure among these disorders supercedes differences. This suggests the possibility of distilling a set of psychological procedures that would comprise a unified intervention for emotional disorders. Based on theory and data emerging from the fields of learning, emotional development and regulation, and cognitive science, we identify three fundamental therapeutic components relevant to the treatment of emotional disorders generally. These three components include (a) altering antecedent cognitive reappraisals; (b) preventing emotional avoidance; and (c) facilitating action tendencies not associated with the emotion that is dysregulated. This treatment takes place in the context of provoking emotional expression (emotional exposure) through situational, internal, and somatic (interoceptive cues), as well as through standard mood-induction exercises, and differs from patient to patient only in the situational cues and exercises utilized. Theory and rationale supporting this new approach are described along with some preliminary experience with the protocol. This unified treatment may

represent a more efficient and possibly a more effective strategy in treating emotional disorders, pending further evaluation.

Barnett: Alcohol sensors and their potential for improving clinical care Barnett-2014

Nancy P. Barnett. "Alcohol sensors and their potential for improving clinical care". In: *Addiction* 110.1 (Dec. 2014), pp. 1–3. ISSN: 1360-0443. DOI: 10.1111/add.12764.

Abstract: Alcohol sensors are used successfully to monitor alcohol offenders in criminal justice, but their potential clinical applications with other populations are untapped. Sensors may improve provider understanding about patients' patterns of alcohol use, augment current treatment approaches and identify when alcohol may be interfering with the treatment of other conditions.

Bell et al.: Explaining fixed effects: Random effects modeling of time-series cross-sectional and panel data Bell-Jones-2014

Andrew Bell and Kelvyn Jones. "Explaining fixed effects: Random effects modeling of time-series cross-sectional and panel data". In: *Political Science Research and Methods* 3.1 (May 2014), pp. 133–153. ISSN: 2049-8489. DOI: 10.1017/psrm.2014.7.

Abstract: This article challenges Fixed Effects (FE) modeling as the 'default' for time-series-cross-sectional and panel data. Understanding different within and between effects is crucial when choosing modeling strategies. The downside of Random Effects (RE) modeling—correlated lower-level covariates and higher-level residuals—is omitted-variable bias, solvable with Mundlak's (1978a) formulation. Consequently, RE can provide everything that FE promises and more, as confirmed by Monte-Carlo simulations, which additionally show problems with Plümper and Troeger's FE Vector Decomposition method when data are unbalanced. As well as incorporating time-invariant variables, RE models are readily extendable, with random coefficients, cross-level interactions and complex variance functions. We argue not simply for technical solutions to endogeneity, but for the substantive importance of context/heterogeneity, modeled using RE. The implications extend beyond political science to all multilevel datasets. However, omitted variables could still bias estimated higher-level variable effects; as with any model, care is required in interpretation.

Beltz et al.: Bridging the nomothetic and idiographic approaches to the analysis of clinical data Beltz-Wright-Sprague-etal-2016

Adriene M. Beltz et al. "Bridging the nomothetic and idiographic approaches to the analysis of clinical data". In: *Assessment* 23.4 (May 2016), pp. 447–458. ISSN: 1552-3489. DOI: 10.1177/1073191116648209.

Abstract: The nomothetic approach (i.e., the study of interindividual variation) dominates analyses of clinical data, even though its assumption of homogeneity across people and time is often violated. The idiographic approach (i.e., the study of intraindividual variation) is best suited for analyses of heterogeneous clinical data, but its person-specific methods and results have been criticized as unwieldy. Group iterative multiple model estimation (GIMME) combines the assets of the nomothetic and idiographic approaches by creating person-specific maps that contain a group-level structure. The maps show how intensively measured variables predict and are predicted by each other at different time scales. In this article, GIMME is introduced conceptually and mathematically, and then applied to an empirical data set containing the negative affect, detachment, disinhibition, and hostility composite ratings from the daily diaries of 25 individuals with personality pathology. Results are discussed with the aim of elucidating GIMME's potential for clinical research and practice.

Bernardo et al.: Pathways from collectivist coping to life satisfaction among Chinese: The roles of locus-of-hope Bernardo-Wang-Pesigan-etal-2017

Allan B.I. Bernardo et al. "Pathways from collectivist coping to life satisfaction among Chinese: The roles of locus-of-hope". In: *Personality and Individual Differences* 106 (Feb. 2017), pp. 253–256. ISSN: 0191-8869. DOI: 10.1016/j.paid.2016.10.059.

Abstract: Collectivist coping styles describe approaches to coping in collectivist cultures, but there is not much research on these coping styles and their relationship to well-being. We propose that two collectivistic coping styles (acceptance/reframing/striving and family support) contribute to life satisfaction by drawing from both personal and relational resources, and these resources are reflected in the distinct roles of internal and external-family loci-of-hope. Chinese students completed scales on collectivist coping, locus-of-hope, and life satisfaction. Path analysis of three models indicates that the most parsimonious model describes distinct pathways where acceptance/reframing/striving relates to life satisfaction through internal locus-of-hope and family support relates to life satisfaction through external-family locus-of-hope. The use of specific collectivist coping styles and reports of life satisfaction involves distinct personal and relational resources for coping that are valued in collectivist societies.

Biesanz et al.: Assessing mediational models: Testing and interval estimation for indirect effects **Biesanz-Falk-Savalei-2010**

Jeremy C. Biesanz, Carl F. Falk, and Victoria Savalei. "Assessing mediational models: Testing and interval estimation for indirect effects". In: *Multivariate Behavioral Research* 45.4 (Aug. 2010), pp. 661–701. DOI: 10.1080/00273171.2010.498292.

Abstract: Theoretical models specifying indirect or mediated effects are common in the social sciences. An indirect effect exists when an independent variable's influence on the dependent variable is mediated through an intervening variable. Classic approaches to assessing such mediational hypotheses (Baron & Kenny, 1986; Sobel, 1982) have in recent years been supplemented by computationally intensive methods such as bootstrapping, the distribution of the product methods, and hierarchical Bayesian Markov chain Monte Carlo (MCMC) methods. These different approaches for assessing mediation are illustrated using data from Dunn, Biesanz, Human, and Finn (2007). However, little is known about how these methods perform relative to each other, particularly in more challenging situations, such as with data that are incomplete and/or nonnormal. This article presents an extensive Monte Carlo simulation evaluating a host of approaches for assessing mediation. We examine Type I error rates, power, and coverage. We study normal and nonnormal data as well as complete and incomplete data. In addition, we adapt a method, recently proposed in statistical literature, that does not rely on confidence intervals (CIs) to test the null hypothesis of no indirect effect. The results suggest that the new inferential method–the partial posterior p value–slightly outperforms existing ones in terms of maintaining Type I error rates while maximizing power, especially with incomplete data. Among confidence interval approaches, the bias-corrected accelerated (BCa) bootstrapping approach often has inflated Type I error rates and inconsistent coverage and is not recommended. In contrast, the bootstrapped percentile confidence interval and the hierarchical Bayesian MCMC method perform best overall, maintaining Type I error rates, exhibiting reasonable power, and producing stable and accurate coverage rates.

Blanca et al.: Skewness and kurtosis in real data samples

Blanca-Arnau-LopezMontiel-etal-2013

Maria J. Blanca et al. "Skewness and kurtosis in real data samples". In: *Methodology* 9.2 (May 2013), pp. 78–84. DOI: 10.1027/1614-2241/a000057.

Abstract: Parametric statistics are based on the assumption of normality. Recent findings suggest that Type I error and power can be adversely affected when data are non-normal. This paper aims to assess the distributional shape of real data by examining the values of the third and fourth central moments as a measurement of skewness and kurtosis in small samples. The analysis concerned 693 distributions with a sample size ranging from 10 to 30. Measures of cognitive ability and of other psychological variables were included. The results showed that skewness ranged between -2.49 and 2.33. The values of kurtosis ranged between -1.92 and 7.41. Considering skewness and kurtosis together the results indicated that only 5.5% of distributions were close to expected values under normality. Although extreme contamination does not seem to be very frequent, the findings are consistent with previous research suggesting that normality is not the rule with real data.

Boettiger et al.: An introduction to Rocker: Docker containers for R **Boettiger-Eddelbuettel-2017**

Carl Boettiger and Dirk Eddelbuettel. "An introduction to Rocker: Docker containers for R". In: *The R Journal* 9.2 (2017), p. 527. DOI: 10.32614/rj-2017-065.

Abstract: We describe the Rocker project, which provides a widely-used suite of Docker images with customized R environments for particular tasks. We discuss how this suite is organized, and how these tools can increase portability, scaling, reproducibility, and convenience of R users and developers.

Bollen et al.: A general panel model with random and fixed effects: A structural equations approach **Bollen-Brand-2010**

Kenneth A. Bollen and Jennie E. Brand. "A general panel model with random and fixed effects: A structural equations approach". In: *Social Forces* 89.1 (Sept. 2010), pp. 1–34. ISSN: 1534-7605. DOI: 10.1353/sof.2010.0072.

Abstract: Fixed- and random-effects models for longitudinal data are common in sociology. Their primary advantage is that they control for time-invariant omitted variables. However, analysts face several issues when they employ these models. One is the choice of which to apply; another is that FEM and REM models as usually implemented might be insufficiently flexible. For example, the effects of variables, including the latent time-invariant variable, might change over time. The latent time-invariant variable might correlate with some variables and not others. Lagged endogenous variables might be necessary. Alternatives that move beyond the classic FEM and REM models are known, but they involve estimators and software that make these extended models difficult to implement and to compare. This article presents a general panel model that includes the standard FEM and REM as special cases. In addition, it provides a sequence of nested models that provide a richer range of models that researchers can easily compare with likelihood ratio tests and fit statistics. Furthermore, researchers can implement our general panel model and its special cases in widely available structural equation models software.

Bond et al.: Adjustments for drink size and ethanol content: New results from a self-report diary and transdermal sensor validation study

Bond-Greenfield-Patterson-etal-2014

Jason C. Bond et al. "Adjustments for drink size and ethanol content: New results from a self-report diary and transdermal sensor validation study". In: *Alcoholism: Clinical and Experimental Research* 38.12 (Dec. 2014), pp. 3060–3067. ISSN: 1530-0277. DOI: 10.1111/acer.12589.

Abstract: Background: Prior studies adjusting self-reported measures of alcohol intake for drink size and ethanol (EtOH) content have relied on single-point assessments. Methods: A prospective 28-day diary study investigated magnitudes of drink-EtOH adjustments and factors associated with these adjustments. Transdermal alcohol sensor (TAS) readings and prediction of alcohol-related problems by number of drinks versus EtOH-adjusted intake were used to validate drink-EtOH adjustments. Self-completed event diaries listed up to 4 beverage types and 4 drinking events/d. Eligible volunteers had $\geq$ weekly drinking and $\geq$ 3+ drinks per occasion with $\geq$ 26 reported days and pre- and postsummary measures ($n = 220$). Event reports included drink types, sizes, brands or spirits contents, venues, drinks consumed, and drinking duration. Results: Wine drinks averaged 1.19, beer 1.09, and spirits 1.54 U.S. standard drinks (14 g EtOH). Mean-adjusted alcohol intake was 22% larger using drink size and strength (brand/EtOH concentration) data. Adjusted drink levels were larger than "raw" drinks in all quantity ranges. Individual-level drink-EtOH adjustment ratios (EtOH adjusted/unadjusted amounts) averaged across all days drinking ranged from 0.73 to 3.33 (mean 1.22). Adjustment ratio was only marginally (and not significantly) positively related to usual quantity, frequency, and heavy drinking (all $ps < 0.10$), independent of gender, age, employment, and education, but those with lower incomes (both $p < 0.01$) drank stronger/bigger drinks. Controlling for raw number of drinks and other covariates, degree of adjustment independently predicted alcohol dependence symptoms ($p < 0.01$) and number of consequences ($p < 0.05$). In 30

respondents with sufficiently high-quality TAS readings, higher correlations ($p = 0.04$) were found between the adjusted versus the raw drinks/event and TAS areas under the curve. Conclusions: Absent drink size and strength data, intake assessments are downward biased by at least 20%. Between-subject variation in typical drink content and pour sizes should be addressed in treatment and epidemiological research.

Bosley et al.: Affect Dynamics as Predictors of Symptom Severity and Treatment Response in Mood and Anxiety Disorders: Evidence for Specificity

Bosley-Soyster-Fisher-2019

Hannah G. Bosley, Peter D. Soyster, and Aaron J. Fisher. "Affect Dynamics as Predictors of Symptom Severity and Treatment Response in Mood and Anxiety Disorders: Evidence for Specificity". In: *Journal for Person-Oriented Research* 5.2 (Dec. 2019), pp. 101–113. ISSN: 2003-0177. DOI: 10.17505/jpor.2019.09.

Abstract: Studies of affect dynamics in psychopathology often focus on the prediction of broad constructs like subjective well-being and psychological health. Less is known about how fluctuation in affect over time relates to specific symptom measures (e.g.,anxiety or depression), or how these domains change in treatment. A clinical sample of 32 adults with mood and anxiety disorders (13 generalized anxiety, 5 major depression, 14 comorbid) completed four daily assessments of positive (PA) and negative affect (NA) for 30 days prior to receiving cognitive behavioral treatment. Anxiety and depression symptom severity were assessed pre-and post-treatment. We calculated three metrics of affect dynamics for each person's PA and NA time series: (1) variability(experiencing emotional extremes, the standard deviation of a person's PA or NA vector); (2)instability(magnitude of point-to-point change in emotion, the vector's mean squared successive difference);and (3) inertia(the extent to which emotions self-perpetuate over time, the lag-1 autocorrelation of the vector). Multipleregression models were run to test dynamics of positive and negative affect as between-subjects predictors of symptom severity and pre-to-posttreatment change in symptoms. Findings suggest NA dynamics are unrelated to depression symptom severity or treatment response, but we observed a specific effect of NA instability (MSSD) on both severity and response of anxiety symptoms. All PA dynamics were unrelated to anxiety or depression symptom severity. However, variability, instability, and inertia of PA were all found to relate to treatment response for both anxiety and depression symptoms. Taken together, our results suggest that affect dynamics have some specificity in their relationship to clinicallyrelevant phenomena such assymptom severity and treatmentoutcomes at the between-subjects level of analysis.

Bou et al.: Univariate versus multivariate modeling of panel data: Model specification and goodness-of-fit testing **Bou-Satorra-2017**

Juan Carlos Bou and Albert Satorra. "Univariate versus multivariate modeling of panel data: Model specification and goodness-of-fit testing". In: *Organizational Research Methods* 21.1 (June 2017), pp. 150–196. ISSN: 1552-7425. DOI: 10.1177/1094428117715509.

Abstract: Two approaches are commonly in use for analyzing panel data: the univariate, which arranges data in long format and estimates just one regression equation; and the multivariate, which arranges data in wide format, and simultaneously estimates a set of regression equations. Although technical articles relating the two approaches exist, they do not seem to have had an impact in organizational research. This article revisits the connection between the univariate and multivariate approaches, elucidating conditions under which they yield the same—or similar—results, and discusses their complementariness. The article is addressed to applied researchers. For those familiar only with the univariate approach, it contributes with conceptual simplicity on goodness-of-fit testing and a variety of tests for misspecification (Hausman test, heteroscedasticity, autocorrelation, etc.), and simplifies expanding the model to time-varying parameters, dynamics, measurement error, and so on. For all practitioners, the comparative and side-by-side analyses of the two approaches on two data sets—demonstration data and empirical data with missing values—contributes to broadening their perspective of panel data modeling and expanding their tools for analyses. Both univariate and multivariate analyses are performed in Stata and R.

Bringmann et al.: Assessing temporal emotion dynamics using networks

Bringmann-Pe-Vissers-etal-2016

Laura F. Bringmann, Madeline L. Pe, et al. "Assessing temporal emotion dynamics using networks". In: *Assessment* 23.4 (May 2016), pp. 425–435. ISSN: 1552-3489. DOI: 10.1177/1073191116645909.

Abstract: Multivariate psychological processes have recently been studied, visualized, and analyzed as networks. In this network approach, psychological constructs are represented as complex systems of interacting components. In addition to insightful visualization of dynamics, a network perspective leads to a new way of thinking about the nature of psychological phenomena by offering new tools for studying dynamical processes in psychology. In this article, we explain the rationale of the network approach, the associated methods and visualization, and illustrate it using an empirical example focusing on the relation between the daily fluctuations of emotions and neuroticism. The results suggest that individuals with high levels of neuroticism had a denser emotion network compared with their less neurotic peers. This effect is especially pronounced for the negative emotion network, which is in line with previous studies that found a denser network in depressed subjects than in healthy subjects. In sum, we show how the network approach may offer new tools for studying dynamical processes in psychology.

Bringmann et al.: A network approach to psychopathology: New insights into clinical longitudinal data

Bringmann-Vissers-Wichers-etal-2013

Laura F. Bringmann, Nathalie Vissers, et al. "A network approach to psychopathology: New insights into clinical longitudinal data". In: *PLoS ONE* 8.4 (Apr. 2013). ISSN: 1932-6203. DOI: 10.1371/journal.pone.0060188.

Abstract: In the network approach to psychopathology, disorders are conceptualized as networks of mutually interacting symptoms (e.g., depressed mood) and transdiagnostic factors (e.g., rumination). This suggests that it is necessary to study how symptoms dynamically interact over time in

a network architecture. In the present paper, we show how such an architecture can be constructed on the basis of time-series data obtained through Experience Sampling Methodology (ESM). The proposed methodology determines the parameters for the interaction between nodes in the network by estimating a multilevel vector autoregression (VAR) model on the data. The methodology allows combining between-subject and within-subject information in a multilevel framework. The resulting network architecture can subsequently be analyzed through network analysis techniques. In the present study, we apply the method to a set of items that assess mood-related factors. We show that the analysis generates a plausible and replicable network architecture, the structure of which is related to variables such as neuroticism; that is, for subjects who score high on neuroticism, worrying plays a more central role in the network. Implications and extensions of the methodology are discussed.

Brody et al.: Is resilience only skin deep?: Rural African Americans' socioeconomic status-related risk and competence in preadolescence and psychological adjustment and allostatic load at age 19 **Brody-Yu-Chen-etal-2013**

Gene H. Brody et al. "Is resilience only skin deep?: Rural African Americans' socioeconomic status-related risk and competence in preadolescence and psychological adjustment and allostatic load at age 19". In: *Psychological Science* 24.7 (May 2013), pp. 1285–1293. ISSN: 1467-9280. DOI: 10.1177/0956797612471954.

Abstract: Many African American youth may develop high levels of allostatic load, a measure of physiological wear and tear on the body, by developing psychosocial competence under conditions of high risk related to socioeconomic status (SES). The current study was designed to test this hypothesis, which is based on John Henryism theory. In a representative sample of 489 African American youth living in the rural South, cumulative SES-related risks and teacher-reported competence were assessed at ages 11 to 13; depressive symptoms, externalizing behavior, and allostatic load were assessed at age 19. The data revealed that rural African American preadolescents who evinced high psychosocial competence under conditions of high cumulative SES-related risk displayed low levels of adjustment problems along with high allostatic load at age 19. These results suggest that, for many rural African Americans, resilience may indeed be only "skin deep."

Castro-Schilo et al.: Comparison of nomothetic versus idiographic-oriented methods for making predictions about distal outcomes from time series data

CastroSchilo-Ferrer-2013

Laura Castro-Schilo and Emilio Ferrer. "Comparison of nomothetic versus idiographic-oriented methods for making predictions about distal outcomes from time series data". In: *Multivariate Behavioral Research* 48.2 (Mar. 2013), pp. 175–207. ISSN: 1532-7906. DOI: 10.1080/00273171.2012.736042.

Abstract: We illustrate the idiographic/nomothetic debate by comparing 3 approaches to using daily self-report data on affect for predicting relationship quality and breakup. The 3 approaches included (a) the first day in the series of daily data; (b) the mean and variability of the daily series; and (c) parameters from dynamic factor analysis, a statistical model that uses all measurement occasions to estimate the structure and dynamics of the data. Our results indicated that data from the first measurement occasion does not provide information about the couples' relationship quality or breakup 1 to 2 years later. The mean and variability of the time series, however, were more informative: females' average positive and negative affect across time was related to relationship quality, whereas males' variability in negative affect across time was predictive of breakup. The dynamic factor analysis, in turn, allowed us to extract information central to the dyadic dynamics. This information proved useful to predict relationship quality but not breakup. The importance of examining intraindividual variability and couple dynamics is highlighted.

Chen et al.: Vector autoregression, structural equation modeling, and their synthesis in neuroimaging data analysis **Chen-Daniel-Ziad-etal-2011**

Gang Chen et al. "Vector autoregression, structural equation modeling, and their synthesis in

neuroimaging data analysis”. In: *Computers in Biology and Medicine* 41.12 (Dec. 2011), pp. 1142–1155. DOI: 10.1016/j.compbiomed.2011.09.004.

Abstract: Vector autoregression (VAR) and structural equation modeling (SEM) are two popular brain-network modeling tools. VAR, which is a data-driven approach, assumes that connected regions exert time-lagged influences on one another. In contrast, the hypothesis-driven SEM is used to validate an existing connectivity model where connected regions have contemporaneous interactions among them. We present the two models in detail and discuss their applicability to FMRI data, and their interpretational limits. We also propose a unified approach that models both lagged and contemporaneous effects. The unifying model, structural vector autoregression (SVAR), may improve statistical and explanatory power, and avoid some prevalent pitfalls that can occur when VAR and SEM are utilized separately.

M. W.-L. Cheung: Multivariate meta-analysis as structural equation models

Cheung-2013

Mike W.-L. Cheung. “Multivariate meta-analysis as structural equation models”. In: *Structural Equation Modeling: A Multidisciplinary Journal* 20.3 (July 2013), pp. 429–454. ISSN: 1532-8007. DOI: 10.1080/10705511.2013.797827.

Abstract: Multivariate meta-analysis has become increasingly popular in the educational, social, and medical sciences. It is because the outcome measures in a meta-analysis can involve more than one effect size. This article proposes 2 mathematically equivalent models to implement multivariate meta-analysis in structural equation modeling (SEM). Specifically, this article shows how multivariate fixed-, random- and mixed-effects meta-analyses can be formulated as structural equation models. metaSEM (a free R package based on OpenMx) and Mplus are used to implement the proposed procedures. A real data set is used to illustrate the procedures. Formulating multivariate meta-analysis as structural equation models provides many new research opportunities for

methodological development in both meta-analysis and SEM. Issues related to and extensions on the SEM-based meta-analysis are discussed.

M. W.-L. Cheung: Modeling dependent effect sizes with three-level meta-analyses: A structural equation modeling approach **Cheung-2014**

Mike W.-L. Cheung. "Modeling dependent effect sizes with three-level meta-analyses: A structural equation modeling approach". In: *Psychological Methods* 19.2 (2014), pp. 211–229. ISSN: 1082-989X. DOI: 10.1037/a0032968.

Abstract: Meta-analysis is an indispensable tool used to synthesize research findings in the social, educational, medical, management, and behavioral sciences. Most meta-analytic models assume independence among effect sizes. However, effect sizes can be dependent for various reasons. For example, studies might report multiple effect sizes on the same construct, and effect sizes reported by participants from the same cultural group are likely to be more similar than those reported by other cultural groups. This article reviews the problems and common methods to handle dependent effect sizes. The objective of this article is to demonstrate how 3-level meta-analyses can be used to model dependent effect sizes. The advantages of the structural equation modeling approach over the multilevel approach with regard to conducting a 3-level meta-analysis are discussed. This article also seeks to extend the key concepts of Q statistics, I2, and R2 from 2-level meta-analyses to 3-level meta-analyses. The proposed procedures are implemented using the open source metaSEM package for the R statistical environment. Two real data sets are used to illustrate these procedures. New research directions related to 3-level meta-analyses are discussed.

M. W.-L. Cheung: Computing multivariate effect sizes and their sampling covariance matrices with structural equation modeling: Theory, examples, and computer simulations **Cheung-2018**

Mike W.-L. Cheung. "Computing multivariate effect sizes and their sampling covariance matrices with structural equation modeling: Theory, examples, and computer simulations". In: *Frontiers in Psychology* 9 (Aug. 2018). ISSN: 1664-1078. DOI: 10.3389/fpsyg.2018.01387.

Abstract: In the social and behavioral sciences, it is recommended that effect sizes and their sampling variances be reported. Formulas for common effect sizes such as standardized and raw mean differences, correlation coefficients, and odds ratios are well known and have been well studied. However, the statistical properties of multivariate effect sizes have received less attention in the literature. This study shows how structural equation modeling (SEM) can be used to compute multivariate effect sizes and their sampling covariance matrices. We focus on the standardized mean difference (multiple-treatment and multiple-endpoint studies) with or without the assumption of the homogeneity of variances (or covariance matrices) in this study. Empirical examples were used to illustrate the procedures in R. Two computer simulation studies were used to evaluate the empirical performance of the SEM approach. The findings suggest that in multiple-treatment and multiple-endpoint studies, when the assumption of the homogeneity of variances (or covariance matrices) is questionable, it is preferable not to impose this assumption when estimating the effect sizes. Implications and further directions are discussed.

M. W.-L. Cheung et al.: Random-effects models for meta-analytic structural equation modeling: Review, issues, and illustrations **Cheung-Cheung-2016**

Mike W.-L. Cheung and Shu Fai Cheung. "Random-effects models for meta-analytic structural equation modeling: Review, issues, and illustrations". In: *Research Synthesis Methods* 7.2 (June 2016), pp. 140–155. ISSN: 1759-2887. DOI: 10.1002/jrsm.1166.

Abstract: Meta-analytic structural equation modeling (MASEM) combines the techniques of meta-analysis and structural equation modeling for the purpose of synthesizing correlation or covariance matrices and fitting structural equation models on the pooled correlation or covariance matrix. Both fixed-effects and random-effects models can be defined in MASEM. Random-effects models are well known in conventional meta-analysis but are less studied in MASEM. The primary objective of this paper was to address issues related to random-effects models in MASEM. Specifically, we compared two different random-effects models in MASEM—correlation-based MASEM and parameter-based MASEM—and explored their strengths and limitations. Two examples were used to illustrate the similarities and differences between these models. We offered some practical guidelines for choosing between these two models. Future directions for research on random-effects models in MASEM were also discussed.

Chow et al.: Equivalence and differences between structural equation modeling and state-space modeling techniques **Chow-Ho-Hamaker-etal-2010**

Sy-Miin Chow, Moon-ho R. Ho, et al. "Equivalence and differences between structural equation modeling and state-space modeling techniques". In: *Structural Equation Modeling: A Multidisciplinary Journal* 17.2 (Apr. 2010), pp. 303–332. DOI: 10.1080/10705511003661553.

Abstract: State-space modeling techniques have been compared to structural equation modeling (SEM) techniques in various contexts but their unique strengths have often been overshadowed by their similarities to SEM. In this article, we provide a comprehensive discussion of these 2 approaches' similarities and differences through analytic comparisons and numerical simulations, with a focus on their use in representing intraindividual dynamics and interindividual differences. To demonstrate the respective strengths and weaknesses of the 2 approaches in representing these 2 aspects, we simulated data under (a) a cross-sectional common factor model, (b) a latent difference score model with random effects in intercept and slope, and (c) a bivariate dynamic factor analysis model with auto- and cross-regression parameters. Possible ways in which SEM and state-space modeling can be utilized as complementary tools in representing human developmental and other related processes are discussed.

Chow et al.: The cusp catastrophe model as cross-sectional and longitudinal mixture structural equation models **Chow-Witkiewitz-Grasman-etal-2015**

Sy-Miin Chow, Katie Witkiewitz, et al. “The cusp catastrophe model as cross-sectional and longitudinal mixture structural equation models”. In: *Psychological Methods* 20.1 (Mar. 2015), pp. 142–164. ISSN: 1082-989X. DOI: 10.1037/a0038962.

Abstract: Catastrophe theory (Thom, 1972, 1993) is the study of the many ways in which continuous changes in a system’s parameters can result in discontinuous changes in 1 or several outcome variables of interest. Catastrophe theory—inspired models have been used to represent a variety of change phenomena in the realm of social and behavioral sciences. Despite their promise, widespread applications of catastrophe models have been impeded, in part, by difficulties in performing model fitting and model comparison procedures. We propose a new modeling framework for testing 1 kind of catastrophe model—the cusp catastrophe model—as a mixture structural equation model (MSEM) when cross-sectional data are available; or alternatively, as an MSEM with regime-switching (MSEM-RS) when longitudinal panel data are available. The proposed models and the advantages offered by this alternative modeling framework are illustrated using 2 empirical examples and a simulation study.

Chow et al.: Nonlinear regime-switching state-space (RSSS) models **Chow-Zhang-2013**

Sy-Miin Chow and Guangjian Zhang. “Nonlinear regime-switching state-space (RSSS) models”. In: *Psychometrika* 78.4 (Oct. 2013), pp. 740–768. ISSN: 1860-0980. DOI: 10.1007/s11336-013-9330-8.

Abstract: Nonlinear dynamic factor analysis models extend standard linear dynamic factor analysis models by allowing time series processes to be nonlinear at the latent level (e.g., involving interaction between two latent processes). In practice, it is often of interest to identify the phases—namely, latent “regimes” or classes—during which a system is characterized by distinctly different dynamics. We propose a new class of models, termed nonlinear regime-switching state-space (RSSS) models,

which subsumes regime-switching nonlinear dynamic factor analysis models as a special case. In nonlinear RSSS models, the change processes within regimes, represented using a state-space model, are allowed to be nonlinear. An estimation procedure obtained by combining the extended Kalman filter and the Kim filter is proposed as a way to estimate nonlinear RSSS models. We illustrate the utility of nonlinear RSSS models by fitting a nonlinear dynamic factor analysis model with regime-specific cross-regression parameters to a set of experience sampling affect data. The parallels between nonlinear RSSS models and other well-known discrete change models in the literature are discussed briefly.

Chow et al.: Dynamic factor analysis models with time-varying parameters

Chow-Zu-Shifren-etal-2011

Sy-Miin Chow, Jiyun Zu, et al. "Dynamic factor analysis models with time-varying parameters". In: *Multivariate Behavioral Research* 46.2 (Apr. 2011), pp. 303–339. ISSN: 1532-7906. DOI: 10.1080/00273171.2011.563697.

Abstract: Dynamic factor analysis models with time-varying parameters offer a valuable tool for evaluating multivariate time series data with time-varying dynamics and/or measurement properties. We use the Dynamic Model of Activation proposed by Zautra and colleagues (Zautra, Potter, & Reich, 1997) as a motivating example to construct a dynamic factor model with vector autoregressive relations and time-varying cross-regression parameters at the factor level. Using techniques drawn from the state-space literature, the model was fitted to a set of daily affect data (over 71 days) from 10 participants who had been diagnosed with Parkinson's disease. Our empirical results lend partial support and some potential refinement to the Dynamic Model of Activation with regard to how the time dependencies between positive and negative affects change over time. A simulation study is conducted to examine the performance of the proposed techniques when (a) changes in the time-varying parameters are represented using the true model of change, (b) supposedly time-invariant parameters are represented as time-varying, and (c) the time-varying parameters show discrete shifts that are approximated using an autoregressive model of differences.

Clapp et al.: Examining the social ecology of a bar-crawl: An exploratory pilot study

Clapp-Madden-Mooney-etal-2017

John D. Clapp et al. "Examining the social ecology of a bar-crawl: An exploratory pilot study". In: *PLOS ONE* 12.9 (Sept. 2017). Ed. by Etsuro Ito, pp. 1–27. ISSN: 1932-6203. DOI: 10.1371/journal.pone.0185238.

Abstract: Many of the problems associated with alcohol occur after a single drinking event (e.g. drink driving, assault). These acute alcohol problems have a huge global impact and account for a large percentage of unintentional and intentional injuries in the world. Nonetheless, alcohol research and preventive interventions rarely focus on drinking at the event-level since drinking events are complex, dynamic, and methodologically challenging to observe. This exploratory study provides an example of how event-level data may be collected, analyzed, and interpreted. The drinking behavior of twenty undergraduate students enrolled at a large Midwestern public university was observed during a single bar crawl event that is organized by students annually. Alcohol use was monitored with transdermal alcohol devices coupled with ecological momentary assessments and geospatial data. "Small N, Big Data" studies have the potential to advance health behavior theory and to guide real-time interventions. However, such studies generate large amounts of within subject data that can be challenging to analyze and present. This study examined how to visually display event-level data and also explored the relationship between some basic indicators and alcohol consumption.

Crocetti et al.: Generalized anxiety symptoms and identity processes in cross-cultural samples of adolescents from the general population

Crocetti-Hale-Dimitrova-etal-2015

Elisabetta Crocetti et al. "Generalized anxiety symptoms and identity processes in cross-cultural samples of adolescents from the general population". In: *Child & Youth Care Forum* 44.2 (2015), pp. 159–174. ISSN: 1573-3319. DOI: 10.1007/s10566-014-9275-9.

Abstract: Background: Approximately 20% of adolescents around the world experience mental health problems, most commonly depression or anxiety. High levels of anxiety disorder symptoms can hinder adolescent development, persist into adulthood, and predict negative mental outcomes, such as suicidal ideation and attempts. Objectives: We analyzed generalized anxiety disorder (GAD) symptoms in cross-cultural samples from the general population. We sought to examine cultural and gender differences, and correlates of GAD symptoms in samples of adolescents from six countries located in three different continents (Europe: Bulgaria, Italy, the Netherlands; Africa: Kenya; Asia: China and Philippines). Methods: Participants were 3,445 (51% male) adolescents aged between 14 and 18 years old. They filled self-report measures of GAD symptoms and identity. Results: First, it was found that the scores on GAD symptoms varied significantly across countries, with Dutch respondents reporting the lowest levels whereas Filipino participants exhibited the highest levels of GAD symptoms. Second, gender differences (i.e., girls reported more GAD symptoms than boys) were significant in each country (as well as in the total sample), with the only exception being that of Kenya. Third, GAD symptoms were significantly related to identity processes and similarities and differences across countries were examined.

Curran et al.: The disaggregation of within-person and between-person effects in longitudinal models of change **Curran-Bauer-2011**

Patrick J. Curran and Daniel J. Bauer. “The disaggregation of within-person and between-person effects in longitudinal models of change”. In: *Annual Review of Psychology* 62.1 (Jan. 2011), pp. 583–619. DOI: 10.1146/annurev.psych.093008.100356.

Abstract: Longitudinal models are becoming increasingly prevalent in the behavioral sciences, with key advantages including increased power, more comprehensive measurement, and establishment of temporal precedence. One particularly salient strength offered by longitudinal data is the ability to disaggregate between-person and within-person effects in the regression of an outcome on a time-varying covariate. However, the ability to disaggregate these effects has not been fully capitalized upon in many social science research applications. Two likely reasons for this omission are the

general lack of discussion of disaggregating effects in the substantive literature and the need to overcome several remaining analytic challenges that limit existing quantitative methods used to isolate these effects in practice. This review explores both substantive and quantitative issues related to the disaggregation of effects over time, with a particular emphasis placed on the multilevel model. Existing analytic methods are reviewed, a general approach to the problem is proposed, and both the existing and proposed methods are demonstrated using several artificial data sets. Potential limitations and directions for future research are discussed, and recommendations for the disaggregation of effects in practice are offered.

Curran et al.: The separation of between-person and within-person components of individual change over time: A latent curve model with structured residuals.

Curran-Howard-Bainter-etal-2014

Patrick J. Curran, Andrea L. Howard, et al. "The separation of between-person and within-person components of individual change over time: A latent curve model with structured residuals." In: *Journal of Consulting and Clinical Psychology* 82.5 (2014), pp. 879–894. ISSN: 0022-006X. DOI: 10.1037/a0035297.

Abstract: Objective: Although recent statistical and computational developments allow for the empirical testing of psychological theories in ways not previously possible, one particularly vexing challenge remains: how to optimally model the prospective, reciprocal relations between 2 constructs as they developmentally unfold over time. Several analytic methods currently exist that attempt to model these types of relations, and each approach is successful to varying degrees. However, none provide the unambiguous separation over time of between-person and within-person components of stability and change, components that are often hypothesized to exist in the psychological sciences. Our goal in this article is to propose and demonstrate a novel extension of the multivariate latent curve model to allow for the disaggregation of these effects. Method: We begin with a review of the standard latent curve models and describe how these primarily capture between-person differences in change. We then extend this model to allow for regression structures among the time-specific residuals to capture within-person differences in change. Results: We demonstrate this model using an artificial data set generated to mimic the developmental relation between alcohol use and depressive symptomatology spanning 5 repeated measures. Conclusions: We obtain a specificity of results from the proposed analytic strategy that is not available from other existing methodologies. We conclude with potential limitations of our approach and directions for future research.

S. de Haan-Rietdijk et al.: On the use of mixed Markov models for intensive longitudinal data deHaanRietdijk-Kuppens-Bergeman-etal-2017

S. de Haan-Rietdijk et al. "On the use of mixed Markov models for intensive longitudinal data". In: *Multivariate Behavioral Research* 52.6 (Sept. 2017), pp. 747–767. ISSN: 1532-7906. DOI: 10.1080/00273171.2017.1370364.

Abstract: Markov modeling presents an attractive analytical framework for researchers who are interested in state-switching processes occurring within a person, dyad, family, group, or other system over time. Markov modeling is flexible and can be used with various types of data to study observed or latent state-switching processes, and can include subject-specific random effects to account for heterogeneity. We focus on the application of mixed Markov models to intensive longitudinal data sets in psychology, which are becoming ever more common and provide a rich description of each subject's process. We examine how specifications of a Markov model change when continuous random effect distributions are included, and how mixed Markov models can be used in the intensive longitudinal research context. Advantages of Bayesian estimation are discussed and the approach is illustrated by two empirical applications.

Silvia de Haan-Rietdijk et al.: What's in a day? A guide to decomposing the variance in intensive longitudinal data deHaanRietdijk-Kuppens-Hamaker-2016

Silvia de Haan-Rietdijk, Peter Kuppens, and Ellen L. Hamaker. "What's in a day? A guide to

decomposing the variance in intensive longitudinal data". In: *Frontiers in Psychology* 7 (June 2016). ISSN: 1664-1078. DOI: 10.3389/fpsyg.2016.00891.

Abstract: In recent years there has been a growing interest in the use of intensive longitudinal research designs to study within-person processes. Examples are studies that use experience sampling data and autoregressive modeling to investigate emotion dynamics and between-person differences therein. Such designs often involve multiple measurements per day and multiple days per person, and it is not clear how this nesting of the data should be accounted for: That is, should such data be considered as two-level data (which is common practice at this point), with occasions nested in persons, or as three-level data with beeps nested in days which are nested in persons. We show that a significance test of the day-level variance in an empty three-level model is not reliable when there is autocorrelation. Furthermore, we show that misspecifying the number of levels can lead to spurious or misleading findings, such as inflated variance or autoregression estimates. Throughout the paper we present instructions and R code for the implementation of the proposed models, which includes a novel three-level AR(1) model that estimates moment-to-moment inertia and day-to-day inertia. Based on our simulations we recommend model selection using autoregressive multilevel models in combination with the AIC. We illustrate this method using empirical emotion data from two independent samples, and discuss the implications and the relevance of the existence of a day level for the field.

Silvia de Haan-Rietdijk et al.: Discrete- vs. continuous-time modeling of unequally spaced experience sampling method data

deHaanRietdijk-Voelkle-Keijsers-Hamaker-2017

Silvia de Haan-Rietdijk, Manuel C. Voelkle, et al. "Discrete- vs. continuous-time modeling of unequally spaced experience sampling method data". In: *Frontiers in Psychology* 8 (Oct. 2017). ISSN: 1664-1078. DOI: 10.3389/fpsyg.2017.01849.

Abstract: The Experience Sampling Method is a common approach in psychological research for collecting intensive longitudinal data with high ecological validity. One characteristic of ESM data is that it is often unequally spaced, because the measurement intervals within a day are deliberately varied, and measurement continues over several days. This poses a problem for discrete-time (DT) modeling approaches, which are based on the assumption that all measurements are equally spaced. Nevertheless, DT approaches such as (vector) autoregressive modeling are often used to analyze ESM data, for instance in the context of affective dynamics research. There are equivalent continuous-time (CT) models, but they are more difficult to implement. In this paper we take a pragmatic approach and evaluate the practical relevance of the violated model assumption in DT AR(1) and VAR(1) models, for the $N = 1$ case. We use simulated data under an ESM measurement design to investigate the bias in the parameters of interest under four different model implementations, ranging from the true CT model that accounts for all the exact measurement times, to the crudest possible DT model implementation, where even the nighttime is treated as a regular interval. An analysis of empirical affect data illustrates how the differences between DT and CT modeling can play out in practice. We find that the size and the direction of the bias in DT (V)AR models for unequally spaced ESM data depend quite strongly on the true parameter in addition to data characteristics. Our recommendation is to use CT modeling whenever possible, especially now that new software implementations have become available.

Deboeck et al.: Integration of stochastic differential equations using structural equation modeling: A method to facilitate model fitting and pedagogy Deboeck-Boulton-2016

Pascal R. Deboeck and Aaron J. Boulton. "Integration of stochastic differential equations using structural equation modeling: A method to facilitate model fitting and pedagogy". In: *Structural Equation Modeling: A Multidisciplinary Journal* 23.6 (Oct. 2016), pp. 888–903. ISSN: 1532-8007. DOI: 10.1080/10705511.2016.1218763.

Abstract: Stochastic differential equation (SDE) models are a promising method for modeling intraindividual change and variability. Applications of SDEs in the social sciences are relatively lim-

ited, as these models present conceptual and programming challenges. This article presents a novel method for conceptualizing SDEs. This method uses structural equation modeling (SEM) conventions to simplify SDE specification, the flexibility of SEM to expand the range of SDEs that can be fit, and SEM diagram conventions to facilitate the teaching of SDE concepts. This method is a variation of latent difference scores (McArdle, 2009; McArdle & Hamagami, 2001) and the oversampling approach (Singer, 2012), and approximates the advantages of analytic methods such as the exact discrete model (Oud & Jansen, 2000) while retaining the modeling flexibility of methods such as latent differential equation modeling (Boker, Neale, & Rausch, 2004). A simulation and empirical example are presented to illustrate that this method can be implemented on current computing hardware, produces good approximations of analytic solutions, and can flexibly accommodate novel models.

Deboeck et al.: No need to be discrete: A method for continuous time mediation analysis — Deboeck-Preacher-2015

Pascal R. Deboeck and Kristopher J. Preacher. "No need to be discrete: A method for continuous time mediation analysis". In: *Structural Equation Modeling: A Multidisciplinary Journal* 23.1 (June 2015), pp. 61–75. DOI: 10.1080/10705511.2014.973960.

Abstract: Mediation is one concept that has shaped numerous theories. The list of problems associated with mediation models, however, has been growing. Mediation models based on cross-sectional data can produce unexpected estimates, so much so that making longitudinal or causal inferences is inadvisable. Even longitudinal mediation models have faults, as parameter estimates produced by these models are specific to the lag between observations, leading to much debate over appropriate lag selection. Using continuous time models (CTMs) rather than commonly employed discrete time models, one can estimate lag-independent parameters. We demonstrate methodology that allows for continuous time mediation analyses, with attention to concepts such as indirect and direct effects, partial mediation, the effect of lag, and the lags at which relations become maximal. A simulation compares common longitudinal mediation methods with CTMs. Reanalysis of a published covariance matrix demonstrates that CTMs can be fit to data used in longitudinal mediation studies.

Dejonckheere et al.: Complex affect dynamics add limited information to the prediction of psychological well-being **Dejonckheere-Mestdagh-Houben-etal-2019**

Egon Dejonckheere et al. "Complex affect dynamics add limited information to the prediction of psychological well-being". In: *Nature Human Behaviour* 3.5 (Apr. 2019), pp. 478–491. ISSN: 2397-3374. DOI: 10.1038/s41562-019-0555-0.

Abstract: Over the years, many studies have demonstrated a relation between emotion dynamics and psychological well-being1. Because our emotional life is inherently time-dynamic, affective scientists argue that, next to how positive or negative we feel on average, patterns of emotional change are informative for mental health. This growing interest initiated a surge in new affect dynamic measures, each claiming to capture a unique dynamical aspect of our emotional life, crucial for understanding well-being. Although this accumulation suggests scientific progress, researchers have not always evaluated (a) how different affect dynamic measures empirically interrelate and (b) what their added value is in the prediction of psychological well-being. Here, we address these questions by analysing affective time series data from 15 studies (n=1,777). We show that (a) considerable interdependencies between measures exist, suggesting that single dynamics often do not convey unique information, and (b) dynamic measures have little added value over mean levels of positive and negative affect (and variance in these affective states) when predicting individual differences in three indicators of well-being (life satisfaction, depressive symptoms and borderline symptoms). Our findings indicate that conventional emotion research is currently unable to demonstrate independent relations between affect dynamics and psychological well-being.

Demeshko et al.: A novel continuous and structural VAR modeling approach and its application to reactor noise analysis **Demeshko-Washio-Kawahara-etal-2015**

Marina Demeshko et al. "A novel continuous and structural VAR modeling approach and its application to reactor noise analysis". In: *ACM Transactions on Intelligent Systems and Technology* 7.2 (Nov. 2015), pp. 1–22. DOI: 10.1145/2710025.

Abstract: A vector autoregressive model in discrete time domain (DVAR) is often used to analyze continuous time, multivariate, linear Markov systems through their observed time series data sampled at discrete timesteps. Based on previous studies, the DVAR model is supposed to be a non-canonical representation of the system, that is, it does not correspond to a unique system bijectively. However, in this article, we characterize the relations of the DVAR model with its corresponding Structural Vector AR (SVAR) and Continuous Time Vector AR (CTVAR) models through a finite difference method across continuous and discrete time domain. We further clarify that the DVAR model of a continuous time, multivariate, linear Markov system is canonical under a highly generic condition. Our analysis shows that we can uniquely reproduce its SVAR and CTVAR models from the DVAR model. Based on these results, we propose a novel Continuous and Structural Vector Autoregressive (CSVAR) modeling approach to derive the SVAR and the CTVAR models from their DVAR model empirically derived from the observed time series of continuous time linear Markov systems. We demonstrate its superior performance through some numerical experiments on both artificial and real-world data.

Devlieger et al.: Hypothesis testing using factor score regression: A comparison of four methods **Devlieger-Mayer-Rosseel-2016**

Ines Devlieger, Axel Mayer, and Yves Rosseel. "Hypothesis testing using factor score regression: A comparison of four methods". In: *Educational and Psychological Measurement* 76.5 (July 2016), pp. 741–770. ISSN: 1552-3888. DOI: 10.1177/0013164415607618.

Abstract: In this article, an overview is given of four methods to perform factor score regression (FSR), namely regression FSR, Bartlett FSR, the bias avoiding method of Skrondal and Laake, and the bias correcting method of Croon. The bias correcting method is extended to include a reliable standard error. The four methods are compared with each other and with structural equation modeling (SEM) by using analytic calculations and two Monte Carlo simulation studies to examine their finite sample characteristics. Several performance criteria are used, such as the bias using the unstandardized and standardized parameterization, efficiency, mean square error, standard error bias, type I error rate, and power. The results show that the bias correcting method, with the newly developed standard error, is the only suitable alternative for SEM. While it has a higher standard error bias than SEM, it has a comparable bias, efficiency, mean square error, power, and type I error rate.

Devlieger et al.: Factor score path analysis: An alternative for SEM?

Devlieger-Rosseel-2017

Ines Devlieger and Yves Rosseel. "Factor score path analysis: An alternative for SEM?" In: *Methodology* 13.Supplement 1 (June 2017), pp. 31–38. ISSN: 1614-2241. DOI: 10.1027/1614-2241/a000130.

Abstract: Theoretical researchers consider Structural Equation Modeling (SEM) to be the preferred method to study the relationships among latent variables. However, SEM has the disadvantage of requiring a large sample size, especially if the model is complex. Furthermore, since SEM estimates all parameters simultaneously, one misspecification in the model may influence the whole model. For these reasons, applied researchers often use a two-step Factor Score Regression (FSR) approach. In the first step, factor scores are calculated for the latent variables, which are used to perform a linear regression in the second step. However, this method results in incorrect regression coefficients. Croon (2002) developed a method that corrects for this bias. We combine this method of Croon (2002) with path analysis, resulting in Factor Score Path Analysis. This method results in correct path coefficients and has some advantages over SEM: it requires smaller sample sizes, can handle more complex models and the method is less sensitive to misspecifications, because of its stepwise

nature. In conclusion, this method can be a suitable alternative for SEM, when one is dealing with a complex model and small sample sizes.

Doñamayor et al.: The involuntary nature of binge drinking: Goal directedness and awareness of intention **Donamayor-Strelchuk-Baek-etal-2017**

Nuria Doñamayor et al. "The involuntary nature of binge drinking: Goal directedness and awareness of intention". In: *Addiction Biology* 23.1 (Apr. 2017), pp. 515–526. ISSN: 1369-1600. DOI: 10.1111/adb.12505.

Abstract: Binge drinking represents a public health issue and is a known risk factor in the development of alcohol use disorders. Previous studies have shown behavioural as well as neuroanatomical alterations associated with binge drinking. Here, we address the question of the automaticity or involuntary nature of the behaviour by assessing goal-directed behaviour and intentionality. In this study, we used a computational two-step task, designed to discern between model-based/goal-directed and model-free/habitual behaviours, and the classic Libet clock task, to study intention awareness, in a sample of 31 severe binge drinkers (BD) and 35 matched healthy volunteers. We observed that BD had impaired goal-directed behaviour in the two-step task compared with healthy volunteers. In the Libet clock task, BD showed delayed intention awareness. Further, we demonstrated that alcohol use severity, as reflected by the alcohol use disorders identification test, correlated with decreased conscious awareness of volitional intention in BD, although it was unrelated to performance on the two-step task. However, the time elapsed since the last drinking binge influenced the model-free scores, with BD showing less habitual behaviour after longer abstinence. Our findings suggest that the implementation of goal-directed strategies and the awareness of volitional intention are affected in current heavy alcohol users. However, the modulation of these impairments by alcohol use severity and abstinence suggests a state effect of alcohol use in these measures and that top-down volitional control might be ameliorated with alcohol use cessation.

Driver et al.: Continuous time structural equation modeling with R package ctsem

Driver-Oud-Voelkle-2017

Charles C. Driver, Johan H. L. Oud, and Manuel C. Voelkle. “Continuous time structural equation modeling with R package ctsem”. In: *Journal of Statistical Software* 77.5 (2017). ISSN: 1548-7660. DOI: 10.18637/jss.v077.i05.

Abstract: We introduce ctsem, an R package for continuous time structural equation modeling of panel ($N > 1$) and time series ($N = 1$) data, using full information maximum likelihood. Most dynamic models (e.g., cross-lagged panel models) in the social and behavioural sciences are discrete time models. An assumption of discrete time models is that time intervals between measurements are equal, and that all subjects were assessed at the same intervals. Violations of this assumption are often ignored due to the difficulty of accounting for varying time intervals, therefore parameter estimates can be biased and the time course of effects becomes ambiguous. By using stochastic differential equations to estimate an underlying continuous process, continuous time models allow for any pattern of measurement occasions. By interfacing to OpenMx, ctsem combines the flexible specification of structural equation models with the enhanced data gathering opportunities and improved estimation of continuous time models. ctsem can estimate relationships over time for multiple latent processes, measured by multiple noisy indicators with varying time intervals between observations. Within and between effects are estimated simultaneously by modeling both observed covariates and unobserved heterogeneity. Exogenous shocks with different shapes, group differences, higher order diffusion effects and oscillating processes can all be simply modeled. We first introduce and define continuous time models, then show how to specify and estimate a range of continuous time models using ctsem.

Driver et al.: Hierarchical Bayesian continuous time dynamic modeling.

Driver-Voelkle-2018

Charles C. Driver and Manuel C. Voelkle. “Hierarchical Bayesian continuous time dynamic modeling." In: *Psychological Methods* 23.4 (Dec. 2018), pp. 774–799. ISSN: 1082-989X. DOI: 10.1037/met0000168.

Abstract: Continuous time dynamic models are similar to popular discrete time models such as autoregressive cross-lagged models, but through use of stochastic differential equations can accurately account for differences in time intervals between measurements, and more parsimoniously specify complex dynamics. As such they offer powerful and flexible approaches to understand ongoing psychological processes and interventions, and allow for measurements to be taken a variable number of times, and at irregular intervals. However, limited developments have taken place regarding the use of continuous time models in a fully hierarchical context, in which all model parameters are allowed to vary over individuals. This has meant that questions regarding individual differences in parameters have had to rely on single-subject time series approaches, which require far more measurement occasions per individual. We present a hierarchical Bayesian approach to estimating continuous time dynamic models, allowing for individual variation in all model parameters. We also describe an extension to the ctsem package for R, which interfaces to the Stan software and allows simple specification and fitting of such models. To demonstrate the approach, we use a subsample from the German socioeconomic panel and relate overall life satisfaction and satisfaction with health.

Dudgeon: Some improvements in confidence intervals for standardized regression coefficients **Dudgeon-2017**

Paul Dudgeon. "Some improvements in confidence intervals for standardized regression coefficients". In: *Psychometrika* 82.4 (Mar. 2017), pp. 928–951. DOI: 10.1007/s11336-017-9563-z.

Abstract: Yuan and Chan (Psychometrika 76:670-690, 2011. doi:10.1007/S11336-011-9224-6) derived consistent confidence intervals for standardized regression coefficients under fixed and random score assumptions. Jones and Waller (Psychometrika 80:365-378, 2015. doi:10.1007/S11336-013-9380-Y) extended these developments to circumstances where data are non-normal by examining

confidence intervals based on Browne's (Br J Math Stat Psychol 37:62-83, 1984. doi:10.1111/j.2044-8317.1984.tb00789.x) asymptotic distribution-free (ADF) theory. Seven different heteroscedastic-consistent (HC) estimators were investigated in the current study as potentially better solutions for constructing confidence intervals on standardized regression coefficients under non-normality. Normal theory, ADF, and HC estimators were evaluated in a Monte Carlo simulation. Findings confirmed the superiority of the HC3 (MacKinnon and White, J Econ 35:305-325, 1985. doi:10.1016/0304-4076(85)90158-7) and HC5 (Cribari-Neto and Da Silva, Adv Stat Anal 95:129-146, 2011. doi:10.1007/s10182-010-0141-2) interval estimators over Jones and Waller's ADF estimator under all conditions investigated, as well as over the normal theory method. The HC5 estimator was more robust in a restricted set of conditions over the HC3 estimator. Some possible extensions of HC estimators to other effect size measures are considered for future developments.

Eddelbuettel et al.: Extending R with C++: A brief introduction to Rcpp

Eddelbuettel-Balamuta-2017

Dirk Eddelbuettel and James Joseph Balamuta. "Extending R with C++: A brief introduction to Rcpp". In: *PeerJ Preprints* 3188v1.3 (Aug. 2017). DOI: `10.7287/peerj.preprints.3188v1`.

Abstract: R has always provided an application programming interface (API) for extensions. Based on the C language, it uses a number of macros and other low-level constructs to exchange data structures between the R process and any dynamically-loaded component modules authors added to it. With the introduction of the Rcpp package, and its later refinements, this process has become considerably easier yet also more robust. By now, Rcpp has become the most popular extension mechanism for R. This article introduces Rcpp, and illustrates with several examples how the Rcpp Attributes mechanism in particular eases the transition of objects between R and C++ code.

Eddelbuettel et al.: Rcpp: Seamless R and C++ integration

Eddelbuettel-Francois-2011

Dirk Eddelbuettel and Romain François. "Rcpp: Seamless R and C++ integration". In: *Journal of Statistical Software* 40.8 (2011). DOI: 10.18637/jss.v040.i08.

Abstract: The Rcpp package simplifies integrating C++ code with R. It provides a consistent C++ class hierarchy that maps various types of R objects (vectors, matrices, functions, environments, ...) to dedicated C++ classes. Object interchange between R and C++ is managed by simple, flexible and extensible concepts which include broad support for C++ Standard Template Library idioms. C++ code can both be compiled, linked and loaded on the fly, or added via packages. Flexible error and exception code handling is provided. Rcpp substantially lowers the barrier for programmers wanting to combine C++ code with R.

Eddelbuettel et al.: RcppArmadillo: Accelerating R with high-performance C++ linear algebra

Eddelbuettel-Sanderson-2014

Dirk Eddelbuettel and Conrad Sanderson. "RcppArmadillo: Accelerating R with high-performance C++ linear algebra". In: *Computational Statistics & Data Analysis* 71 (Mar. 2014), pp. 1054–1063. DOI: 10.1016/j.csda.2013.02.005.

Abstract: The R statistical environment and language has demonstrated particular strengths for interactive development of statistical algorithms, as well as data modelling and visualisation. Its current implementation has an interpreter at its core which may result in a performance penalty in comparison to directly executing user algorithms in the native machine code of the host CPU. In contrast, the C++ language has no built-in visualisation capabilities, handling of linear algebra or even basic statistical algorithms; however, user programs are converted to high-performance machine code, ahead of execution. A new method avoids possible speed penalties in R by using the Rcpp extension package in conjunction with the Armadillo C++ matrix library. In addition to the inherent performance advantages of compiled code, Armadillo provides an easy-to-use template-based meta-programming framework, allowing the automatic pooling of several linear algebra operations into one, which in turn can lead to further speedups. With the aid of Rcpp and Armadillo, conversion of linear algebra centred algorithms from R to C++ becomes straightforward. The algorithms retain the overall structure as well as readability, all while maintaining a bidirectional link with the host R environment. Empirical timing comparisons of R and C++ implementations of a Kalman filtering algorithm indicate a speedup of several orders of magnitude.

Efron: Bayesian inference and the parametric bootstrap **Efron-2012**

Bradley Efron. "Bayesian inference and the parametric bootstrap". In: *The Annals of Applied Statistics* 6.4 (Dec. 2012). DOI: 10.1214/12-aoas571.

Abstract: The parametric bootstrap can be used for the efficient computation of Bayes posterior distributions. Importance sampling formulas take on an easy form relating to the deviance in exponential families and are particularly simple starting from Jeffreys invariant prior. Because of the i.i.d. nature of bootstrap sampling, familiar formulas describe the computational accuracy of the Bayes estimates. Besides computational methods, the theory provides a connection between Bayesian and frequentist analysis. Efficient algorithms for the frequentist accuracy of Bayesian inferences are developed and demonstrated in a model selection example.

Eisenberg et al.: Emotion-related self-regulation and its relation to children's maladjustment **Eisenberg-Spinrad-Eggum-2010**

Nancy Eisenberg, Tracy L. Spinrad, and Natalie D. Eggum. "Emotion-related self-regulation and its relation to children's maladjustment". In: *Annual Review of Clinical Psychology* 6.1 (Mar. 2010), pp. 495–525. ISSN: 1548-5951. DOI: 10.1146/annurev.clinpsy.121208.131208.

Abstract: The development of children's emotion-related self-regulation appears to be related to, and likely involved in, many aspects of children's development. In this review, the distinction between

effortful self-regulatory processes and those that are somewhat less voluntary is discussed, and literature on the former capacities is reviewed. Emotion-related self-regulation develops rapidly in the early years of life and improves more slowly into adulthood. Individual differences in children's self-regulation are fairly stable after the first year or two of life. Such individual differences are inversely related to at least some types of externalizing problems. Findings for internalizing problems are less consistent and robust, although emotion-related self-regulation appears to be inversely related to internalizing problems after the early years. Self-regulatory capacities have been related to both genetic and environmental factors and their interaction. Some interventions designed to foster self-regulation and, hence, reduce maladjustment, have proved to be at least partially effective.

Enders et al.: A Bayesian approach for estimating mediation effects with missing data

Enders-Fairchild-MacKinnon-2013

Craig K. Enders, Amanda J. Fairchild, and David P. MacKinnon. "A Bayesian approach for estimating mediation effects with missing data". In: *Multivariate Behavioral Research* 48.3 (May 2013), pp. 340–369. ISSN: 1532-7906. DOI: 10.1080/00273171.2013.784862.

Abstract: Methodologists have developed mediation analysis techniques for a broad range of substantive applications, yet methods for estimating mediating mechanisms with missing data have been understudied. This study outlined a general Bayesian missing data handling approach that can accommodate mediation analyses with any number of manifest variables. Computer simulation studies showed that the Bayesian approach produced frequentist coverage rates and power estimates that were comparable to those of maximum likelihood with the bias-corrected bootstrap. We share a SAS macro that implements Bayesian estimation and use two data analysis examples to demonstrate its use.

Epskamp et al.: Estimating psychological networks and their accuracy: A tutorial paper

Epskamp-Borsboom-Fried-2017

Sacha Epskamp, Denny Borsboom, and Eiko I. Fried. "Estimating psychological networks and their accuracy: A tutorial paper". In: *Behavior Research Methods* 50.1 (Mar. 2017), pp. 195–212. ISSN: 1554-3528. DOI: 10.3758/s13428-017-0862-1.

Abstract: The usage of psychological networks that conceptualize behavior as a complex interplay of psychological and other components has gained increasing popularity in various research fields. While prior publications have tackled the topics of estimating and interpreting such networks, little work has been conducted to check how accurate (i.e., prone to sampling variation) networks are estimated, and how stable (i.e., interpretation remains similar with less observations) inferences from the network structure (such as centrality indices) are. In this tutorial paper, we aim to introduce the reader to this field and tackle the problem of accuracy under sampling variation. We first introduce the current state-of-the-art of network estimation. Second, we provide a rationale why researchers should investigate the accuracy of psychological networks. Third, we describe how bootstrap routines can be used to (A) assess the accuracy of estimated network connections, (B) investigate the stability of centrality indices, and (C) test whether network connections and centrality estimates for different variables differ from each other. We introduce two novel statistical methods: for (B) the correlation stability coefficient, and for (C) the bootstrapped difference test for edge-weights and centrality indices. We conducted and present simulation studies to assess the performance of both methods. Finally, we developed the free R-package bootnet that allows for estimating psychological networks in a generalized framework in addition to the proposed bootstrap methods. We showcase bootnet in a tutorial, accompanied by R syntax, in which we analyze a dataset of 359 women with posttraumatic stress disorder available online.

Epskamp et al.: qgraph: Network visualizations of relationships in psychometric data **Epskamp-Cramer-Waldorp-etal-2012**

Sacha Epskamp, Angélique O. J. Cramer, et al. "qgraph: Network visualizations of relationships in psychometric data". In: *Journal of Statistical Software* 48.4 (2012). ISSN: 1548-7660. DOI: 10.18637/jss.v048.i04.

Abstract: We present the qgraph package for R, which provides an interface to visualize data through network modeling techniques. For instance, a correlation matrix can be represented as a network in which each variable is a node and each correlation an edge; by varying the width of the edges according to the magnitude of the correlation, the structure of the correlation matrix can be visualized. A wide variety of matrices that are used in statistics can be represented in this fashion, for example matrices that contain (implied) covariances, factor loadings, regression parameters and p values. qgraph can also be used as a psychometric tool, as it performs exploratory and confirmatory factor analysis, using sem and lavaan; the output of these packages is automatically visualized in qgraph, which may aid the interpretation of results. In this article, we introduce qgraph by applying the package functions to data from the NEO-PI-R, a widely used personality questionnaire.

Epskamp et al.: The Gaussian Graphical Model in Cross-Sectional and Time-Series Data **Epskamp-Waldorp-Mottus-etal-2018**

Sacha Epskamp, Lourens J. Waldorp, et al. "The Gaussian Graphical Model in Cross-Sectional and Time-Series Data". In: *Multivariate Behavioral Research* 53.4 (Apr. 2018), pp. 453–480. ISSN: 1532-7906. DOI: 10.1080/00273171.2018.1454823.

Abstract: We discuss the Gaussian graphical model (GGM; an undirected network of partial correlation coefficients) and detail its utility as an exploratory data analysis tool. The GGM shows which variables predict one-another, allows for sparse modeling of covariance structures, and may highlight potential causal relationships between observed variables. We describe the utility in three kinds of psychological data sets: data sets in which consecutive cases are assumed independent (e.g., cross-sectional data), temporally ordered data sets (e.g., n = 1 time series), and a mixture of the 2 (e.g., n ¿ 1 time series). In time-series analysis, the GGM can be used to model the residual structure of a vector-autoregression analysis (VAR), also termed graphical VAR. Two network models can then be obtained: a temporal network and a contemporaneous network. When analyzing data from multiple subjects, a GGM can also be formed on the covariance structure of stationary means-the between-subjects network. We discuss the interpretation of these models and propose estimation methods to obtain these networks, which we implement in the R packages graphicalVAR and mlVAR. The methods are showcased in two empirical examples, and simulation studies on these methods are included in the supplementary materials.

Evans et al.: Childhood poverty, chronic stress, self-regulation, and coping

Evans-Kim-2012

Gary W. Evans and Pilyoung Kim. "Childhood poverty, chronic stress, self-regulation, and coping". In: *Child Development Perspectives* 7.1 (Dec. 2012), pp. 43–48. ISSN: 1750-8606. DOI: 10.1111/cdep.12013.

Abstract: Poverty is a powerful factor that can alter lifetime developmental trajectories in cognitive, socioemotional, and physical health outcomes. Most explanatory work on the underlying psychological processes of how poverty affects development has focused on parental investment and parenting practices, principally responsiveness. Our primary objective in this article was to describe a third, complementary pathway—chronic stress and coping—that may also prove helpful in understanding the developmental impacts of early childhood poverty throughout life. Disadvantaged children are more likely than their wealthier peers to confront a wide array of physical stressors (e.g., substandard housing, chaotic environments) and psychosocial stressors (e.g., family turmoil, separation from adult caregivers). As exposure to stressors accumulates, physiological response systems that are designed to handle relatively infrequent, acute environmental demands are overwhelmed.

Chronic cumulative stressors also disrupt the self-regulatory processes that help children cope with external demands.

Finlay et al.: Leisure activities, the social weekend, and alcohol use: Evidence from a daily study of first-year college students **Finlay-Ram-Maggs-etal-2012**

Andrea K. Finlay et al. "Leisure activities, the social weekend, and alcohol use: Evidence from a daily study of first-year college students". In: *Journal of Studies on Alcohol and Drugs* 73.2 (Mar. 2012), pp. 250–259. ISSN: 1938-4114. DOI: 10.15288/jsad.2012.73.250.

Abstract: Objective: The aim of this study was to document within-person and between-persons associations between the duration of day-to-day activities (volunteering, spiritual activities, media use, socializing, entertainment/campus events and clubs, athletics, classes, working for pay) and alcohol use (quantity and heavy drinking) and to examine whether these associations differed by gender and the time of week. Method: First-semester college students ($N = 717$ persons; 51.6% female) provided up to 14 consecutive days of data ($N = 9,431$ days) via daily web-based surveys. Multilevel analyses tested whether alcohol use was associated with activity duration, gender, and time of week. Results: Between-persons associations indicated that alcohol use was higher among individuals who spent more time involved in athletics and socializing and lower among students who spent more time in spiritual and volunteer activities. Within-person associations indicated that students consumed more alcohol and were more likely to drink heavily on weekends, on days they spent more time than usual socializing, and on days they spent less time than usual in spiritual activities and using media. Conclusions: Select activities and days were linked with less alcohol use at both the between- and within-person levels, suggesting that attention should be paid to both selection effects and social context to understand the mechanisms linking activity duration and student drinking.

Fisher et al.: Enhancing the personalization of psychotherapy with dynamic assessment and modeling **Fisher-Boswell-2016**

Aaron J. Fisher and James F. Boswell. "Enhancing the personalization of psychotherapy with dynamic assessment and modeling". In: *Assessment* 23.4 (Mar. 2016), pp. 496–506. ISSN: 1552-3489. DOI: 10.1177/1073191116638735.

Abstract: Clinicians have long recognized the importance of tailoring psychotherapy interventions to the needs and characteristics of the individual patient. However, traditional approaches to clinical assessment, service delivery, and intervention research have not been conducive to such personalization. Contrary to traditional nomothetic approaches, idiographic assessment and modeling of intraindividual dynamic processes holds tremendous promise for tailoring the implementation of psychotherapy to the individual patient. In this article, we (a) present an argument for assessing person-specific dynamics, (b) provide a detailed description of a method that harnesses person-specific dynamic assessment and modeling for use in routine psychotherapy, (c) present exemplar clinical cases illustrating these methods, and (d) discuss how these methods can be translated into routine clinical assessment and psychotherapy.

Fisher et al.: Lack of group-to-individual generalizability is a threat to human subjects research **Fisher-Medaglia-Jeronimus-2018**

Aaron J. Fisher, John D. Medaglia, and Bertus F. Jeronimus. "Lack of group-to-individual generalizability is a threat to human subjects research". In: *Proceedings of the National Academy of Sciences* 115.27 (June 2018). ISSN: 1091-6490. DOI: 10.1073/pnas.1711978115.

Abstract: Only for ergodic processes will inferences based on group-level data generalize to individual experience or behavior. Because human social and psychological processes typically have an individually variable and time-varying nature, they are unlikely to be ergodic. In this paper, six studies with a repeated-measure design were used for symmetric comparisons of interindividual and intraindividual variation. Our results delineate the potential scope and impact of nonergodic data

in human subjects research. Analyses across six samples (with 87-94 participants and an equal number of assessments per participant) showed some degree of agreement in central tendency estimates (mean) between groups and individuals across constructs and data collection paradigms. However, the variance around the expected value was two to four times larger within individuals than within groups. This suggests that literatures in social and medical sciences may overestimate the accuracy of aggregated statistical estimates. This observation could have serious consequences for how we understand the consistency between group and individual correlations, and the generalizability of conclusions between domains. Researchers should explicitly test for equivalence of processes at the individual and group level across the social and medical sciences.

Fisher et al.: A quantitative method for the analysis of nomothetic relationships between idiographic structures: Dynamic patterns create attractor states for sustained posttreatment change **Fisher-Newman-Molenaar-2011**

Aaron J. Fisher, Michelle G. Newman, and Peter C. M. Molenaar. "A quantitative method for the analysis of nomothetic relationships between idiographic structures: Dynamic patterns create attractor states for sustained posttreatment change". In: *Journal of Consulting and Clinical Psychology* 79.4 (Aug. 2011), pp. 552–563. ISSN: 0022-006X. DOI: 10.1037/a0024069.

Abstract: Objective: The present article aimed to demonstrate that the establishment of dynamic patterns during the course of psychotherapy can create attractor states for continued adaptive change following the conclusion of treatment. Method: This study is a secondary analysis of T. D. Borkovec and E. Costello (1993). Of the 55 participants in the original study, 33 were retained for the present analysis due to the homogeneity of psychotherapy outcome among these participants. Of these 33, the majority were White (88%) and female (70%), and the average age was 35.44 years (SD = 14.46). Participants participated in 12 weeks of either cognitive behavioral therapy or applied relaxation. Daily diary entries from the treatment period were subjected to time series analyses in order to determine the degree of order versus disorder present within individual dynamic systems. These idiographic data were then aggregated for nomothetic analysis of treatment outcome via linear

mixed effect models. Results: Spectral power due to daily to intradaily oscillations in thrice-daily diary data significantly moderated reliable change over posttreatment follow-up such that lesser power predicted increases in reliable change over the 1-year follow-up period. Additionally, residual variance for dynamic factor models significantly moderated the slope for change over the follow-up period, such that lesser variance—and thus greater order in dynamic systems—predicted increases in reliable change. Conclusions: The degree of order in dynamic systems established during therapy acted as an adaptive attractor state, promoting continued positive gains 1 year after the conclusion of therapy. The present study represents an important innovation in the study of dynamic systems in psychotherapy.

Fisher et al.: Exploring the idiographic dynamics of mood and anxiety via network analysis **Fisher-Reeves-Lawyer-etal-2017**

Aaron J. Fisher, Jonathan W. Reeves, et al. "Exploring the idiographic dynamics of mood and anxiety via network analysis". In: *Journal of Abnormal Psychology* 126.8 (Nov. 2017), pp. 1044–1056. ISSN: 0021-843X. DOI: 10.1037/abn0000311.

Abstract: Individual variation is increasingly recognized as important to psychopathology research. Concurrently, new methods of analysis based on network models are bringing new perspectives on mental (dys)function. This current work analyzed idiographic multivariate time series data using a novel network methodology that incorporates contemporaneous and lagged associations in mood and anxiety symptomatology. Data were taken from 40 individuals with generalized anxiety disorder (GAD), major depressive disorder (MDD), or comorbid GAD and MDD, who answered questions about 21 descriptors of mood and anxiety symptomatology 4 times a day over a period of approximately 30 days. The model provided an excellent fit to the intraindividual symptom dynamics of all 40 individuals. The most central symptoms in contemporaneous systems were those related to positive and negative mood. The temporal networks highlighted the importance of anger to symptomatology, while also finding that depressed mood and worry—the principal diagnostic criteria for GAD and MDD—were the least influential nodes across the sample. The method's potential for analysis of individual symptom patterns is demonstrated by 3 exemplar participants. Idiographic network-based analysis may fundamentally alter the way psychopathology is assessed, classified, and treated, allowing researchers and clinicians to better understand individual symptom dynamics.

Fredrickson et al.: Positive emotion correlates of meditation practice: A comparison of mindfulness meditation and loving-kindness meditation

Fredrickson-Boulton-Firestine-etal-2017

Barbara L. Fredrickson et al. "Positive emotion correlates of meditation practice: A comparison of mindfulness meditation and loving-kindness meditation". In: *Mindfulness* 8.6 (May 2017), pp. 1623–1633. ISSN: 1868-8535. DOI: 10.1007/s12671-017-0735-9.

Abstract: The purpose of this study was to uncover the day-to-day emotional profiles and dose-response relations, both within persons and between persons, associated with initiating one of two meditation practices, either mindfulness meditation or loving-kindness meditation. Data were pooled across two studies of midlife adults ($N = 339$) who were randomized to learn either mindfulness meditation or loving-kindness meditation in a 6-week workshop. The duration and frequency of meditation practice was measured daily for 9 weeks, commencing with the first workshop session. Likewise, positive and negative emotions were also measured daily, using the modified Differential Emotions Scale (Fredrickson, Advances in Experimental Social Psychology 47:1-53, 2013). Analysis of daily emotion reports over the targeted 9-week period showed significant gains in positive emotions and no change in negative emotions, regardless of meditation type. Multilevel models also revealed significant dose-response relations between duration of meditation practice and positive emotions, both within persons and between persons. Moreover, the within-person dose-response relation was stronger for loving-kindness meditation than for mindfulness meditation. Similar dose-response relations were observed for the frequency of meditation practice. In the context of prior research on the mental and physical health benefits produced by subtle increases in day-to-day experiences of positive emotions, the present research points to evidence-based practices—both mindfulness meditation and loving-kindness meditation—that can improve emotional well-being.

Fritz et al.: Explanation of two anomalous results in statistical mediation analysis

Fritz-Taylor-MacKinnon-2012

Matthew S. Fritz, Aaron B. Taylor, and David P. MacKinnon. "Explanation of two anomalous results in statistical mediation analysis". In: *Multivariate Behavioral Research* 47.1 (Feb. 2012), pp. 61–87. DOI: 10.1080/00273171.2012.640596.

Abstract: Previous studies of different methods of testing mediation models have consistently found two anomalous results. The first result is elevated Type I error rates for the bias-corrected and accelerated bias-corrected bootstrap tests not found in nonresampling tests or in resampling tests that did not include a bias correction. This is of special concern as the bias-corrected bootstrap is often recommended and used due to its higher statistical power compared with other tests. The second result is statistical power reaching an asymptote far below 1.0 and in some conditions even declining slightly as the size of the relationship between X and M, a, increased. Two computer simulations were conducted to examine these findings in greater detail. Results from the first simulation found that the increased Type I error rates for the bias-corrected and accelerated bias-corrected bootstrap are a function of an interaction between the size of the individual paths making up the mediated effect and the sample size, such that elevated Type I error rates occur when the sample size is small and the effect size of the nonzero path is medium or larger. Results from the second simulation found that stagnation and decreases in statistical power as a function of the effect size of the a path occurred primarily when the path between M and Y, b, was small. Two empirical mediation examples are provided using data from a steroid prevention and health promotion program aimed at high school football players (Athletes Training and Learning to Avoid Steroids; Goldberg et al., 1996), one to illustrate a possible Type I error for the bias-corrected bootstrap test and a second to illustrate a loss in power related to the size of a. Implications of these findings are discussed.

Gates et al.: Automatic search for fMRI connectivity mapping: An alternative to Granger causality testing using formal equivalences among SEM path modeling, VAR, and unified SEM **Gates-Molenaar-Hillary-etal-2010**

Kathleen M. Gates et al. "Automatic search for fMRI connectivity mapping: An alternative to Granger causality testing using formal equivalences among SEM path modeling, VAR, and unified SEM". In: *NeuroImage* 50.3 (Apr. 2010), pp. 1118–1125. DOI: 10.1016/j.neuroimage.2009.12.117.

Abstract: Modeling the relationships among brain regions of interest (ROIs) carries unique potential to explicate how the brain orchestrates information processing. However, hurdles arise when using functional MRI data. Variation in ROI activity contains sequential dependencies and shared influences on synchronized activation. Consequently, both lagged and contemporaneous relationships must be considered for unbiased statistical parameter estimation. Identifying these relationships using a data-driven approach could guide theory-building regarding integrated processing. The present paper demonstrates how the unified SEM attends to both lagged and contemporaneous influences on ROI activity. Additionally, this paper offers an approach akin to Granger causality testing, Lagrange multiplier testing, for statistically identifying directional influence among ROIs and employs this approach using an automatic search procedure to arrive at the optimal model. Rationale for this equivalence is offered by explicating the formal relationships among path modeling, vector autoregression, and unified SEM. When applied to simulated data, biases in estimates which do not consider both lagged and contemporaneous paths become apparent. Finally, the use of unified SEM with the automatic search procedure is applied to an empirical data example.

George et al.: Concurrent and subsequent associations between daily digital technology use and high-risk adolescents' mental health symptoms **George-Russell-Piontak-etal-2017**

Madeleine J. George et al. "Concurrent and subsequent associations between daily digital technology use and high-risk adolescents' mental health symptoms". In: *Child Development* 89.1 (May 2017), pp. 78–88. ISSN: 1467-8624. DOI: 10.1111/cdev.12819.

Abstract: Adolescents are spending an unprecedented amount of time using digital technologies (especially mobile technologies), and there are concerns that adolescents' constant connectivity is associated with poor mental health, particularly among at-risk adolescents. Participants included 151 adolescents at risk for mental health problems ($M_{\text{age}} = 13.1$) who completed a baseline assessment, 30-day ecological momentary assessment, and 18 month follow-up assessment. Results from multilevel regression models showed that daily reports of both time spent using digital technologies and the number of text messages sent were associated with increased same-day attention deficit hyperactivity disorder (ADHD) and conduct disorder (CD) symptoms. Adolescents' reported digital technology usage and text messaging across the ecological momentary assessment (EMA) period was also associated with poorer self-regulation and increases in conduct problem symptoms between the baseline and follow-up assessments.

Gistelinck et al.: The actor-partner interdependence model for longitudinal dyadic data: An implementation in the SEM framework **Gistelinck-Loeys-2019**

Fien Gistelinck and Tom Loeys. "The actor-partner interdependence model for longitudinal dyadic data: An implementation in the SEM framework". In: *Structural Equation Modeling: A Multidisciplinary Journal* 26.3 (2019), pp. 329–347. ISSN: 1532-8007. DOI: 10.1080/10705511.2018.1527223.

Abstract: In dyadic research, the actor-partner interdependence model (APIM) is widely used to model the effect of a predictor measured across dyad members on one's own and one's partner outcome. When such dyadic data are measured repeatedly over time, both the non-independence within couples and the non-independence over time need to be accounted for. In this paper, we present a longitudinal extension of the APIM, the L-APIM, that allows for both stable and time-varying sources of non-independence. Its implementation is readily available in multilevel software, such as proc mixed in SAS, but is lacking in the structural equation modeling (SEM) framework. We tackle

the computational challenges associated with its SEM-implementation and propose a user-friendly free application for the L-APIM, which can be found at http://fgisteli.shinyapps.io/Shiny_LDD. As an illustration, we explore the actor and partner effects of positive relationship feelings on next day's intimacy using 3-week diary data of 66 heterosexual couples.

Greenfield et al.: Risks of alcohol use disorders related to drinking patterns in the U.S. general population **Greenfield-Ye-Bond-etal-2014**

Thomas K. Greenfield et al. "Risks of alcohol use disorders related to drinking patterns in the U.S. general population". In: *Journal of Studies on Alcohol and Drugs* 75.2 (Mar. 2014), pp. 319–327. ISSN: 1938-4114. DOI: 10.15288/jsad.2014.75.319.

Abstract: Objective: The purpose of this study was to examine the relations between drinking (mean quantity and heavy drinking patterns) and alcohol use disorders (AUDs) in the U.S. general population. Method: Data from three telephone National Alcohol Surveys (in 2000, 2005, and 2010) were pooled, with separate analyses for men and women restricted to current drinkers ($ns =$ $5,922$ men, 6,270 women). Predictors were 12-month volume (mean drinks per day), rates of heavy drinking (5+/4+ drinks in a day for men/women), and very heavy drinking (8+, 12+, and 24+ drinks in a day). Outcomes were negative alcohol-related consequences constituting abuse (1+ of 4 DSM-IV-based domains assessed by 13 items) and alcohol dependence (symptoms in 3+ of 7 DSM-IV-based domains), together taken to indicate an AUD. Segmentation analyses were used to model risks of problem outcomes from drinking patterns separately by gender. Results: In the general population, men and women who consumed ≤ 1 drink/day on average with no heavy drinking days did not incur substantial risks of an AUD ($< 10\%$). Men who drank from 1 to 2 drinks/day on average but never 5+ incurred a 16% risk of reporting an AUD (3.5% alcohol dependence). At higher volumes, men and women who indicated higher rates of drinking larger amounts per day and/or involving 8+ and 12+ drinks/day (and even 24+ drinks/day for men) showed much higher risks of experiencing AUDs. Conclusions: The findings provide quantitative guidance for primary care practitioners who wish to make population-based recommendations to patients who might

benefit by reducing both overall intake and amounts per occasion in an effort to lower their risks of developing AUDs.

Grimm et al.: Nonlinear structured growth mixture models in Mplus and OpenMx

Grimm-Ram-Estabrook-2010

Kevin J. Grimm, Nilam Ram, and Ryne Estabrook. "Nonlinear structured growth mixture models in Mplus and OpenMx". In: *Multivariate Behavioral Research* 45.6 (Nov. 2010), pp. 887–909. ISSN: 1532-7906. DOI: 10.1080/00273171.2010.531230.

Abstract: Growth mixture models (GMMs; B. O. Muthen & Muthen, 2000; B. O. Muthen & Shedden, 1999) are a combination of latent curve models (LCMs) and finite mixture models to examine the existence of latent classes that follow distinct developmental patterns. GMMs are often fit with linear, latent basis, multiphase, or polynomial change models because of their common use, flexibility in modeling many types of change patterns, the availability of statistical programs to fit such models, and the ease of programming. In this article, we present additional ways of modeling nonlinear change patterns with GMMs. Specifically, we show how LCMs that follow specific nonlinear functions can be extended to examine the presence of multiple latent classes using the Mplus and OpenMx computer programs. These models are fit to longitudinal reading data from the Early Childhood Longitudinal Study-Kindergarten Cohort to illustrate their use.

Gu et al.: A state space modeling approach to mediation analysis

Gu-Preacher-Ferrer-2014

Fei Gu, Kristopher J. Preacher, and Emilio Ferrer. "A state space modeling approach to mediation analysis". In: *Journal of Educational and Behavioral Statistics* 39.2 (Apr. 2014), pp. 117–143. ISSN: 1935-1054. DOI: 10.3102/1076998614524823.

Abstract: Mediation is a causal process that evolves over time. Thus, a study of mediation requires data collected throughout the process. However, most applications of mediation analysis use cross-sectional rather than longitudinal data. Another implicit assumption commonly made in longitudinal designs for mediation analysis is that the same mediation process universally applies to all members of the population under investigation. This assumption ignores the important issue of ergodicity before aggregating the data across subjects. We first argue that there exists a discrepancy between the concept of mediation and the research designs that are typically used to investigate it. Second, based on the concept of ergodicity, we argue that a given mediation process probably is not equally valid for all individuals in a population. Therefore, the purpose of this article is to propose a two-faceted solution. The first facet of the solution is that we advocate a single-subject time-series design that aligns data collection with researchers' conceptual understanding of mediation. The second facet is to introduce a flexible statistical method—the state space model—as an ideal technique to analyze single-subject time series data in mediation studies. We provide an overview of the state space method and illustrative applications using both simulated and real time series data. Finally, we discuss additional issues related to research design and modeling.

Guo et al.: Bayesian Lasso for Semiparametric Structural Equation Models

Guo-Zhu-Chow-etal-2012

Ruixin Guo et al. "Bayesian Lasso for Semiparametric Structural Equation Models". In: *Biometrics* 68.2 (Feb. 2012), pp. 567–577. ISSN: 1541-0420. DOI: 10.1111/j.1541-0420.2012.01751.x.

Abstract: There has been great interest in developing nonlinear structural equation models and associated statistical inference procedures, including estimation and model selection methods. In this paper a general semiparametric structural equation model (SSEM) is developed in which the structural equation is composed of nonparametric functions of exogenous latent variables and fixed covariates on a set of latent endogenous variables. A basis representation is used to approximate these nonparametric functions in the structural equation and the Bayesian Lasso method coupled with a Markov Chain Monte Carlo (MCMC) algorithm is used for simultaneous estimation and model selection. The proposed method is illustrated using a simulation study and data from the

Affective Dynamics and Individual Differences (ADID) study. Results demonstrate that our method can accurately estimate the unknown parameters and correctly identify the true underlying model.

E. L. Hamaker et al.: At the frontiers of modeling intensive longitudinal data: Dynamic structural equation models for the affective measurements from the COGITO study

Hamaker-Asparouhov-Brose-etal-2018

Ellen L. Hamaker, Tihomir Asparouhov, et al. "At the frontiers of modeling intensive longitudinal data: Dynamic structural equation models for the affective measurements from the COGITO study". In: *Multivariate Behavioral Research* 53.6 (Apr. 2018), pp. 820–841. ISSN: 1532-7906. DOI: 10.1080/00273171.2018.1446819.

Abstract: With the growing popularity of intensive longitudinal research, the modeling techniques and software options for such data are also expanding rapidly. Here we use dynamic multilevel modeling, as it is incorporated in the new dynamic structural equation modeling (DSEM) toolbox in Mplus, to analyze the affective data from the COGITO study. These data consist of two samples of over 100 individuals each who were measured for about 100 days. We use composite scores of positive and negative affect and apply a multilevel vector autoregressive model to allow for individual differences in means, autoregressions, and cross-lagged effects. Then we extend the model to include random residual variances and covariance, and finally we investigate whether prior depression affects later depression scores through the random effects of the daily diary measures. We end with discussing several urgent—but mostly unresolved—issues in the area of dynamic multilevel modeling.

E. L. Hamaker et al.: Modeling affect dynamics: State of the art and future challenges

Hamaker-Ceulemans-Grasman-etal-2015

Ellen L. Hamaker, Eva Ceulemans, et al. "Modeling affect dynamics: State of the art and future

challenges". In: *Emotion Review* 7.4 (July 2015), pp. 316–322. ISSN: 1754-0747. DOI: 10.1177/1754073915590619.

Abstract: The current article aims to provide an up-to-date synopsis of available techniques to study affect dynamics using intensive longitudinal data (ILD). We do so by introducing the following eight dichotomies that help elucidate what kind of data one has, what process aspects are of interest, and what research questions are being considered: (1) single- versus multiple-person data; (2) univariate versus multivariate models; (3) stationary versus nonstationary models; (4) linear versus nonlinear models; (5) discrete time versus continuous time models; (6) discrete versus continuous variables; (7) time versus frequency domain; and (8) modeling the process versus computing descriptives. In addition, we discuss what we believe to be the most urging future challenges regarding the modeling of affect dynamics.

E. L. Hamaker et al.: To center or not to center? Investigating inertia with a multilevel autoregressive model — Hamaker-Grasman-2015

Ellen L. Hamaker and Raoul P. P. P. Grasman. "To center or not to center? Investigating inertia with a multilevel autoregressive model". In: *Frontiers in Psychology* 5 (Jan. 2015). ISSN: 1664-1078. DOI: 10.3389/fpsyg.2014.01492.

Abstract: Whether level 1 predictors should be centered per cluster has received considerable attention in the multilevel literature. While most agree that there is no one preferred approach, it has also been argued that cluster mean centering is desirable when the within-cluster slope and the between-cluster slope are expected to deviate, and the main interest is in the within-cluster slope. However, we show in a series of simulations that if one has a multilevel autoregressive model in which the level 1 predictor is the lagged outcome variable (i.e., the outcome variable at the previous occasion), cluster mean centering will in general lead to a downward bias in the parameter estimate of the within-cluster slope (i.e., the autoregressive relationship). This is particularly relevant if the main question is whether there is on average an autoregressive effect. Nonetheless, we show that if the main interest is in estimating the effect of a level 2 predictor on the autoregressive parameter (i.e., a cross-level interaction), cluster mean centering should be preferred over other forms of centering. Hence, researchers should be clear on what is considered the main goal of their study, and base their choice of centering method on this when using a multilevel autoregressive model.

E. L. Hamaker et al.: A critique of the cross-lagged panel model

Hamaker-Kuiper-Grasman-2015

Ellen L. Hamaker, Rebecca M. Kuiper, and Raoul P. P. P. Grasman. "A critique of the cross-lagged panel model". In: *Psychological Methods* 20.1 (2015), pp. 102–116. DOI: 10.1037/a0038889.

Abstract: The cross-lagged panel model (CLPM) is believed by many to overcome the problems associated with the use of cross-lagged correlations as a way to study causal influences in longitudinal panel data. The current article, however, shows that if stability of constructs is to some extent of a trait-like, timeinvariant nature, the autoregressive relationships of the CLPM fail to adequately account for this. As a result, the lagged parameters that are obtained with the CLPM do not represent the actual within-person relationships over time, and this may lead to erroneous conclusions regarding the presence, predominance, and sign of causal influences. In this article we present an alternative model that separates the within-person process from stable between-person differences through the inclusion of random intercepts, and we discuss how this model is related to existing structural equation models that include cross-lagged relationships. We derive the analytical relationship between the cross-lagged parameters from the CLPM and the alternative model, and use simulations to demonstrate the spurious results that may arise when using the CLPM to analyze data that include stable, trait-like individual differences. We also present a modeling strategy to avoid this pitfall and illustrate this using an empirical data set. The implications for both existing and future cross-lagged panel research are discussed.

E. L. Hamaker et al.: Using a few snapshots to distinguish mountains from waves: Weak factorial invariance in the context of trait-state research

Hamaker-Schuurman-Zijlmans-2016

Ellen L. Hamaker, Noémi K. Schuurman, and Eva A. O. Zijlmans. "Using a few snapshots to distinguish mountains from waves: Weak factorial invariance in the context of trait-state research". In: *Multivariate Behavioral Research* 52.1 (Nov. 2016), pp. 47–60. ISSN: 1532-7906. DOI: 10.1080/00273171.2016.1251299.

Abstract: In this article, we show that the underlying dimensions obtained when factor analyzing cross-sectional data actually form a mix of within-person state dimensions and between-person trait dimensions. We propose a factor analytical model that distinguishes between four independent sources of variance: common trait, unique trait, common state, and unique state. We show that by testing whether there is weak factorial invariance across the trait and state factor structures, we can tackle the fundamental question first raised by Cattell; that is, are within-person state dimensions qualitatively the same as between-person trait dimensions? Furthermore, we discuss how this model is related to other trait-state factor models, and we illustrate its use with two empirical data sets. We end by discussing the implications for cross-sectional factor analysis and suggest potential future developments.

E. L. Hamaker et al.: No time like the present: Discovering the hidden dynamics in intensive longitudinal data **Hamaker-Wichers-2017**

Ellen L. Hamaker and Marieke Wichers. "No time like the present: Discovering the hidden dynamics in intensive longitudinal data". In: *Current Directions in Psychological Science* 26.1 (Feb. 2017), pp. 10–15. ISSN: 1467-8721. DOI: 10.1177/0963721416666518.

Abstract: There has been a strong increase in the number of studies based on intensive longitudinal data, such as those obtained with experience sampling and daily diaries. These data contain a wealth of information regarding the dynamics of processes as they unfold within individuals over time. In

this article, we discuss how combining intensive longitudinal data with either time-series analysis, which consists of modeling the temporal dependencies in the data for a single individual, or dynamic multilevel modeling, which consists of using a time-series model at Level 1 to describe the within-person process while allowing for individual differences in the parameters of these processes at Level 2, has led to new insights in clinical psychology. In addition, we discuss several methodological and statistical challenges that researchers face when they are interested in studying the dynamics of psychological processes using intensive longitudinal data.

Hayes et al.: The relative trustworthiness of inferential tests of the indirect effect in statistical mediation analysis — Hayes-Scharkow-2013

Andrew F. Hayes and Michael Scharkow. "The relative trustworthiness of inferential tests of the indirect effect in statistical mediation analysis". In: *Psychological Science* 24.10 (Aug. 2013), pp. 1918–1927. DOI: 10.1177/0956797613480187.

Abstract: A content analysis of 2 years of Psychological Science articles reveals inconsistencies in how researchers make inferences about indirect effects when conducting a statistical mediation analysis. In this study, we examined the frequency with which popularly used tests disagree, whether the method an investigator uses makes a difference in the conclusion he or she will reach, and whether there is a most trustworthy test that can be recommended to balance practical and performance considerations. We found that tests agree much more frequently than they disagree, but disagreements are more common when an indirect effect exists than when it does not. We recommend the bias-corrected bootstrap confidence interval as the most trustworthy test if power is of utmost concern, although it can be slightly liberal in some circumstances. Investigators concerned about Type I errors should choose the Monte Carlo confidence interval or the distribution-of-the-product approach, which rarely disagree. The percentile bootstrap confidence interval is a good compromise test.

Hecht et al.: Continuous-time modeling in prevention research: An illustration

Hecht-Voelkle-2019

Martin Hecht and Manuel C. Voelkle. "Continuous-time modeling in prevention research: An illustration". In: *International Journal of Behavioral Development* 45.1 (Nov. 2019), pp. 19–27. ISSN: 1464-0651. DOI: 10.1177/0165025419885026.

Abstract: The analysis of cross-lagged relationships is a popular approach in prevention research to explore the dynamics between constructs over time. However, a limitation of commonly used cross-lagged models is the requirement of equally spaced measurement occasions that prevents the usage of flexible longitudinal designs and complicates cross-study comparisons. Continuous-time modeling overcomes these limitations. In this article, we illustrate the use of continuous-time models using Bayesian and frequentist approaches to model estimation. As an empirical example, we study the dynamic interplay of physical activity and health, a classic research topic in prevention science, using data from the "Midlife in the United States (MIDUS 2): Daily Stress Project, 2004-2009." To help prevention researchers in adopting the approach, we provide annotated R scripts and a simulated data set based on the results from analyzing the MIDUS 2 data.

Hedeker et al.: Modeling between-subject and within-subject variances in ecological momentary assessment data using mixed-effects location scale models

Hedeker-Mermelstein-Demirtas-2012

Donald Hedeker, Robin J. Mermelstein, and Hakan Demirtas. "Modeling between-subject and within-subject variances in ecological momentary assessment data using mixed-effects location scale models". In: *Statistics in Medicine* 31.27 (Mar. 2012), pp. 3328–3336. ISSN: 1097-0258. DOI: 10.1002/sim.5338.

Abstract: Ecological momentary assessment and/or experience sampling methods are increasingly used in health studies to study subjective experiences within changing environmental contexts. In

these studies, up to 30 or 40 observations are often obtained for each subject. Because there are so many measurements per subject, one can characterize a subject's mean and variance and can specify models for both. In this article, we focus on an adolescent smoking study using ecological momentary assessment where interest is on characterizing changes in mood variation. We describe how covariates can influence the mood variances and also extend the statistical model by adding a subject-level random effect to the within-subject variance specification. This permits subjects to have influence on the mean, or location, and variability, or (square of the) scale, of their mood responses. These mixed-effects location scale models have useful applications in many research areas where interest centers on the joint modeling of the mean and variance structure.

Hedges et al.: Overlap between treatment and control distributions as an effect size measure in experiments **Hedges-Olkin-2016**

Larry V. Hedges and Ingram Olkin. "Overlap between treatment and control distributions as an effect size measure in experiments". In: *Psychological Methods* 21.1 (2016), pp. 61–68. ISSN: 1082-989X. DOI: 10.1037/met0000042.

Abstract: The proportion π of treatment group observations that exceed the control group mean has been proposed as an effect size measure for experiments that randomly assign independent units into 2 groups. We give the exact distribution of a simple estimator of π based on the standardized mean difference and use it to study the small sample bias of this estimator. We also give the minimum variance unbiased estimator of π under 2 models, one in which the variance of the mean difference is known and one in which the variance is unknown. We show how to use the relation between the standardized mean difference and the overlap measure to compute confidence intervals for π and show that these results can be used to obtain unbiased estimators, large sample variances, and confidence intervals for 3 related effect size measures based on the overlap. Finally, we show how the effect size π can be used in a meta-analysis.

Herring et al.: Utility of the Athlete Drinking Scale for assessing drinking motives among high school athletes — Herring-Zamboanga-Olthuis-etal-2016

Tracy E. Herring et al. "Utility of the Athlete Drinking Scale for assessing drinking motives among high school athletes". In: *Addictive Behaviors* 60 (Sept. 2016), pp. 18–23. ISSN: 0306-4603. DOI: 10.1016/j.addbeh.2016.03.026.

Abstract: Research suggests that high school athletes are at greater risk for heavy alcohol use and alcohol-related problems than their non-athlete peers. Drinking motives unique to the athletic experience may contribute to elevated use. The Athlete Drinking Scale (ADS) was designed to assess sport-related motives for alcohol use, but has not yet been validated among high school athletes. The purpose of this study was to examine the psychometric properties of the ADS among a sample of high school athletes. Participants were 216 high school student-athlete drinkers who completed anonymous self-report surveys. A confirmatory factor analysis resulted in a revised three-factor solution with a satisfactory overall model fit. Path analyses indicated that the Positive Reinforcement motives subscale was the only ADS subscale that was significantly associated with alcohol use and alcohol-related problems when controlling for the effects of the other factors (i.e., age and gender) in this population. The ADS may be a valuable assessment tool for researchers and clinicians involved in alcohol prevention efforts for high school athletes.

Hesterberg: What teachers should know about the bootstrap: Resampling in the undergraduate statistics curriculum — Hesterberg-2015

Tim C. Hesterberg. "What teachers should know about the bootstrap: Resampling in the undergraduate statistics curriculum". In: *The American Statistician* 69.4 (Oct. 2015), pp. 371–386. DOI: 10.1080/00031305.2015.1089789.

Abstract: Bootstrapping has enormous potential in statistics education and practice, but there are subtle issues and ways to go wrong. For example, the common combination of nonparametric bootstrapping and bootstrap percentile confidence intervals is less accurate than using t-intervals

for small samples, though more accurate for larger samples. My goals in this article are to provide a deeper understanding of bootstrap methods–how they work, when they work or not, and which methods work better-and to highlight pedagogical issues. Supplementary materials for this article are available online.

Hingson et al.: Magnitude and trends in heavy episodic drinking, alcohol-impaired driving, and alcohol-related mortality and overdose hospitalizations among emerging adults of college ages 18-24 in the United States, 1998-2014 Hingson-Zha-Smyth-2017

Ralph Hingson, Wenxing Zha, and Daniel Smyth. "Magnitude and trends in heavy episodic drinking, alcohol-impaired driving, and alcohol-related mortality and overdose hospitalizations among emerging adults of college ages 18-24 in the United States, 1998-2014". In: *Journal of Studies on Alcohol and Drugs* 78.4 (July 2017), pp. 540–548. ISSN: 1938-4114. DOI: 10.15288/jsad.2017.78.540.

Abstract: Objective: This article estimates percentages of U.S. emerging adults ages 18-24 engaging in past-month heavy episodic drinking and past-year alcohol-impaired driving, and numbers experiencing alcohol-related unintentional injury deaths and overdose hospitalizations between 1998 and 2014. Method: We analyzed national injury mortality data from coroner, census, and college enrollment statistics, the National Survey on Drug Use and Health, and the Nationwide Inpatient Sample. Results: From 1999 to 2005, percentages of emerging adults ages 18-24 reporting past-month heavy episodic drinking rose from 37.1% to 43.1% and then declined to 38.8% in 2014. Alcohol-impaired driving rose from 24% to 25.5% and then declined to 16.0%. Alcohol-related unintentional injury deaths increased from 4,807 in 1998 to 5,531 in 2005 and then declined to 4,105 in 2014, a reduction of 29% per 100,000 since 1998. Alcohol-related traffic deaths increased from 3,783 in 1998 to 4,114 in 2005 and then declined to 2,614 in 2014, down 43% per 100,000 since 1998. Alcohol-related overdose deaths increased from 207 in 1998 to 891 in 2014, a 254% increase per 100,000. Other types of nontraffic unintentional injury deaths declined. Alcohol-overdose hospitalizations rose 26% per 100,000 from 1998 to 2014, especially from increases in alcohol/other drug overdoses, up 61% (alcohol/opioid overdoses up 197%). Conclusions: Among emerging adults, a trend toward increased

alcohol-related unintentional injury deaths, heavy episodic drinking, and alcohol-impaired driving between 1998 and 2005 was reversed by 2014. Persistent high levels of heavy episodic drinking and related problems among emerging adults underscore a need to expand individually oriented interventions, college/community collaborative programs, and evidence-supported policies to reduce their drinking and related problems.

Hollenstein: This time, it's real: Affective flexibility, time scales, feedback loops, and the regulation of emotion Hollenstein-2015

Tom Hollenstein. "This time, it's real: Affective flexibility, time scales, feedback loops, and the regulation of emotion". In: *Emotion Review* 7.4 (July 2015), pp. 308–315. ISSN: 1754-0747. DOI: 10.1177/1754073915590621.

Abstract: Because both emotional arousal and regulation are continuous, ongoing processes, it is difficult, if not impossible, to separate them. Thus, affective dynamics can reveal the regulation of emotion as it occurs in real time. One way that this can be done is through the examination of intra- and interpersonal flexibility or the transitions into and out of affective states. The present article reviews and then expands upon the Flex3 model of real-time dynamic and reactive flexibility, specifying the ways in which individual differences in emotion regulation manifest as differences in flexibility. The differences in results at the real-time scale versus diurnal variability are also discussed within an emotion regulation framework.

Houben et al.: The relation between short-term emotion dynamics and psychological well-being: A meta-analysis Houben-VandenNoortgate-Kuppens-2015

Marlies Houben, Wim Van den Noortgate, and Peter Kuppens. "The relation between short-term emotion dynamics and psychological well-being: A meta-analysis". In: *Psychological Bulletin* 141.4 (July 2015), pp. 901–930. ISSN: 0033-2909. DOI: 10.1037/a0038822.

Abstract: Not only how good or bad people feel on average, but also how their feelings fluctuate across time is crucial for psychological health. The last 2 decades have witnessed a surge in research linking various patterns of short-term emotional change to adaptive or maladaptive psychological functioning, often with conflicting results. A meta-analysis was performed to identify consistent relationships between patterns of short-term emotion dynamics—including patterns reflecting emotional variability (measured in terms of within-person standard deviation of emotions across time), emotional instability (measured in terms of the magnitude of consecutive emotional changes), and emotional inertia of emotions over time (measured in terms of autocorrelation)—and relatively stable indicators of psychological well-being or psychopathology. We determined how such relationships are moderated by the type of emotional change, type of psychological well-being or psychopathology involved, valence of the emotion, and methodological factors. A total of 793 effect sizes were identified from 79 articles ($N = 11,381$) and were subjected to a 3-level meta-analysis. The results confirmed that overall, low psychological well-being co-occurs with more variable (overall $\hat{\rho} = -.178$), unstable (overall $\hat{\rho} = -.205$), but also more inert (overall $\hat{\rho} = -.151$) emotions. These effect sizes were stronger when involving negative compared with positive emotions. Moreover, the results provided evidence for consistency across different types of psychological well-being and psychopathology in their relation with these dynamical patterns, although specificity was also observed. The findings demonstrate that psychological flourishing is characterized by specific patterns of emotional fluctuations across time, and provide insight into what constitutes optimal and suboptimal emotional functioning.

Hunter: State space modeling in an open source, modular, structural equation modeling environment **Hunter-2017**

Michael D. Hunter. "State space modeling in an open source, modular, structural equation modeling environment". In: *Structural Equation Modeling: A Multidisciplinary Journal* 25.2 (Oct. 2017), pp. 307–324. DOI: 10.1080/10705511.2017.1369354.

Abstract: State space models (SSMs) are introduced in the context of structural equation modeling (SEM). In particular, the OpenMx implementation of SSMs using the Kalman filter and prediction error decomposition is discussed. In reflection of modularity, the implementation uses the same full information maximum likelihood missing data procedures for SSMs and SEMs. Similarly, generic OpenMx features such as likelihood ratio tests, profile likelihood confidence intervals, Hessian-based standard errors, definition variables, and the matrix algebra interface are all supported. Example scripts for specification of autoregressive models, multiple lag models (VAR(p)), multiple lag moving average models (VARMA(p, q)), multiple subject models, and latent growth models are provided. Additionally, latent variable calculation based on the Kalman filter and raw data generation based on a model are all included. Finally, future work for extending SSMs to allow for random effects and for presenting them in diagrams is discussed.

Hussong et al.: Integrative data analysis in clinical psychology research

Hussong-Curran-Bauer-2013

Andrea M. Hussong, Patrick J. Curran, and Daniel J. Bauer. “Integrative data analysis in clinical psychology research”. In: *Annual Review of Clinical Psychology* 9.1 (Mar. 2013), pp. 61–89. ISSN: 1548-5951. DOI: 10.1146/annurev-clinpsy-050212-185522.

Abstract: Integrative data analysis (IDA), a novel framework for conducting the simultaneous analysis of raw data pooled from multiple studies, offers many advantages including economy (i.e., reuse of extant data), power (i.e., large combined sample sizes), the potential to address new questions not answerable by a single contributing study (e.g., combining longitudinal studies to cover a broader swath of the lifespan), and the opportunity to build a more cumulative science (i.e., examining the similarity of effects across studies and potential reasons for dissimilarities). There are also methodological challenges associated with IDA, including the need to account for sampling heterogeneity across studies, to develop commensurate measures across studies, and to account for multiple sources of study differences as they impact hypothesis testing. In this review, we outline potential solutions to these challenges and describe future avenues for developing IDA as a framework for studies in clinical psychology.

Hussong et al.: An internalizing pathway to alcohol use and disorder.

Hussong-Jones-Stein-etal-2011

Andrea M. Hussong, Deborah J. Jones, et al. "An internalizing pathway to alcohol use and disorder." In: *Psychology of Addictive Behaviors* 25.3 (Sept. 2011), pp. 390–404. ISSN: 0893-164X. DOI: 10.1037/a0024519.

Abstract: Research emanating from the field of developmental science indicates that initial risk factors for alcohol use and disorder can be evident in early childhood. One dominant developmental pathway connecting these initial risk factors with subsequent alcohol involvement focuses on the central role of disinhibited or externalizing behaviors. In the current paper, we delineate a second pathway that focuses on internalizing symptomatology. Several studies indicate that internalizing symptoms in early and middle childhood predict alcohol involvement in adolescence and young adulthood. We use a developmental psychopathology framework to describe a risk model that traces the potential developmental markers of this internalizing pathway and to consider the relation between the internalizing pathway and the more widely researched externalizing pathway. We outline the markers of risk in this pathway and conclude with a discussion of the implications of this model for prevention efforts and future research. In this manner, we strive for a translational goal, linking our existing understanding of internalizing processes and alcohol use and disorder with our efforts to develop effective prevention programs.

Hutton et al.: Longitudinal Multi-Trait-State-Method Model Using Ordinal Data

Hutton-Chow-2014

R. Shane Hutton and Sy-Miin Chow. "Longitudinal Multi-Trait-State-Method Model Using Ordinal

Data". In: *Multivariate Behavioral Research* 49.3 (May 2014), pp. 269–282. ISSN: 1532-7906. DOI: 10.1080/00273171.2014.903832.

Abstract: Multi-trait multi-method (MTMM) models provide a way to assess convergent and discriminant validity when multiple traits are measured by multiple methods. In recent years, longitudinal extensions of MTMM models have been proposed in the structural equation modeling framework to evaluate whether and how the trait as well as method factors change over time. We propose a novel longitudinal ordinal MTMM model that can be used to effectively distinguish volatile "state" processes from "trait" processes that tend to remain stable and invariant over time. The proposed model, termed a longitudinal multi-trait-state-method (LM-TSM) model, combines 3 key modeling components: (a) a measurement model for ordinal data, (b) a vector autoregressive moving average model at the latent level to examine changes in the state as well as the method factors over time, and (c) a second-order factor-analytic model to capture time-invariant traits as shared variances among the state factors across all measurement occasions. Data from the Affective Dynamics and Individual Differences (ADID; Emotions and Dynamic Systems Laboratory, Citation2010) study was used to illustrate the proposed longitudinal LM-TSM model. Methodological issues associated with fitting the LM-TSM model are discussed.

Jackson et al.: Multivariate meta-analysis: Potential and promise

Jackson-Riley-White-2011

Dan Jackson, Richard D. Riley, and Ian R. White. "Multivariate meta-analysis: Potential and promise". In: *Statistics in Medicine* 30.20 (Jan. 2011), pp. 2481–2498. ISSN: 1097-0258. DOI: 10.1002/sim.4172.

Abstract: The multivariate random effects model is a generalization of the standard univariate model. Multivariate meta-analysis is becoming more commonly used and the techniques and related computer software, although continually under development, are now in place. In order to raise awareness of the multivariate methods, and discuss their advantages and disadvantages, we

organized a one day 'Multivariate meta-analysis' event at the Royal Statistical Society. In addition to disseminating the most recent developments, we also received an abundance of comments, concerns, insights, critiques and encouragement. This article provides a balanced account of the day's discourse. By giving others the opportunity to respond to our assessment, we hope to ensure that the various view points and opinions are aired before multivariate meta-analysis simply becomes another widely used de facto method without any proper consideration of it by the medical statistics community. We describe the areas of application that multivariate meta-analysis has found, the methods available, the difficulties typically encountered and the arguments for and against the multivariate methods, using four representative but contrasting examples. We conclude that the multivariate methods can be useful, and in particular can provide estimates with better statistical properties, but also that these benefits come at the price of making more assumptions which do not result in better inference in every case. Although there is evidence that multivariate meta-analysis has considerable potential, it must be even more carefully applied than its univariate counterpart in practice.

Jackson et al.: Quantifying the impact of between-study heterogeneity in multivariate meta-analyses **Jackson-White-Riley-2012**

Dan Jackson, Ian R. White, and Richard D. Riley. "Quantifying the impact of between-study heterogeneity in multivariate meta-analyses". In: *Statistics in Medicine* 31.29 (July 2012), pp. 3805–3820. ISSN: 1097-0258. DOI: 10.1002/sim.5453.

Abstract: Measures that quantify the impact of heterogeneity in univariate meta-analysis, including the very popular I^2 statistic, are now well established. Multivariate meta-analysis, where studies provide multiple outcomes that are pooled in a single analysis, is also becoming more commonly used. The question of how to quantify heterogeneity in the multivariate setting is therefore raised. It is the univariate R^2 statistic, the ratio of the variance of the estimated treatment effect under the random and fixed effects models, that generalises most naturally, so this statistic provides our basis. This statistic is then used to derive a multivariate analogue of I^2, which we call I^2_R. We

also provide a multivariate H^2 statistic, the ratio of a generalisation of Cochran's heterogeneity statistic and its associated degrees of freedom, with an accompanying generalisation of the usual I^2 statistic, I_H^2. Our proposed heterogeneity statistics can be used alongside all the usual estimates and inferential procedures used in multivariate meta-analysis. We apply our methods to some real datasets and show how our statistics are equally appropriate in the context of multivariate meta-regression, where study level covariate effects are included in the model. Our heterogeneity statistics may be used when applying any procedure for fitting the multivariate random effects model.

Jensen et al.: Contributions of psychology to the understanding and treatment of people with chronic pain: Why it matters to ALL psychologists Jensen-Turk-2014

Mark P. Jensen and Dennis C. Turk. "Contributions of psychology to the understanding and treatment of people with chronic pain: Why it matters to ALL psychologists". In: *American Psychologist* 69.2 (2014), pp. 105–118. ISSN: 0003-066X. DOI: 10.1037/a0035641.

Abstract: Chronic pain is a prevalent problem with significant costs to individuals, significant others, and society. In this article, which introduces the American Psychologist special issue on chronic pain, we provide an overview of the seminal contributions made by psychologists to our current understanding of this important problem. We also describe the primary treatments that have been developed based on psychological principles and models of pain, many of which have demonstrated efficacy for reducing pain and its impact on psychological and physical functioning. The article ends with an enumeration of directions for future research and clinical practice. We believe that the chronicle of psychology's role in improving our understanding and treatment of pain provides a model for how psychologists can have a significant influence on many fields, and that the models and approaches developed for understanding and treating pain may be of use to psychologists working in other areas. Thus, we think that chronic pain is an important area of study that offers insights about translational research for ALL psychologists.

Jeff A. Jones and Niels G. Waller. "Computing confidence intervals for standardized regression coefficients." In: *Psychological Methods* 18.4 (2013), pp. 435–453. DOI: 10.1037/a0033269.

Abstract: With fixed predictors, the standard method (Cohen, Cohen, West, & Aiken, 2003, p. 86; Harris, 2001, p. 80; Hays, 1994, p. 709) for computing confidence intervals (CIs) for standardized regression coefficients fails to account for the sampling variability of the criterion standard deviation. With random predictors, this method also fails to account for the sampling variability of the predictor standard deviations. Nevertheless, under some conditions the standard method will produce CIs with accurate coverage rates. To delineate these conditions, we used a Monte Carlo simulation to compute empirical CI coverage rates in samples drawn from 36 populations with a wide range of data characteristics. We also computed the empirical CI coverage rates for 4 alternative methods that have been discussed in the literature: noncentrality interval estimation, the delta method, the percentile bootstrap, and the bias-corrected and accelerated bootstrap. Our results showed that for many data-parameter configurations–for example, sample size, predictor correlations, coefficient of determination (R^2), orientation of β with respect to the eigenvectors of the predictor correlation matrix, R_X–the standard method produced coverage rates that were close to their expected values. However, when population R^2 was large and when β approached the last eigenvector of R_X, then the standard method coverage rates were frequently below the nominal rate (sometimes by a considerable amount). In these conditions, the delta method and the 2 bootstrap procedures were consistently accurate. Results using noncentrality interval estimation were inconsistent. In light of these findings, we recommend that researchers use the delta method to evaluate the sampling variability of standardized regression coefficients.

J. A. Jones et al.: The normal-theory and asymptotic distribution-free (ADF) covariance matrix of standardized regression coefficients: Theoretical extensions and finite sample behavior Jones-Waller-2015

Jeff A. Jones and Niels G. Waller. "The normal-theory and asymptotic distribution-free (ADF) covariance matrix of standardized regression coefficients: Theoretical extensions and finite sample behavior". In: *Psychometrika* 80.2 (June 2015), pp. 365–378. DOI: 10.1007/s11336-013-9380-y.

Abstract: Yuan and Chan (Psychometrika, 76, 670-690, 2011) recently showed how to compute the covariance matrix of standardized regression coefficients from covariances. In this paper, we describe a method for computing this covariance matrix from correlations. Next, we describe an asymptotic distribution-free (ADF; Browne in British Journal of Mathematical and Statistical Psychology, 37, 62-83, 1984) method for computing the covariance matrix of standardized regression coefficients. We show that the ADF method works well with nonnormal data in moderate-to-large samples using both simulated and real-data examples. R code (R Development Core Team, 2012) is available from the authors or through the Psychometrika online repository for supplementary materials.

Jongerling et al.: A multilevel AR(1) model: Allowing for inter-individual differences in trait-scores, inertia, and innovation variance Jongerling-Laurenceau-Hamaker-2015

Joran Jongerling, Jean-Philippe Laurenceau, and Ellen L. Hamaker. "A multilevel AR(1) model: Allowing for inter-individual differences in trait-scores, inertia, and innovation variance". In: *Multivariate Behavioral Research* 50.3 (May 2015), pp. 334–349. ISSN: 1532-7906. DOI: 10.1080/00273171.2014.1003772.

Abstract: In this article we consider a multilevel first-order autoregressive [AR(1)] model with random intercepts, random autoregression, and random innovation variance (i.e., the level 1 residual variance). Including random innovation variance is an important extension of the multilevel AR(1) model for two reasons. First, between-person differences in innovation variance are important from a substantive point of view, in that they capture differences in sensitivity and/or exposure to unmea-

sured internal and external factors that influence the process. Second, using simulation methods we show that modeling the innovation variance as fixed across individuals, when it should be modeled as a random effect, leads to biased parameter estimates. Additionally, we use simulation methods to compare maximum likelihood estimation to Bayesian estimation of the multilevel AR(1) model and investigate the trade-off between the number of individuals and the number of time points. We provide an empirical illustration by applying the extended multilevel AR(1) model to daily positive affect ratings from 89 married women over the course of 42 consecutive days.

Kazak: Editorial: Journal article reporting standards **Kazak-2018**

Anne E. Kazak. "Editorial: Journal article reporting standards". In: *American Psychologist* 73.1 (Jan. 2018), pp. 1–2. ISSN: 0003-066X. DOI: 10.1037/amp0000263.

Kelley et al.: On effect size **Kelley-Preacher-2012**

Ken Kelley and Kristopher J. Preacher. "On effect size". In: *Psychological Methods* 17.2 (2012), pp. 137–152. ISSN: 1082-989X. DOI: 10.1037/a0028086.

Abstract: The call for researchers to report and interpret effect sizes and their corresponding confidence intervals has never been stronger. However, there is confusion in the literature on the definition of effect size, and consequently the term is used inconsistently. We propose a definition for effect size, discuss 3 facets of effect size (dimension, measure/index, and value), outline 10 corollaries that follow from our definition, and review ideal qualities of effect sizes. Our definition of effect size is general and subsumes many existing definitions of effect size. We define effect size as *a quantitative reflection of the magnitude of some phenomenon that is used for the purpose of addressing a question of interest*. Our definition of effect size is purposely more inclusive than the way many have defined and conceptualized effect size, and it is unique with regard to linking effect size to a question of interest. Additionally, we review some important developments in the effect size literature and discuss the importance of accompanying an effect size with an interval estimate that acknowledges the uncertainty with which the population value of the effect size has been estimated. We hope that this article will facilitate discussion and improve the practice of reporting and interpreting effect sizes.

Kenny et al.: Power anomalies in testing mediation **Kenny-Judd-2013**

David A. Kenny and Charles M. Judd. "Power anomalies in testing mediation". In: *Psychological Science* 25.2 (Dec. 2013), pp. 334–339. ISSN: 1467-9280. DOI: 10.1177/0956797613502676.

Abstract: Two rather surprising anomalies relating to statistical power occur in testing mediation. First, in a model with no direct effect for which the total effect and indirect effect are identical, the power for the test of the total effect can be dramatically smaller than the power for the test of the indirect effect. Second, when there is a direct effect of a causal variable on the outcome controlling for the mediator, the power of the test of the indirect effect is often considerably greater than the power of the test of the direct effect, even when the two are of the same magnitude. We try to explain the reasons for these anomalies and how they affect practice.

Kisbu-Sakarya et al.: The distribution of the product explains normal theory mediation confidence interval estimation **KisbuSakarya-MacKinnon-Miocevic-2014**

Yasemin Kisbu-Sakarya, David P. MacKinnon, and Milica Miočević. "The distribution of the product explains normal theory mediation confidence interval estimation". In: *Multivariate Behavioral Research* 49.3 (May 2014), pp. 261–268. DOI: 10.1080/00273171.2014.903162.

Abstract: The distribution of the product has several useful applications. One of these applications is its use to form confidence intervals for the indirect effect as the product of 2 regression coefficients. The purpose of this article is to investigate how the moments of the distribution of the product explain normal theory mediation confidence interval coverage and imbalance. Values of the critical ratio for each random variable are used to demonstrate how the moments of the distribution of the product change across values of the critical ratio observed in research studies. Results of the simulation study showed that as skewness in absolute value increases, coverage decreases. And as skewness in absolute value and kurtosis increases, imbalance increases. The difference between testing the significance of the indirect effect using the normal theory versus the asymmetric distribution of the product is further illustrated with a real data example. This article is the first study to show the direct link between the distribution of the product and indirect effect confidence intervals and clarifies the results of previous simulation studies by showing why normal theory confidence intervals for indirect effects are often less accurate than those obtained from the asymmetric distribution of the product or from resampling methods.

Koopman et al.: Small sample mediation testing: Misplaced confidence in bootstrapped confidence intervals **Koopman-Howe-Hollenbeck-etal-2015**

Joel Koopman et al. "Small sample mediation testing: Misplaced confidence in bootstrapped confidence intervals". In: *Journal of Applied Psychology* 100.1 (2015), pp. 194–202. DOI: 10.1037/a0036635.

Abstract: Bootstrapping is an analytical tool commonly used in psychology to test the statistical significance of the indirect effect in mediation models. Bootstrapping proponents have particularly advocated for its use for samples of 20-80 cases. This advocacy has been heeded, especially in the Journal of Applied Psychology, as researchers are increasingly utilizing bootstrapping to test mediation with samples in this range. We discuss reasons to be concerned with this escalation, and in a simulation study focused specifically on this range of sample sizes, we demonstrate not only that bootstrapping has insufficient statistical power to provide a rigorous hypothesis test in most conditions but also that bootstrapping has a tendency to exhibit an inflated Type I error rate. We then extend our simulations to investigate an alternative empirical resampling method as well as a Bayesian approach and demonstrate that they exhibit comparable statistical power to bootstrapping in small samples without the associated inflated Type I error. Implications for researchers testing mediation hypotheses in small samples are presented. For researchers wishing to use these methods in their own research, we have provided R syntax in the online supplemental materials.

Kossakowski et al.: Data from 'Critical slowing down as a personalized early warning signal for depression' Kossakowski-Groot-Haslbeck-2017

Jolanda J. Kossakowski et al. "Data from 'Critical slowing down as a personalized early warning signal for depression'". In: *Journal of Open Psychology Data* 5 (Feb. 2017). ISSN: 2050-9863. DOI: 10.5334/jopd.29.

Abstract: We present a dataset of a single (N = 1) participant diagnosed with major depressive disorder, who completed 1478 measurements over the course of 239 consecutive days in 2012 and 2013. The experiment included a double-blind phase in which the dosage of anti-depressant medication was gradually reduced. The entire study looked at momentary affective states in daily life before, during, and after the double-blind phase. The items, which were asked ten times a day, cover topics like mood, physical condition and social contacts. Also, depressive symptoms were measured on a weekly basis using the Symptom Checklist Revised (SCL-90-R). The data are suitable for various time-series analyses and studies in complex dynamical systems.

Koval et al.: Emotional inertia is associated with lower well-being when controlling for differences in emotional context Koval-Sutterlin-Kuppens-2016

Peter Koval, Stefan Sütterlin, and Peter Kuppens. "Emotional inertia is associated with lower well-being when controlling for differences in emotional context". In: *Frontiers in Psychology* 6 (Jan. 2016). ISSN: 1664-1078. DOI: 10.3389/fpsyg.2015.01997.

Abstract: Previous studies have linked higher emotional inertia (i.e., a stronger autoregressive slope of emotions) with lower well-being. We aimed to replicate these findings, while extending upon previous research by addressing a number of unresolved issues and controlling for potential confounds. Specifically, we report results from two studies (Ns = 100 and 202) examining how emotional inertia, assessed in response to a standardized sequence of emotional stimuli in the lab, correlates with several measures of well-being. The current studies build on previous research by examining how inertia of both positive emotions (PE) and negative emotions (NE) relates to positive (e.g., life satisfaction) and negative (e.g., depressive symptoms) indicators of well-being, while controlling for between-person differences in the mean level and variability of emotions. Our findings replicated previous research and further revealed that (a) NE inertia was more strongly associated with lower well-being than PE inertia; (b) emotional inertia correlated more consistently with negative indicators (e.g., depressive symptoms) than positive indicators (e.g., life satisfaction) of well-being; and (c) these relationships were independent of individual differences in mean level and variability of emotions. We conclude, in line with recent findings, that higher emotional inertia, particularly of NE, may be an indicator of increased vulnerability to depression.

Kuiper et al.: Drawing conclusions from cross-lagged relationships: Re-considering the role of the time-interval **Kuiper-Oisin-2018**

Rebecca M. Kuiper and Oisin Ryan. "Drawing conclusions from cross-lagged relationships: Re-considering the role of the time-interval". In: *Structural Equation Modeling: A Multidisciplinary Journal* 25.5 (Mar. 2018), pp. 809–823. DOI: 10.1080/10705511.2018.1431046.

Abstract: The cross-lagged panel model (CLPM), a discrete-time (DT) SEM model, is frequently used to gather evidence for (reciprocal) Granger-causal relationships when lacking an experimental design. However, it is well known that CLPMs can lead to different parameter estimates depending on the time-interval of observation. Consequently, this can lead to researchers drawing conflicting conclusions regarding the sign and/or dominance of relationships. Multiple authors have suggested the use of continuous-time models to address this issue. In this article, we demonstrate the exact circumstances under which such conflicting conclusions occur. Specifically, we show that such conflicts are only avoided in general in the case of bivariate, stable, nonoscillating, first-order systems, when comparing models with uniform time-intervals between observations. In addition, we provide a range of tools, proofs, and guidelines regarding the comparison of discrete- and continuous-time parameter estimates.

Kuppens: It's about time: A special section on affect dynamics Kuppens-2015

Peter Kuppens. "It's about time: A special section on affect dynamics". In: *Emotion Review* 7.4 (July 2015), pp. 297–300. ISSN: 1754-0747. DOI: 10.1177/1754073915590947.

Abstract: The study of affect dynamics aims to discover the patterns and regularities with which emotions and affective experiences and components change across time, the underlying mechanisms involved, and their potential relevance for healthy psychological functioning. The intention of this special section is to serve as a mini handbook covering the contemporary state of research into affect dynamics. Contributions address theoretical viewpoints on the origins and functions of emotional change, methodological and modeling approaches, biological and social perspectives on affect dynamics, and the downstream consequences for well-being and psychopathology.

Kuppens et al.: Emotional inertia and psychological maladjustment Kuppens-Allen-Sheeber-2010

Peter Kuppens, Nicholas B. Allen, and Lisa B. Sheeber. "Emotional inertia and psychological maladjustment". In: *Psychological Science* 21.7 (May 2010), pp. 984–991. ISSN: 1467-9280. DOI: 10.1177/0956797610372634.

Abstract: In this article, we examine the concept of emotional inertia as a fundamental property of the emotion dynamics that characterize psychological maladjustment. Emotional inertia refers to the degree to which emotional states are resistant to change. Because psychological maladjustment has been associated with both emotional underreactivity and ineffective emotion-regulation skills, we hypothesized that its overall emotion dynamics would be characterized by high levels of inertia. We provide evidence from two naturalistic studies that, using different methods, showed that the emotional fluctuations of individuals who exhibited low self-esteem (Study 1) and depression (Study

2) were characterized by higher levels of inertia in both positive and negative emotions than the emotional fluctuations of people who did not exhibit low self-esteem and depression. We also discuss the usefulness of the concept of emotional inertia as a hallmark of maladaptive emotion dynamics.

Kuppens et al.: Feelings change: Accounting for individual differences in the temporal dynamics of affect **Kuppens-Oravecz-Tuerlinckx-2010**

Peter Kuppens, Zita Oravecz, and Francis Tuerlinckx. "Feelings change: Accounting for individual differences in the temporal dynamics of affect". In: *Journal of Personality and Social Psychology* 99.6 (Dec. 2010), pp. 1042–1060. ISSN: 0022-3514. DOI: 10.1037/a0020962.

Abstract: People display a remarkable variability in the patterns and trajectories with which their feelings change over time. In this article, we present a theoretical account for the dynamics of affect (DynAffect) that identifies the major processes underlying individual differences in the temporal dynamics of affective experiences. It is hypothesized that individuals are characterized by an affective home base, a baseline attractor state around which affect fluctuates. These fluctuations vary as the result of internal or external processes to which an individual is more or less sensitive and are regulated and tied back to the home base by the attractor strength. Individual differences in these 3 processes—affective home base, variability, and attractor strength—are proposed to underlie individual differences in affect dynamics. The DynAffect account is empirically evaluated by means of a diffusion modeling approach in 2 extensive experience-sampling studies on people's core affective experiences. The findings show that the model is capable of adequately capturing the observed dynamics in core affect across both large (Study 1) and shorter time scales (Study 2) and illuminate how the key processes are related to personality and emotion dispositions. Implications for the understanding of affect dynamics and affective dysfunctioning in psychopathology are also discussed.

Kuppens et al.: Emotional inertia prospectively predicts the onset of depressive disorder in adolescence **Kuppens-Sheeber-Yap-etal-2012**

Peter Kuppens, Lisa B. Sheeber, et al. "Emotional inertia prospectively predicts the onset of depressive disorder in adolescence". In: *Emotion* 12.2 (Apr. 2012), pp. 283–289. ISSN: 1528-3542. DOI: 10.1037/a0025046.

Abstract: Emotional inertia refers to the degree to which a person's current emotional state is predicted by their prior emotional state, reflecting how much it carries over from one moment to the next. Recently, in a cross-sectional study, we showed that high inertia is an important characteristic of the emotion dynamics observed in psychological maladjustment such as depression. In the present study, we examined whether emotional inertia prospectively predicts the onset of first-episode depression during adolescence. Emotional inertia was assessed in a sample of early adolescents ($N = 165$) based on second-to-second behavioral coding of videotaped naturalistic interactions with a parent. Greater inertia of both negative and positive emotional behaviors predicted the emergence of clinical depression 2.5 years later. The implications of these findings for the understanding of the etiology and early detection of depression are discussed.

Kurtzer et al.: Singularity: Scientific containers for mobility of compute **Kurtzer-Sochat-Bauer-2017**

Gregory M. Kurtzer, Vanessa Sochat, and Michael W. Bauer. "Singularity: Scientific containers for mobility of compute". In: *PLOS ONE* 12.5 (May 2017), e0177459. DOI: 10.1371/journal.pone.0177459.

Abstract: Here we present Singularity, software developed to bring containers and reproducibility to scientific computing. Using Singularity containers, developers can work in reproducible environments of their choosing and design, and these complete environments can easily be copied and executed on other platforms. Singularity is an open source initiative that harnesses the expertise of system and software engineers and researchers alike, and integrates seamlessly into common work-

flows for both of these groups. As its primary use case, Singularity brings mobility of computing to both users and HPC centers, providing a secure means to capture and distribute software and compute environments. This ability to create and deploy reproducible environments across these centers, a previously unmet need, makes Singularity a game changing development for computational science.

Kwan et al.: Comparing standardized coefficients in structural equation modeling: A model reparameterization approach **Kwan-Chan-2011**

Joyce L. Y. Kwan and Wai Chan. "Comparing standardized coefficients in structural equation modeling: A model reparameterization approach". In: *Behavior Research Methods* 43.3 (Apr. 2011), pp. 730–745. DOI: 10.3758/s13428-011-0088-6.

Abstract: We propose a two-stage method for comparing standardized coefficients in structural equation modeling (SEM). At stage 1, we transform the original model of interest into the standardized model by model reparameterization, so that the model parameters appearing in the standardized model are equivalent to the standardized parameters of the original model. At stage 2, we impose appropriate linear equality constraints on the standardized model and use a likelihood ratio test to make statistical inferences about the equality of standardized coefficients. Unlike other existing methods for comparing standardized coefficients, the proposed method does not require specific modeling features (e.g., specification of nonlinear constraints), which are available only in certain SEM software programs. Moreover, this method allows researchers to compare two or more standardized coefficients simultaneously in a standard and convenient way. Three real examples are given to illustrate the proposed method, using EQS, a popular SEM software program. Results show that the proposed method performs satisfactorily for testing the equality of standardized coefficients.

Kwan et al.: Comparing squared multiple correlation coefficients using structural equation modeling **Kwan-Chan-2014**

Joyce L. Y. Kwan and Wai Chan. "Comparing squared multiple correlation coefficients using structural equation modeling". In: *Structural Equation Modeling: A Multidisciplinary Journal* 21.2 (Apr. 2014), pp. 225–238. DOI: 10.1080/10705511.2014.882673.

Abstract: In social science research, a common topic in multiple regression analysis is to compare the squared multiple correlation coefficients in different populations. Existing methods based on asymptotic theories (Olkin & Finn, 1995) and bootstrapping (Chan, 2009) are available but these can only handle a 2-group comparison. Another method based on structural equation modeling (SEM) has been proposed recently. However, this method has three disadvantages. First, it requires the user to explicitly specify the sample R2 as a function in terms of the basic SEM model parameters, which is sometimes troublesome and error prone. Second, it requires the specification of nonlinear constraints, which is not available in some popular SEM software programs. Third, it is for a 2-group comparison primarily. In this article, a 2-stage SEM method is proposed as an alternative. Unlike all other existing methods, the proposed method is simple to use, and it does not require any specific programming features such as the specification of nonlinear constraints. More important, the method allows a simultaneous comparison of 3 or more groups. A real example is given to illustrate the proposed method using EQS, a popular SEM software program.

Lachowicz et al.: A novel measure of effect size for mediation analysis **Lachowicz-Preacher-Kelley-2018**

Mark J. Lachowicz, Kristopher J. Preacher, and Ken Kelley. "A novel measure of effect size for mediation analysis". In: *Psychological Methods* 23.2 (June 2018), pp. 244–261. ISSN: 1082-989X. DOI: 10.1037/met0000165.

Abstract: Mediation analysis has become one of the most popular statistical methods in the social sciences. However, many currently available effect size measures for mediation have limitations that

restrict their use to specific mediation models. In this article, we develop a measure of effect size that addresses these limitations. We show how modification of a currently existing effect size measure results in a novel effect size measure with many desirable properties. We also derive an expression for the bias of the sample estimator for the proposed effect size measure and propose an adjusted version of the estimator. We present a Monte Carlo simulation study conducted to examine the finite sampling properties of the adjusted and unadjusted estimators, which shows that the adjusted estimator is effective at recovering the true value it estimates. Finally, we demonstrate the use of the effect size measure with an empirical example. We provide freely available software so that researchers can immediately implement the methods we discuss. Our developments here extend the existing literature on effect sizes and mediation by developing a potentially useful method of communicating the magnitude of mediation.

Leffingwell et al.: Continuous Objective Monitoring of Alcohol Use: Twenty-First Century Measurement Using Transdermal Sensors Leffingwell-Cooney-Murphy-etal-2012

Thad R. Leffingwell et al. "Continuous Objective Monitoring of Alcohol Use: Twenty-First Century Measurement Using Transdermal Sensors". In: *Alcoholism: Clinical and Experimental Research* 37.1 (July 2012), pp. 16–22. ISSN: 1530-0277. DOI: 10.1111/j.1530-0277.2012.01869.x.

Abstract: Transdermal alcohol sensors continuously collect reliable and valid data on alcohol consumption in vivo over the course of hours to weeks. Transdermal alcohol readings are highly correlated with breath alcohol measurements, but transdermal alcohol levels lag behind breath alcohol levels by one or more hours owing to the longer time required for alcohol to be expelled through perspiration. By providing objective information about alcohol consumption, transdermal alcohol sensors can validate self-report and provide important information not previously available. In this article, we describe the development and evaluation of currently available transdermal alcohol sensors, present the strengths and limitations of the technology, and give examples of recent research using the sensors.

Heidi M. Levitt et al. "Journal article reporting standards for qualitative primary, qualitative meta-analytic, and mixed methods research in psychology: The APA Publications and Communications Board task force report". In: *American Psychologist* 73.1 (Jan. 2018), pp. 26–46. ISSN: 0003-066X. DOI: 10.1037/amp0000151.

Abstract: The American Psychological Association Publications and Communications Board Working Group on Journal Article Reporting Standards for Qualitative Research (JARS-Qual Working Group) was charged with examining the state of journal article reporting standards as they applied to qualitative research and with generating recommendations for standards that would be appropriate for a wide range of methods within the discipline of psychology. These standards describe what should be included in a research report to enable and facilitate the review process. This publication marks a historical moment—the first inclusion of qualitative research in APA Style, which is the basis of both the Publication Manual of the American Psychological Association (APA, 2010) and APA Style CENTRAL, an online program to support APA Style. In addition to the general JARS-Qual guidelines, the Working Group has developed standards for both qualitative meta-analysis and mixed methods research. The reporting standards were developed for psychological qualitative research but may hold utility for a broad range of social sciences. They honor a range of qualitative traditions, methods, and reporting styles. The Working Group was composed of a group of researchers with backgrounds in varying methods, research topics, and approaches to inquiry. In this article, they present these standards and their rationale, and they detail the ways that the standards differ from the quantitative research reporting standards. They describe how the standards can be used by authors in the process of writing qualitative research for submission as well as by reviewers and editors in the process of reviewing research.

Haiyan Liu et al.: Comparison of inverse Wishart and separation-strategy priors for Bayesian estimation of covariance parameter matrix in growth curve analysis

Liu-Zhang-Grimm-2015

Haiyan Liu, Zhiyong Zhang, and Kevin J. Grimm. "Comparison of inverse Wishart and separation-strategy priors for Bayesian estimation of covariance parameter matrix in growth curve analysis". In: *Structural Equation Modeling: A Multidisciplinary Journal* 23.3 (Sept. 2015), pp. 354–367. ISSN: 1532-8007. DOI: 10.1080/10705511.2015.1057285.

Abstract: Growth curve modeling provides a general framework for analyzing longitudinal data from social, behavioral, and educational sciences. Bayesian methods have been used to estimate growth curve models, in which priors need to be specified for unknown parameters. For the covariance parameter matrix, the inverse Wishart prior is most commonly used due to its proper and conjugate properties. However, many researchers have pointed out that the inverse Wishart prior might not work as expected. The purpose of this study is to investigate the influence of the inverse Wishart prior and compare it with a class of separation-strategy priors on the parameter estimates of growth curve models. In this article, we illustrate the use of different types of priors with 2 real data analyses, and then conduct simulation studies to evaluate and compare these priors in estimating both linear and nonlinear growth curve models. For the linear model, the simulation study shows that both the inverse Wishart and the separation-strategy priors work well for the fixed effects parameters. For the Level 1 residual variance estimate, the separation-strategy prior performs better than the inverse Wishart prior. For the covariance matrix, the results are mixed. Overall, the inverse Wishart prior is suggested if the population correlation coefficient and at least 1 of the 2 marginal variances are large. Otherwise, the separation-strategy prior is preferred. For the nonlinear growth curve model, the separation-strategy priors work better than the inverse Wishart prior.

Huiying Liu et al.: Everyday social interactions and intra-individual variability in affect: A systematic review and meta-analysis of ecological momentary assessment studies **Liu-Xie-Lou-2019**

Huiying Liu, Qian Wen Xie, and Vivian W. Q. Lou. "Everyday social interactions and intra-individual variability in affect: A systematic review and meta-analysis of ecological momentary assessment studies". In: *Motivation and Emotion* 43.2 (2019), pp. 339–353. ISSN: 1573-6644. DOI: 10.1007/s11031-018-9735-x.

Abstract: Intra-individual variability in affect has been associated with aspects of individuals' lives, such as everyday social interactions. Ecological momentary assessment (EMA) has been tailored to assess the dynamics of everyday events and feelings and in recent decades there has been a burgeoning of EMA research on the intra-individual links between social interactions and momentary affect in daily life. This systematic review and meta-analysis was conducted to identify within-person relationships between social interactions with positive affect (PA) and negative affect (NA). Both quantitative and qualitative features of social interaction (i.e., uniqueness of the partner, interpersonal perceptions of an interaction) predict within-person variance in PA and NA. The results of meta-analysis indicated small-sized associations between quantitative features of social interactions with PA and NA, and moderate-sized associations between qualitative features of social interactions with PA and NA, which were only somewhat moderated by methodological factors. We conclude that EMA is a promising method of investigating intra-individual variability in affect unfolding in everyday social environments and offer suggestions for substantiating the within-person perspective to both researchers and clinicians.

S. Liu: Person-specific versus multilevel autoregressive models: Accuracy in parameter estimates at the population and individual levels **Liu-2017**

Siwei Liu. "Person-specific versus multilevel autoregressive models: Accuracy in parameter estimates at the population and individual levels". In: *British Journal of Mathematical and Statistical Psychology* 70.3 (Feb. 2017), pp. 480–498. ISSN: 2044-8317. DOI: 10.1111/bmsp.12096.

Abstract: This paper compares the multilevel modelling (MLM) approach and the person-specific (PS) modelling approach in examining autoregressive (AR) relations with intensive longitudinal data. Two simulation studies are conducted to examine the influences of sample heterogeneity, time series length, sample size, and distribution of individual level AR coefficients on the accuracy of AR estimates, both at the population level and at the individual level. It is found that MLM generally outperforms the PS approach under two conditions: when the sample has a homogeneous AR pattern, namely, when all individuals in the sample are characterized by AR processes with the same order; and when the sample has heterogeneous AR patterns, but a multilevel model with a sufficiently high order (i.e., an order equal to or higher than the maximum order of individual AR patterns in the sample) is fitted and successfully converges. If a lower-order multilevel model is chosen for heterogeneous samples, the higher-order lagged effects are misrepresented, resulting in bias at the population level and larger prediction errors at the individual level. In these cases, the PS approach is preferable, given sufficient measurement occasions ($T \geq 50$). In addition, sample size and distribution of individual level AR coefficients do not have a large impact on the results. Implications of these findings on model selection and research design are discussed.

Maxwell et al.: Bias in cross-sectional analyses of longitudinal mediation: Partial and complete mediation under an autoregressive model Maxwell-Cole-Mitchell-2011

Scott E. Maxwell, David A. Cole, and Melissa A. Mitchell. "Bias in cross-sectional analyses of longitudinal mediation: Partial and complete mediation under an autoregressive model". In: *Multivariate Behavioral Research* 46.5 (Sept. 2011), pp. 816–841. DOI: 10.1080/00273171.2011.606716.

Abstract: Maxwell and Cole (2007) showed that cross-sectional approaches to mediation typically generate substantially biased estimates of longitudinal parameters in the special case of complete mediation. However, their results did not apply to the more typical case of partial mediation. We

extend their previous work by showing that substantial bias can also occur with partial mediation. In particular, cross-sectional analyses can imply the existence of a substantial indirect effect even when the true longitudinal indirect effect is zero. Thus, a variable that is found to be a strong mediator in a cross-sectional analysis may not be a mediator at all in a longitudinal analysis. In addition, we show that very different combinations of longitudinal parameter values can lead to essentially identical cross-sectional correlations, raising serious questions about the interpretability of cross-sectional mediation data. More generally, researchers are encouraged to consider a wide variety of possible mediation models beyond simple cross-sectional models, including but not restricted to autoregressive models of change.

Merkel: Docker: Lightweight Linux containers for consistent development and deployment **Merkel-2014**

Dirk Merkel. "Docker: Lightweight Linux containers for consistent development and deployment". In: *Linux Journal* 2014.239 (2014), p. 2. URL: https://www.linuxjournal.com/content/docker-lightweight-linux-containers-consistent-development-and-deployment.

Miocevic et al.: A tutorial in Bayesian potential outcomes mediation analysis **Miocevic-Gonzalez-Valente-etal-2017**

Milica Miocevic et al. "A tutorial in Bayesian potential outcomes mediation analysis". In: *Structural Equation Modeling: A Multidisciplinary Journal* 25.1 (July 2017), pp. 121–136. ISSN: 1532-8007. DOI: 10.1080/10705511.2017.1342541.

Abstract: Statistical mediation analysis is used to investigate intermediate variables in the relation between independent and dependent variables. Causal interpretation of mediation analyses is challenging because randomization of subjects to levels of the independent variable does not rule out the possibility of unmeasured confounders of the mediator to outcome relation. Furthermore, commonly used frequentist methods for mediation analysis compute the probability of the data given the null

hypothesis, which is not the probability of a hypothesis given the data as in Bayesian analysis. Under certain assumptions, applying the potential outcomes framework to mediation analysis allows for the computation of causal effects, and statistical mediation in the Bayesian framework gives indirect effects probabilistic interpretations. This tutorial combines causal inference and Bayesian methods for mediation analysis so the indirect and direct effects have both causal and probabilistic interpretations. Steps in Bayesian causal mediation analysis are shown in the application to an empirical example.

Moeyaert et al.: Methods for dealing with multiple outcomes in meta-analysis:a comparison between averaging effect sizes, robust variance estimation and multilevel meta-analysis **Moeyaert-Ugille-Beretvas-2016**

Mariola Moeyaert et al. "Methods for dealing with multiple outcomes in meta-analysis:a comparison between averaging effect sizes, robust variance estimation and multilevel meta-analysis". In: *International Journal of Social Research Methodology* 20.6 (Nov. 2016), pp. 559–572. ISSN: 1464-5300. DOI: 10.1080/13645579.2016.1252189.

Abstract: This study investigates three methods to handle dependency among effect size estimates in meta-analysis arising from studies reporting multiple outcome measures taken on the same sample. The three-level approach is compared with the method of robust variance estimation, and with averaging effects within studies. A simulation study is performed, and the fixed and random effect estimates of the three methods are compared with each other. Both the robust variance estimation and three-level approach result in unbiased estimates of the fixed effects, corresponding standard errors and variances. Averaging effect sizes results in overestimated standard errors when the effect sizes within studies are truly independent. Although the robust variance and three-level approach are more complicated to use, they have the advantage that they do not require an estimate of the correlation between outcomes, and they still result in unbiased parameter estimates.

Molenaar: Equivalent Dynamic Models Molenaar-2017

Peter C. M. Molenaar. "Equivalent Dynamic Models". In: *Multivariate Behavioral Research* 52.2 (Feb. 2017), pp. 242–258. ISSN: 1532-7906. DOI: 10.1080/00273171.2016.1277681.

Abstract: Equivalences of two classes of dynamic models for weakly stationary multivariate time series are discussed: dynamic factor models and autoregressive models. It is shown that exploratory dynamic factor models can be rotated, yielding an infinite set of equivalent solutions for any observed series. It also is shown that dynamic factor models with lagged factor loadings are not equivalent to the currently popular state-space models, and that restriction of attention to the latter type of models may yield invalid results. The known equivalent vector autoregressive model types, standard and structural, are given a new interpretation in which they are conceived of as the extremes of an innovating type of hybrid vector autoregressive models. It is shown that consideration of hybrid models solves many problems, in particular with Granger causality testing.

Moneta et al.: Causal search in structural vector autoregressive models Moneta-Chlas-Entner-etal-2011

Alessio Moneta et al. "Causal search in structural vector autoregressive models". In: *Journal of Machine Learning Research - Proceedings Track* 12 (Jan. 2011), pp. 95–114.

Abstract: This paper reviews a class of methods to perform causal inference in the framework of a structural vector autoregressive model. We consider three different settings. In the first setting the underlying system is linear with normal disturbances and the structural model is identified by exploiting the information incorporated in the partial correlations of the estimated residuals. Zero partial correlations are used as input of a search algorithm formalized via graphical causal models. In the second, semi-parametric, setting the underlying system is linear with non-Gaussian disturbances. In this case the structural vector autoregressive model is identified through a search procedure based on independent component analysis. Finally, we explore the possibility of causal

search in a nonparametric setting by studying the performance of conditional independence tests based on kernel density estimations.

Neale et al.: OpenMx 2.0: Extended structural equation and statistical modeling
Neale-Hunter-Pritikin-etal-2015

Michael C. Neale et al. "OpenMx 2.0: Extended structural equation and statistical modeling". In: *Psychometrika* 81.2 (Jan. 2015), pp. 535–549. DOI: 10.1007/s11336-014-9435-8.

Abstract: The new software package OpenMx 2.0 for structural equation and other statistical modeling is introduced and its features are described. OpenMx is evolving in a modular direction and now allows a mix-and-match computational approach that separates model expectations from fit functions and optimizers. Major backend architectural improvements include a move to swappable open-source optimizers such as the newly written CSOLNP. Entire new methodologies such as item factor analysis and state space modeling have been implemented. New model expectation functions including support for the expression of models in LISREL syntax and a simplified multigroup expectation function are available. Ease-of-use improvements include helper functions to standardize model parameters and compute their Jacobian-based standard errors, access to model components through standard R $ mechanisms, and improved tab completion from within the R Graphical User Interface.

Northcote et al.: Accuracy of self-reported drinking: Observational verification of 'last occasion' drink estimates of young adults **Northcote-Livingston-2011**

Jeremy Northcote and Michael Livingston. "Accuracy of self-reported drinking: Observational verification of 'last occasion' drink estimates of young adults". In: *Alcohol and Alcoholism* 46.6 (Sept. 2011), pp. 709–713. ISSN: 0735-0414. DOI: 10.1093/alcalc/agr138.

Abstract: Aims: As a formative step towards determining the accuracy of self-reported drinking levels commonly used for estimating population alcohol use, the validity of a 'last occasion' selfreporting approach is tested with corresponding field observations of participants' drinking quantity. This study is the first known attempt to validate the accuracy of self-reported alcohol consumption using data from a natural setting. Methods: A total of 81 young adults (aged 18-25 years) were purposively selected in Perth, Western Australia. Participants were asked to report the number of alcoholic drinks consumed at nightlife venues 1-2 days after being observed by peer-based researchers on 239 occasions. Complete observation data and self-report estimates were available for 129 sessions, which were fitted with multi-level models assessing the relationship between observed and reported consumption. Results: Participants accurately estimated their consumption when engaging in light to moderate drinking (eight or fewer drinks in a single session), with no significant difference between the mean reported consumption and the mean observed consumption. In contrast, participants underestimated their own consumption by increasing amounts when engaging in heavy drinking of more than eight drinks. Conclusion: It is suggested that recent recall methods in self-report surveys are potentially reasonably accurate measures of actual drinking levels for light to moderate drinkers, but that underestimating of alcohol consumption increases with heavy consumption. Some of the possible reasons for underestimation of heavy drinking are discussed, with both cognitive and socio-cultural factors considered.

O'Laughlin et al.: Cross-sectional analysis of longitudinal mediation processes

OLaughlin-Martin-Ferrer-2018

Kristine D. O'Laughlin, Monica J. Martin, and Emilio Ferrer. "Cross-sectional analysis of longitudinal mediation processes". In: *Multivariate Behavioral Research* 53.3 (Apr. 2018), pp. 375–402. ISSN: 1532-7906. DOI: 10.1080/00273171.2018.1454822.

Abstract: Statistical mediation analysis can help to identify and explain the mechanisms behind psychological processes. Examining a set of variables for mediation effects is a ubiquitous process in the social sciences literature; however, despite evidence suggesting that cross-sectional data can misrepresent the mediation of longitudinal processes, cross-sectional analyses continue to be used in this manner. Alternative longitudinal mediation models, including those rooted in a structural equation modeling framework (cross-lagged panel, latent growth curve, and latent difference score models) are currently available and may provide a better representation of mediation processes for longitudinal data. The purpose of this paper is twofold: first, we provide a comparison of cross-sectional and longitudinal mediation models; second, we advocate using models to evaluate mediation effects that capture the temporal sequence of the process under study. Two separate empirical examples are presented to illustrate differences in the conclusions drawn from cross-sectional and longitudinal mediation analyses. Findings from these examples yielded substantial differences in interpretations between the cross-sectional and longitudinal mediation models considered here. Based on these observations, researchers should use caution when attempting to use cross-sectional data in place of longitudinal data for mediation analyses.

O'Rourke et al.: Reasons for testing mediation in the absence of an intervention effect: A research imperative in prevention and intervention research

ORourke-MacKinnon-2018

Holly P. O'Rourke and David P. MacKinnon. "Reasons for testing mediation in the absence of an intervention effect: A research imperative in prevention and intervention research". In: *Journal of Studies on Alcohol and Drugs* 79.2 (Mar. 2018), pp. 171–181. DOI: 10.15288/jsad.2018.79.171.

Abstract: Objective: Mediation models are used in prevention and intervention research to assess the mechanisms by which interventions influence outcomes. However, researchers may not investigate mediators in the absence of intervention effects on the primary outcome variable. There is emerging evidence that in some situations, tests of mediated effects can be statistically significant when the total intervention effect is not statistically significant. In addition, there are important conceptual and practical reasons for investigating mediation when the intervention effect is non-significant. Method: This article discusses the conditions under which mediation may be present when an intervention effect does not have a statistically significant effect and why mediation should always be considered important. Results: Mediation may be present in the following conditions: when the total and mediated effects are equal in value, when the mediated and direct effects have

opposing signs, when mediated effects are equal across single and multiple-mediator models, and when specific mediated effects have opposing signs. Mediation should be conducted in every study because it provides the opportunity to test known and replicable mediators, to use mediators as an intervention manipulation check, and to address action and conceptual theory in intervention models. Conclusions: Mediators are central to intervention programs, and mediators should be investigated for the valuable information they provide about the success or failure of interventions.

Odgers et al.: Violence exposure is associated with adolescents' same- and next-day mental health symptoms **Odgers-Russell-2017**

Candice L. Odgers and Michael A. Russell. "Violence exposure is associated with adolescents' same- and next-day mental health symptoms". In: *Journal of Child Psychology and Psychiatry* 58.12 (July 2017), pp. 1310–1318. ISSN: 1469-7610. DOI: 10.1111/jcpp.12763.

Abstract: Background: Young people exposed to violence are at increased risk for mental health and behavioral problems. However, very little is known about the immediate, or same-day, associations between violence exposure and adolescents' mental health symptoms or whether daily symptom or behavioral reactivity marks future problems. Methods: Young adolescents were assessed three times a day for 30 consecutive days using mobile-phone-based Ecological Momentary Assessment (EMA) (N = 151 adolescents). Over 12,500 assessments and 4,329 person days were obtained via the EMA. Adolescents were recruited from low-income neighborhoods based on parent-reported risk for externalizing symptoms. Mental health symptoms were assessed via parent and child report at baseline, multiple times per day via EMA assessments of the adolescents, and again 18 months later when 93% of the adolescents were reinterviewed. Results: Results from multilevel models illustrated that young adolescents were more likely to experience symptoms of anger (OR = 1.74, CI: 1.31-2.30), depression (OR = 1.66, CI: 1.26-2.19), and conduct problems (OR = 2.63, CI: 1.71-4.04) on days that they were exposed versus not exposed to violence. Increases in depressive symptoms were also observed on days following violence exposure (OR = 1.46, CI: 1.09-1.97). Adolescents with the highest levels of violence exposure across the 30-day EMA were less behaviorally reactive to

violence exposures in daily life, and heightened behavioral reactivity predicted increased risk for substance use across early adolescence. Conclusions: Findings support the need to focus on both the immediate and long-term associations between violence exposure and adolescents' mental health and behavior. Results also suggest that heightened behavioral reactivity during early adolescence may signal emerging substance use problems.

Oravecz et al.: A hierarchical latent stochastic differential equation model for affective dynamics **Oravecz-Tuerlinckx-Vandekerckhove-2011**

Zita Oravecz, Francis Tuerlinckx, and Joachim Vandekerckhove. "A hierarchical latent stochastic differential equation model for affective dynamics". In: *Psychological Methods* 16.4 (2011), pp. 468–490. DOI: 10.1037/a0024375.

Abstract: In this article a continuous-time stochastic model (the Ornstein-Uhlenbeck process) is presented to model the perpetually altering states of the core affect, which is a 2-dimensional concept underlying all our affective experiences. The process model that we propose can account for the temporal changes in core affect on the latent level. The key parameters of the model are the average position (also called home base), the variances and covariances of the process, and the regulatory mechanisms that keep the process in the vicinity of the average position. To account for individual differences, the model is extended hierarchically. A particularly novel contribution is that in principle all parameters of the stochastic process (not only the mean but also its variance and the regulatory parameters) are allowed to differ between individuals. In this way, the aim is to understand the affective dynamics of single individuals and at the same time investigate how these individuals differ from one another. The final model is a continuous-time state-space model for repeated measurement data taken at possibly irregular time points. Both time-invariant and time-varying covariates can be included to investigate sources of individual differences. As an illustration, the model is applied to a diary study measuring core affect repeatedly for several individuals (thereby generating intensive longitudinal data).

Ou et al.: What's for dynr: A package for linear and nonlinear dynamic modeling in R **Ou-Hunter-Chow-2019**

Lu Ou, Michael D. Hunter, and Sy-Miin Chow. "What's for dynr: A package for linear and nonlinear dynamic modeling in R". In: *The R Journal* 11.1 (2019), p. 91. DOI: 10.32614/rj-2019-012.

Abstract: Intensive longitudinal data in the behavioral sciences are often noisy, multivariate in nature, and may involve multiple units undergoing regime switches by showing discontinuities interspersed with continuous dynamics. Despite increasing interest in using linear and nonlinear differential/difference equation models with regime switches, there has been a scarcity of software packages that are fast and freely accessible. We have created an R package called dynr that can handle a broad class of linear and nonlinear discreteand continuous-time models, with regime-switching properties and linear Gaussian measurement functions, in C, while maintaining simple and easy-to learn model specification functions in R. We present the mathematical and computational bases used by the dynr R package, and present two illustrative examples to demonstrate the unique features of dynr.

Pastor et al.: On the multilevel nature of meta-analysis: A tutorial, comparison of software programs, and discussion of analytic choices **Pastor-Lazowski-2017**

Dena A. Pastor and Rory A. Lazowski. "On the multilevel nature of meta-analysis: A tutorial, comparison of software programs, and discussion of analytic choices". In: *Multivariate Behavioral Research* 53.1 (Sept. 2017), pp. 74–89. ISSN: 1532-7906. DOI: 10.1080/00273171.2017.1365684.

Abstract: The term "multilevel meta-analysis" is encountered not only in applied research studies, but in multilevel resources comparing traditional meta-analysis to multilevel meta-analysis. In this tutorial, we argue that the term "multilevel meta-analysis" is redundant since all meta-analysis can be formulated as a special kind of multilevel model. To clarify the multilevel nature of meta-analysis the four standard meta-analytic models are presented using multilevel equations and fit to an example data set using four software programs: two specific to meta-analysis (metafor in R

and SPSS macros) and two specific to multilevel modeling (PROC MIXED in SAS and HLM). The same parameter estimates are obtained across programs underscoring that all meta-analyses are multilevel in nature. Despite the equivalent results, not all software programs are alike and differences are noted in the output provided and estimators available. This tutorial also recasts distinctions made in the literature between traditional and multilevel meta-analysis as differences between meta-analytic choices, not between meta-analytic models, and provides guidance to inform choices in estimators, significance tests, moderator analyses, and modeling sequence. The extent to which the software programs allow flexibility with respect to these decisions is noted, with metafor emerging as the most favorable program reviewed.

Patrick et al.: High-intensity drinking by underage young adults in the United States: Underage high-intensity drinking **Patrick-TerryMcElrath-2016**

Megan E. Patrick and Yvonne M. Terry-McElrath. "High-intensity drinking by underage young adults in the United States: Underage high-intensity drinking". In: *Addiction* 112.1 (Sept. 2016), pp. 82–93. ISSN: 0965-2140. DOI: 10.1111/add.13556.

Abstract: Aims. To estimate (1) the prevalence of underage binge drinking, high-intensity drinking and intoxication among young adults aged 19/20 years; (2) change in these behaviors across the transition out of high school and across historical time; and (3) associations between these behaviors and key covariates, including college status. Design, Setting, Participants. Longitudinal data from the US nationally representative Monitoring the Future study included 1657 respondents first surveyed as 12th graders (modal age 18 years) in 2005-13 and again at modal age 19/20 years in 2006-14. Measurements. Self-reported measures of alcohol use, demographics, college attendance and living situation. Findings. Binge drinking (5+ drinks on one occasion) was reported by 24.2% [95% confidence interval (CI) = 22.0, 26.5] of young adults aged 19/20; 10.3% (CI = 8.7, 11.9) reported high-intensity drinking of 10+ drinks; 4.2% (CI = 3.1, 5.2) reported 15+ drinks. Usual moderate/high intoxication when drinking was reported by 33.1% (CI = 30.6, 35.6); 29.6% (CI = 27.2, 32.0) reported usual sustained intoxication of 3+ hours. Significant variability (P ¡ 0.001) in

these behaviors from ages 18 to 19/20 was observed. Significant decreases (P ¡ 0.05) across historical time in 5+ and 10+ drinking were found. Four-year college students not residing with parents had significantly higher odds of moderate/high intoxication, binge drinking and high-intensity drinking compared with other groups (P ¡ 0.001). Conclusions. Young adult underage binge drinking (5+ drinks on one occasion), high-intensity drinking (10+ and 15+ drinks) and intoxication are relatively common in the United States, and show meaningful variability across the transition out of high school. Four-year college students and those who do not live with their parents are more likely to engage in high-intensity drinking than their peers.

Pe et al.: Updating in working memory predicts greater emotion reactivity to and facilitated recovery from negative emotion-eliciting stimuli

Pe-Koval-Houben-etal-2015

Madeline L. Pe et al. "Updating in working memory predicts greater emotion reactivity to and facilitated recovery from negative emotion-eliciting stimuli". In: *Frontiers in Psychology* 6 (Apr. 2015). ISSN: 1664-1078. DOI: 10.3389/fpsyg.2015.00372.

Abstract: That emotions change in response to emotion-eliciting events is a natural part of human life. However, it is equally important for emotions to return to baseline once the emotion-eliciting events have passed. This suggests that the ability to emotionally react to and recover from emotion-eliciting events is critical for healthy psychological functioning. But why do individuals differ in their emotion reactivity and recovery? The present work postulates that the ability to update emotional information in working memory (WM) may explain individual differences in emotion reactivity and recovery. Two studies are presented, which examined whether updating ability was related to emotion reactivity and recovery. In Study 1, we assessed participants' self-reported affect as they viewed negative and positive films. Our results revealed that better updating ability was related to greater emotion reactivity and facilitated (i.e., quicker) recovery from watching negative films. In Study 2, participants recalled a recent angering event, and were then instructed to either ruminate about or reappraise the event. Results revealed that updating ability was again related to greater

emotion reactivity and facilitated (i.e., successful) emotion recovery in response to the angering event, and that this was unrelated to the emotion regulation strategy used. These findings identify the ability to update emotional information in WM as a possible mechanism in emotion responding.

Pesigan et al.: Brief report: Identity processes in Filipino late adolescents and young adults: Parental influences and mental health outcomes

Pesigan-Luyckx-Alampay-2014

Ivan Jacob Agaloos Pesigan, Koen Luyckx, and Liane Peña Alampay. "Brief report: Identity processes in Filipino late adolescents and young adults: Parental influences and mental health outcomes". In: *Journal of Adolescence* 37.5 (May 2014), pp. 599–604. ISSN: 1095-9254. DOI: 10.1016/j.adolescence.2014.04.012.

Abstract: This study focused on a process-oriented approach to identity formation using a sample of Filipino late adolescents and young adults (17-30 years; $N = 779$). Indirect relations between parenting and mental health via identity formation processes were examined. Two parenting dimensions (psychological control and support), two types of mental health outcomes (depression and psychological well-being), and five identity dimensions (commitment making (CM), identification with commitment (IC), exploration in breadth (EB), exploration in depth (ED), and ruminative exploration (RE)) were assessed. Recursive path analysis showed indirect relations between parenting and mental health via EB, ED, RE, and IC. Model differences between late adolescents (17-21 year olds) and young adults (22-30 year olds) were examined using multigroup path analysis. Results showed that the direct effect of psychological control on RE, and its indirect effect on depression through RE differed between the age groups. Implications and suggestions for future research are provided.

Piasecki: Assessment of alcohol use in the natural environment **Piasecki-2019**

Thomas M. Piasecki. "Assessment of alcohol use in the natural environment". In: *Alcoholism:*

Clinical and Experimental Research 43.4 (Mar. 2019), pp. 564–577. ISSN: 1530-0277. DOI: 10.1111/acer.13975.

Abstract: The current article critically reviews 3 methodological options for assessing drinking episodes in the natural environment. Ecological momentary assessment (EMA) typically involves using mobile devices to collect self-report data from participants in daily life. This technique is now widely used in alcohol research, but investigators have implemented diverse assessment strategies. This article focuses on "high-resolution" EMA protocols that oversample experiences and behaviors within individual drinking episodes. A number of approaches have been used to accomplish this, including using signaled follow-ups tied to drinking initiation, asking participants to log entries before and after individual drinks or drinking episodes, and delivering frequent signaled assessments during periods of the day when alcohol use is most common. Transdermal alcohol sensors (TAS) are devices that are worn continuously and are capable of detecting alcohol eliminated through the skin. These methods are appealing because they do not rely upon drinkers' self-report. Studies using TAS have been appearing with greater frequency over the past several years. New methods are making the use of TAS more tractable by permitting back-translation of transdermal alcohol concentration data to more familiar estimates of blood alcohol concentration or breath alcohol concentration. However, the current generation of devices can have problems with missing data and tend to be relatively insensitive to low-level drinking. An emerging area of research investigates the possibility of using mobile device data and machine learning to passively detect the user's drinking, with promising early findings. EMA, TAS, and sensor-based approaches are all valid, and tend to produce convergent information when used in conjunction with one another. Each has a unique profile of advantages, disadvantages, and threats to validity. Therefore, the nature of the underlying research question must dictate the method(s) investigators select.

Preacher et al.: Effect size measures for mediation models: Quantitative strategies for communicating indirect effects **Preacher-Kelley-2011**

Kristopher J. Preacher and Ken Kelley. "Effect size measures for mediation models: Quantitative strategies for communicating indirect effects". In: *Psychological Methods* 16.2 (2011), pp. 93–115. ISSN: 1082-989X. DOI: 10.1037/a0022658.

Abstract: The statistical analysis of mediation effects has become an indispensable tool for helping scientists investigate processes thought to be causal. Yet, in spite of many recent advances in the estimation and testing of mediation effects, little attention has been given to methods for communicating effect size and the practical importance of those effect sizes. Our goals in this article are to (a) outline some general desiderata for effect size measures, (b) describe current methods of expressing effect size and practical importance for mediation, (c) use the desiderata to evaluate these methods, and (d) develop new methods to communicate effect size in the context of mediation analysis. The first new effect size index we describe is a residual-based index that quantifies the amount of variance explained in both the mediator and the outcome. The second new effect size index quantifies the indirect effect as the proportion of the maximum possible indirect effect that could have been obtained, given the scales of the variables involved. We supplement our discussion by offering easy-to-use R tools for the numerical and visual communication of effect size for mediation effects.

Preacher et al.: Advantages of Monte Carlo confidence intervals for indirect effects Preacher-Selig-2012

Kristopher J. Preacher and James P. Selig. "Advantages of Monte Carlo confidence intervals for indirect effects". In: *Communication Methods and Measures* 6.2 (Apr. 2012), pp. 77–98. DOI: 10.1080/19312458.2012.679848.

Abstract: Monte Carlo simulation is a useful but underutilized method of constructing confidence intervals for indirect effects in mediation analysis. The Monte Carlo confidence interval method has several distinct advantages over rival methods. Its performance is comparable to other widely accepted methods of interval construction, it can be used when only summary data are available, it can be used in situations where rival methods (e.g., bootstrapping and distribution of the product

methods) are difficult or impossible, and it is not as computer-intensive as some other methods. In this study we discuss Monte Carlo confidence intervals for indirect effects, report the results of a simulation study comparing their performance to that of competing methods, demonstrate the method in applied examples, and discuss several software options for implementation in applied settings.

Prince et al.: Trajectories of college alcohol involvement and their associations with later alcohol use disorder symptoms **Prince-Read-Colder-2019**

Mark A. Prince, Jennifer P. Read, and Craig R. Colder. "Trajectories of college alcohol involvement and their associations with later alcohol use disorder symptoms". In: *Prevention Science* 20.5 (Jan. 2019), pp. 741–752. ISSN: 1573-6695. DOI: 10.1007/s11121-018-0974-6.

Abstract: Little is known about what differentiates individuals whose drinking patterns escalate into problematic use following the transition out of college compared to those who learn to drink in a way that is consistent with independent adult roles. Patterns of alcohol use and problems during college may pre-sage progression toward problem drinking in adulthood. The present study sought to examine such patterns in an effort to delineate those at greatest risk. Research has not yet elucidated whether, when, and how these groups diverge. Our results indicate that students who report AUD symptoms one year following graduation reported greater alcohol involvement from the first semester and escalated their involvement with alcohol at a more rapid pace. We observed marked and measurable differences in drinking patterns between those who go on to exhibit AUD symptoms following college and those who do not. A close inspection of these differences reveals that relatively small absolute differences in alcohol consumption add up to large differences in alcohol-related consequences. Thus, markers of longer-term risk are present early in college, and greater escalation of drinking across college is an indicator that intervention is needed. Brief Motivational Interventions could help students to anticipate some of the challenges ahead as they transition from the college environment, as well as the potential deleterious effects of immoderate alcohol use on

making a successful transition into adult roles. In addition to the beginning of college, our findings also point to critical periods during which screening and brief intervention may be optimally timed.

Pritikin et al.: Many-level multilevel structural equation modeling: An efficient evaluation strategy **Pritikin-Hunter-vonOertzen-etal-2017**

Joshua N. Pritikin et al. "Many-level multilevel structural equation modeling: An efficient evaluation strategy". In: *Structural Equation Modeling: A Multidisciplinary Journal* 24.5 (Mar. 2017), pp. 684–698. ISSN: 1532-8007. DOI: 10.1080/10705511.2017.1293542.

Abstract: Structural equation models are increasingly used for clustered or multilevel data in cases where mixed regression is too inflexible. However, when there are many levels of nesting, these models can become difficult to estimate. We introduce a novel evaluation strategy, Rampart, that applies an orthogonal rotation to the parts of a model that conform to commonly met requirements. This rotation dramatically simplifies fit evaluation in a way that becomes more potent as the size of the data set increases. We validate and evaluate the implementation using a 3-level latent regression simulation study. Then we analyze data from a statewide child behavioral health measure administered by the Oklahoma Department of Human Services. We demonstrate the efficiency of Rampart compared to other similar software using a latent factor model with a 5-level decomposition of latent variance. Rampart is implemented in OpenMx, a free and open source software package.

Rast et al.: A mixed-effects location scale model for dyadic interactions **Rast-Ferrer-2018**

Philippe Rast and Emilio Ferrer. "A mixed-effects location scale model for dyadic interactions". In: *Multivariate Behavioral Research* 53.5 (Sept. 2018), pp. 756–775. ISSN: 1532-7906. DOI: 10.1080/00273171.2018.1477577.

Abstract: We present a mixed-effects location scale model (MELSM) for examining the daily dynamics of affect in dyads. The MELSM includes person and time-varying variables to predict the

location, or individual means, and the scale, or within-person variances. It also incorporates a submodel to account for between-person variances. The dyadic specification can accommodate individual and partner effects in both the location and the scale components, and allows random effects for all location and scale parameters. All covariances among the random effects, within and across the location and the scale are also estimated. These covariances offer new insights into the interplay of individual mean structures, intra-individual variability, and the influence of partner effects on such factors. To illustrate the model, we use data from 274 couples who provided daily ratings on their positive and negative emotions toward their relationship–up to 90 consecutive days. The model is fit using Hamiltonian Monte Carlo methods, and includes subsets of predictors in order to demonstrate the flexibility of this approach. We conclude with a discussion on the usefulness and the limitations of the MELSM for dyadic research.

Reichardt: Commentary: Are three waves of data sufficient for assessing mediation? Reichardt-2011

Charles S. Reichardt. "Commentary: Are three waves of data sufficient for assessing mediation?" In: *Multivariate Behavioral Research* 46.5 (Sept. 2011), pp. 842–851. ISSN: 1532-7906. DOI: 10.1080/00273171.2011.606740.

Abstract: Maxwell, Cole, and Mitchell (2011) demonstrated that simple structural equation models, when used with cross-sectional data, generally produce biased estimates of meditated effects. I extend those results by showing how simple structural equation models can produce biased estimates of meditated effects when used even with longitudinal data. Even with longitudinal data, simple autoregressive structural equation models can imply the existence of indirect effects when only direct effects exist and the existence of direct effects when only indirect effects exist.

John D. Roache et al. "Processing transdermal alcohol concentration (TAC) data to detect low-level drinking". In: *Alcohol* 81 (Dec. 2019), pp. 101–110. ISSN: 0741-8329. DOI: 10.1016/j.alcohol.2018.08.014.

Abstract: Background: Several studies have objectively quantified drinking through the use of Alcohol Monitoring System's (AMS) transdermal alcohol concentration (TAC) device known as SCRAM CAM. Criteria that AMS uses to detect drinking are known to be conservative and only reliably detect heavy drinking equivalent to 5 or more standard drinks. Our group has developed Research Rules used to process TAC data in a manner that will detect low-level and moderate drinking even though it is below the AMS criteria for detection. Methods: Sixteen male and 14 female paid research volunteers wore TAC monitors for 28 days in their natural environments and responded daily to text message prompts to self-report the previous day's drinking. Current analyses describe the Research Rules that we developed and how use of those rules impacts the detection of self-reported drinking treated as the standard in sensitivity/specificity analysis. Results: We observed 606 occurrences of positive TAC events over a total of 867 days and processed the TAC data to retain 345 as possible drinking events, even though AMS criteria confirmed drinking for only 163 of these events. The kinds of TAC events removed or retained by our rules are illustrated as cases of low and moderate drinking days that were detected by our rules but not by the conservative AMS criteria. AMS-confirmed TAC events have a high specificity (99.8%) to detect primarily heavy drinking, but have a poor sensitivity to detect lower-level drinking and a poor specificity as an indicator of alcohol abstinence. In contrast, our Research Rules detected 100% of TAC events detected by AMS but also detected 31% of the lower-level drinking events not detected by AMS, with 91% specificity. Conclusions: Reliance upon the AMS criteria for alcohol detection affords a high specificity for detection of heavy drinking but is a poor indicator of abstinence rates. In contrast, use of our Research Rules provides more sensitive means to quantify either any drinking or low-moderate levels of drinking while still maintaining good specificity.

Rosseel: lavaan: An R package for structural equation modeling **Rosseel-2012**

Yves Rosseel. "lavaan: An R package for structural equation modeling". In: *Journal of Statistical Software* 48.2 (2012). DOI: 10.18637/jss.v048.i02.

Abstract: Structural equation modeling (SEM) is a vast field and widely used by many applied researchers in the social and behavioral sciences. Over the years, many software packages for structural equation modeling have been developed, both free and commercial. However, perhaps the best state-of-the-art software packages in this field are still closed-source and/or commercial. The R package lavaan has been developed to provide applied researchers, teachers, and statisticians, a free, fully open-source, but commercial-quality package for latent variable modeling. This paper explains the aims behind the development of the package, gives an overview of its most important features, and provides some examples to illustrate how lavaan works in practice.

Rush et al.: Modeling long-term changes in daily within-person associations: An application of multilevel SEM **Rush-Rast-Almeida-etal-2019**

Jonathan Rush et al. "Modeling long-term changes in daily within-person associations: An application of multilevel SEM". In: *Psychology and Aging* 34.2 (Mar. 2019), pp. 163–176. ISSN: 0882-7974. DOI: 10.1037/pag0000331.

Abstract: Short-term within-person associations are considered to reflect unique dynamic characteristics of an individual and are frequently used to predict distal outcomes. These effects are typically examined with a 2-step statistical process. The present research demonstrates how long-term changes in short-term within-person associations can be modeled simultaneously within a multilevel structural equation modeling framework. We demonstrate the utility of this model using measurement burst data from the National Study of Daily Experiences (NSDE) embedded within the Midlife in the United States (MIDUS) longitudinal study. Two measurement bursts were separated by 9 years, with each containing daily measures of stress and affect across 8 consecutive days.

Measures of life satisfaction and psychological well-being were also assessed across the 9-year period. Three-level structural equation models were fit to simultaneously model short-term within-person associations between stress and negative affect and long-term changes in these associations over the 9-year period. Individual differences in long-term changes of the short-term dynamics between stress and affect predicted well-being levels. We highlight how characterizing individuals based on the strength of their within-person associations across multiple time scales can be informative in predicting distal outcomes.

Russell et al.: Adolescents' subjective social status predicts day-to-day mental health and future substance use — Russell-Odgers-2019

Michael A. Russell and Candice L. Odgers. "Adolescents' subjective social status predicts day-to-day mental health and future substance use". In: *Journal of Research on Adolescence* 30.S2 (Apr. 2019), pp. 532–544. ISSN: 1532-7795. DOI: 10.1111/jora.12496.

Abstract: Adolescents' subjective social status (SSS) is associated with mental and behavioral health outcomes, independent of socioeconomic status (SES). Many previous findings, however, come from cross-sectional studies. We report results from a longitudinal study with 151 adolescents identified as at risk for early substance use and behavioral problems sampled from low-SES neighborhoods. We examined whether adolescent's SSS predicted mental health (depression, anxiety, and inattention/impulsivity) measured over 30 days via ecological momentary assessment and risk for substance use at an 18-month follow-up. Results showed that with each perceived step "up" the SSS ladder, adolescents experienced fewer mental health symptoms in daily life and lower future substance use risk after adjusting for objective SES and previous psychopathology. Implications of these findings are discussed.

Russell et al.: Witnessing substance use increases same-day antisocial behavior among at-risk adolescents: Gene-environment interaction in a 30-day ecological momentary assessment study **Russell-Wang-Odgers-2015**

Michael A. Russell, Lin Wang, and Candice L. Odgers. "Witnessing substance use increases same-day antisocial behavior among at-risk adolescents: Gene-environment interaction in a 30-day ecological momentary assessment study". In: *Development and Psychopathology* 28.4pt2 (Dec. 2015), pp. 1441–1456. ISSN: 1469-2198. DOI: 10.1017/s0954579415001182.

Abstract: Many young adolescents are embedded in neighborhoods, schools, and homes where alcohol and drugs are frequently used. However, little is known about (a) how witnessing others' substance use affects adolescents in their daily lives and (b) which adolescents will be most affected. The current study used ecological momentary assessment with 151 young adolescents (ages 11-15) to examine the daily association between witnessing substance use and antisocial behavior across 38 consecutive days. Results from multilevel logistic regression models indicated that adolescents were more likely to engage in antisocial behavior on days when they witnessed others using substances, an association that held when substance use was witnessed inside the home as well as outside the home (e.g., at school or in their neighborhoods). A significant Gene × Environment interaction suggested that the same-day association between witnessing substance use and antisocial behavior was significantly stronger among adolescents with, versus without, the dopamine receptor D4 seven repeat (DRD4-7R) allele. The implications of the findings for theory and research related to adolescent antisocial behavior are discussed.

Sacks et al.: 2010 national and state costs of excessive alcohol consumption **Sacks-Gonzales-Bouchery-etal-2015**

Jeffrey J. Sacks et al. "2010 national and state costs of excessive alcohol consumption". In: *American Journal of Preventive Medicine* 49.5 (Nov. 2015), e73–e79. ISSN: 0749-3797. DOI: 10.1016/j.amepre.2015.05.031.

Abstract: Introduction: Excessive alcohol use cost the U.S. 223.5*billionin*2006.*GiveneconomicshiftsintheU.S.since*2006,
currentestimatesareneededtohelpinformtheplanningofpreventionstrategies.Methods : *FromMarch*2012*toMarch*201
attributabledeaths)*andprice*(*e.g.*, *inflationrateincostofmedicalcare*).*Thetotalcost*, *costtogovernment*, *andcostsforbing*
*ExcessivedrinkingcosttheU.S.*249.0 billion in 2010, or about 2.05*perdrink.Governmentpaidfor*100.7 billion (40.4%) of these costs. Binge drinking accounted for 191.1*billion*(76.7%)*ofcosts*; *underagedrinking*24.3 billion (9.7%) of costs; and drinking while pregnant 5.5*billion*(2.2%)*ofcosts.Themediancostperstatewas*3.5 billion. Binge drinking was responsible for ¿70% of these costs in all states, and ¿40% of the binge drinking-related costs were paid by government. Conclusions: Excessive drinking cost the nation almost 250*billionin*2010.*Twoofevery*5 of the total cost was paid by government, and three quarters of the costs were due to binge drinking. Several evidence-based strategies can help reduce excessive drinking and related costs, including increasing alcohol excise taxes, limiting alcohol outlet density, and commercial host liability.

Schermerhorn et al.: Developmental family processes and interparental conflict: Patterns of microlevel influences. Schermerhorn-Chow-Cummings-2010

Alice C. Schermerhorn, Sy-Miin Chow, and E. Mark Cummings. "Developmental family processes and interparental conflict: Patterns of microlevel influences." In: *Developmental Psychology* 46.4 (2010), pp. 869–885. ISSN: 0012-1649. DOI: 10.1037/a0019662.

Abstract: Although there are frequent calls for the study of effects of children on families and mutual influence processes within families, little empirical progress has been made. We address these questions at the level of microprocesses during marital conflict, including children's influence on marital conflict and parents' influence on each other. Participants were 111 cohabiting couples with a child (55 male, 56 female) age 8-16 years. Data were drawn from parents' diary reports of interparental conflict over 15 days and were analyzed with dynamic systems modeling tools. Child emotions and behavior during conflicts were associated with interparental positivity, negativity, and resolution at the end of the same conflicts. For example, children's agentic behavior was associated with more marital conflict resolution, whereas child negativity was linked with more marital negativity. Re-

garding parents' influence on each other, among the findings, husbands' and wives' influence on themselves from one conflict to the next was indicated, and total number of conflicts predicted greater influence of wives' positivity on husbands' positivity. Contributions of these findings to the understanding of developmental family processes are discussed, including implications for advanced understanding of interrelations between child and adult functioning and development.

Schouten et al.: Generating missing values for simulation purposes: A multivariate amputation procedure **Schouten-Lugtig-Vink-2018**

Rianne Margaretha Schouten, Peter Lugtig, and Gerko Vink. "Generating missing values for simulation purposes: A multivariate amputation procedure". In: *Journal of Statistical Computation and Simulation* 88.15 (July 2018), pp. 2909–2930. DOI: 10.1080/00949655.2018.1491577.

Abstract: Missing data form a ubiquitous problem in scientific research, especially since most statistical analyses require complete data. To evaluate the performance of methods dealing with missing data, researchers perform simulation studies. An important aspect of these studies is the generation of missing values in a simulated, complete data set: the amputation procedure. We investigated the methodological validity and statistical nature of both the current amputation practice and a newly developed and implemented multivariate amputation procedure. We found that the current way of practice may not be appropriate for the generation of intuitive and reliable missing data problems. The multivariate amputation procedure, on the other hand, generates reliable amputations and allows for a proper regulation of missing data problems. The procedure has additional features to generate any missing data scenario precisely as intended. Hence, the multivariate amputation procedure is an efficient method to accurately evaluate missing data methodology.

Schultzberg et al.: Number of subjects and time points needed for multilevel time-series analysis: A simulation study of dynamic structural equation modeling

Schultzberg-Muthen-2017

Mårten Schultzberg and Bengt Muthén. "Number of subjects and time points needed for multilevel time-series analysis: A simulation study of dynamic structural equation modeling". In: *Structural Equation Modeling: A Multidisciplinary Journal* 25.4 (Dec. 2017), pp. 495–515. ISSN: 1532-8007. DOI: 10.1080/10705511.2017.1392862.

Abstract: Dynamic structural equation modeling (DSEM) is a novel, intensive longitudinal data (ILD) analysis framework. DSEM models intraindividual changes over time on Level 1 and allows the parameters of these processes to vary across individuals on Level 2 using random effects. DSEM merges time series, structural equation, multilevel, and time-varying effects models. Despite the well-known properties of these analysis areas by themselves, it is unclear how their sample size requirements and recommendations transfer to the DSEM framework. This article presents the results of a simulation study that examines the estimation quality of univariate 2-level autoregressive models of order 1, AR(1), using Bayesian analysis in Mplus Version 8. Three features are varied in the simulations: complexity of the model, number of subjects, and number of time points per subject. Samples with many subjects and few time points are shown to perform substantially better than samples with few subjects and many time points.

Schuurman et al.: How to compare cross-lagged associations in a multilevel autoregressive model **Schuurman-Ferrer-deBoerSonnenschein-etal-2016**

Noémi K. Schuurman, Emilio Ferrer, et al. "How to compare cross-lagged associations in a multilevel autoregressive model". In: *Psychological Methods* 21.2 (2016), pp. 206–221. ISSN: 1082-989X. DOI: 10.1037/met0000062.

Abstract: By modeling variables over time it is possible to investigate the Granger-causal cross-lagged associations between variables. By comparing the standardized cross-lagged coefficients, the

relative strength of these associations can be evaluated in order to determine important driving forces in the dynamic system. The aim of this study was twofold: first, to illustrate the added value of a multilevel multivariate autoregressive modeling approach for investigating these associations over more traditional techniques; and second, to discuss how the coefficients of the multilevel autoregressive model should be standardized for comparing the strength of the cross-lagged associations. The hierarchical structure of multilevel multivariate autoregressive models complicates standardization, because subject-based statistics or group-based statistics can be used to standardize the coefficients, and each method may result in different conclusions. We argue that in order to make a meaningful comparison of the strength of the cross-lagged associations, the coefficients should be standardized within persons. We further illustrate the bivariate multilevel autoregressive model and the standardization of the coefficients, and we show that disregarding individual differences in dynamics can prove misleading, by means of an empirical example on experienced competence and exhaustion in persons diagnosed with burnout.

Schuurman et al.: A comparison of inverse-Wishart prior specifications for covariance matrices in multilevel autoregressive models Schuurman-Grasman-Hamaker-2016

Noémi K. Schuurman, Raoul P. P. P. Grasman, and Ellen L. Hamaker. "A comparison of inverse-Wishart prior specifications for covariance matrices in multilevel autoregressive models". In: *Multivariate Behavioral Research* 51.2-3 (Mar. 2016), pp. 185–206. ISSN: 1532-7906. DOI: 10.1080/00273171.2015.1065398.

Abstract: Multilevel autoregressive models are especially suited for modeling between-person differences in within-person processes. Fitting these models with Bayesian techniques requires the specification of prior distributions for all parameters. Often it is desirable to specify prior distributions that have negligible effects on the resulting parameter estimates. However, the conjugate prior distribution for covariance matrices—the Inverse-Wishart distribution—tends to be informative when variances are close to zero. This is problematic for multilevel autoregressive models, because autoregressive parameters are usually small for each individual, so that the variance of these parameters will be small. We performed a simulation study to compare the performance of three Inverse-Wishart prior specifications suggested in the literature, when one or more variances for the random effects in the multilevel autoregressive model are small. Our results show that the prior specification that uses plug-in ML estimates of the variances performs best. We advise to always include a sensitivity analysis for the prior specification for covariance matrices of random parameters, especially in autoregressive models, and to include a data-based prior specification in this analysis. We illustrate such an analysis by means of an empirical application on repeated measures data on worrying and positive affect.

Schuurman et al.: Measurement error and person-specific reliability in multilevel autoregressive modeling **Schuurman-Hamaker-2019**

Noémi K. Schuurman and Ellen L. Hamaker. "Measurement error and person-specific reliability in multilevel autoregressive modeling". In: *Psychological Methods* 24.1 (Feb. 2019), pp. 70–91. ISSN: 1082-989X. DOI: 10.1037/met0000188.

Abstract: An increasing number of researchers in psychology are collecting intensive longitudinal data in order to study psychological processes on an intraindividual level. An increasingly popular way to analyze these data is autoregressive time series modeling; either by modeling the repeated measures for a single individual using classic n = 1 autoregressive models, or by using multilevel extensions of these models, with the dynamics for each individual modeled at Level 1 and interindividual differences in these dynamics modeled at Level 2. However, while it is widely accepted in psychology that psychological measurements usually contain a certain amount of measurement error, the issue of measurement error is largely neglected in applied psychological (autoregressive) time series modeling: The regular autoregressive model incorporates innovations, or "dynamic errors," but not measurement error. In this article we discuss the concepts of reliability and measurement error in the context of dynamic (VAR(1)) models, and the consequences of disregarding measurement error variance in the data. For this purpose, we present a preliminary model that accounts for measurement error for constructs that are measured with a single indicator. We further discuss how this model could be used to investigate the between-person reliability of the measurements, as well as the (person-specific) within-person reliabilities and any individual differences in these reliabilities. We illustrate the consequences of assuming perfect reliability, the preliminary model, and reliabilities, using an empirical application in which we relate women's general positive affect to their positive affect concerning their romantic relationship.

Schuurman et al.: Incorporating measurement error in n = 1 psychological autoregressive modeling **Schuurman-Houtveen-Hamaker-2015**

Noémi K. Schuurman, Jan H. Houtveen, and Ellen L. Hamaker. "Incorporating measurement error in n = 1 psychological autoregressive modeling". In: *Frontiers in Psychology* 6 (July 2015). ISSN: 1664-1078. DOI: 10.3389/fpsyg.2015.01038.

Abstract: Measurement error is omnipresent in psychological data. However, the vast majority of applications of autoregressive time series analyses in psychology do not take measurement error into account. Disregarding measurement error when it is present in the data results in a bias of the autoregressive parameters. We discuss two models that take measurement error into account: An autoregressive model with a white noise term (AR+WN), and an autoregressive moving average (ARMA) model. In a simulation study we compare the parameter recovery performance of these models, and compare this performance for both a Bayesian and frequentist approach. We find that overall, the AR+WN model performs better. Furthermore, we find that for realistic (i.e., small) sample sizes, psychological research would benefit from a Bayesian approach in fitting these models. Finally, we illustrate the effect of disregarding measurement error in an AR(1) model by means of an empirical application on mood data in women. We find that, depending on the person, approximately 30-50% of the total variance was due to measurement error, and that disregarding this measurement error results in a substantial underestimation of the autoregressive parameters.

Shrout: Commentary: Mediation analysis, causal process, and cross-sectional data
Shrout-2011

Patrick E. Shrout. “Commentary: Mediation analysis, causal process, and cross-sectional data”. In: *Multivariate Behavioral Research* 46.5 (Sept. 2011), pp. 852–860. DOI: 10.1080/00273171.2011.606718.

Abstract: Maxwell, Cole, and Mitchell (2011) extended the work of Maxwell and Cole (2007), which raised important questions about whether mediation analyses based on cross-sectional data can shed light on longitudinal mediation process. The latest article considers longitudinal processes that can only be partially explained by an intervening variable, and Maxwell et al. showed that the same general conclusions are obtained, namely that analyses of cross-sectional data will not reveal the longitudinal mediation process. While applauding the advances of the target article, this comment encourages the detailed exploration of alternate causal models in psychology beyond the autoregressive model considered by Maxwell et al. When inferences based on cross-sectional analyses are compared to alternate models, different patterns of bias are likely to be observed. I illustrate how different models of the causal process can be derived using examples from research on psychopathology.

Silvia et al.: Planned missing-data designs in experience-sampling research: Monte Carlo simulations of efficient designs for assessing within-person constructs
Silvia-Kwapil-Walsh-etal-2014

Paul J. Silvia et al. “Planned missing-data designs in experience-sampling research: Monte Carlo simulations of efficient designs for assessing within-person constructs”. In: *Behavior Research Methods* 46.1 (2014), pp. 41–54. ISSN: 1554-3528. DOI: 10.3758/s13428-013-0353-y.

Abstract: Experience-sampling research involves trade-offs between the number of questions asked per signal, the number of signals per day, and the number of days. By combining planned missing-

data designs and multilevel latent variable modeling, we show how to reduce the items per signal without reducing the number of items. After illustrating different designs using real data, we present two Monte Carlo studies that explored the performance of planned missing-data designs across different within-person and between-person sample sizes and across different patterns of response rates. The missing-data designs yielded unbiased parameter estimates but slightly higher standard errors. With realistic sample sizes, even designs with extensive missingness performed well, so these methods are promising additions to an experience-sampler's toolbox.

Simons et al.: Event-level associations between affect, alcohol intoxication, and acute depenence symptoms: Effects of urgency, self-control, and drinking experience

Simons-Dvorak-Batien-etal-2010

Jeffrey S. Simons, Robert D. Dvorak, et al. "Event-level associations between affect, alcohol intoxication, and acute depenence symptoms: Effects of urgency, self-control, and drinking experience". In: *Addictive Behaviors* 35.12 (Dec. 2010), pp. 1045–1053. ISSN: 0306-4603. DOI: 10.1016/j.addbeh.2010.07.001.

Abstract: This study used experience sampling to examine within-person associations between positive affect, anxiety, sadness, and hostility and two outcomes: alcohol intoxication and acute dependence symptoms. We examined the role of urgency, premeditation, and perseverance in predicting the alcohol outcomes and tested whether the affective associations varied as a function of urgency. Participants completed baseline assessments and 21 days of experience sampling on PDAs. Hypotheses were partially confirmed. Positive affect was positively, and sadness inversely, associated with intoxication. Hostility was associated with intoxication for men but not women. Negative urgency moderated the association between anxiety and intoxication, making it stronger. However, positive urgency did not moderate the effect of positive affect. Heavier drinkers exhibited the greatest number of symptoms, yet the association between intoxication and acute signs of alcohol disorder was attenuated among these individuals. Results support the use of experience sampling to study acute signs and symptoms of high risk drinking and dependence.

Jeffrey S. Simons, Thomas A. Wills, et al. "Quantifying alcohol consumption: Self-report, transdermal assessment, and prediction of dependence symptoms". In: *Addictive Behaviors* 50 (Nov. 2015), pp. 205–212. ISSN: 0306-4603. DOI: 10.1016/j.addbeh.2015.06.042.

Abstract: Research on alcohol use depends heavily on the validity of self-reported drinking. The present paper presents data from 647 days of self-monitoring with a transdermal alcohol sensor by 60 young adults. We utilized a biochemical measure, transdermal alcohol assessment with the WrisTAS, to examine the convergent validity of three approaches to collecting daily self-report drinking data: experience sampling, daily morning reports of the previous night, and 1-week timeline follow-back (TLFB) assessments. We tested associations between three pharmacokinetic indices (peak concentration, area under the curve (AUC), and time to reach peak concentration) derived from the transdermal alcohol signal and within- and between- person variation in alcohol dependence symptoms. The WrisTAS data corroborated 85.74% of self-reported drinking days based on the experience sampling data. The TLFB assessment and combined experience sampling and morning reports agreed on 87.27% of drinking days. Drinks per drinking day did not vary as a function of wearing or not wearing the sensor; this indicates that participants provided consistent reports of their drinking regardless of biochemical verification. In respect to self-reported alcohol dependence symptoms, the AUC of the WrisTAS alcohol signal was associated with dependence symptoms at both the within- and between- person level. Furthermore, alcohol dependence symptoms at baseline predicted drinking episodes characterized in biochemical data by both higher peak alcohol concentration and faster time to reach peak concentration. The results support the validity of self-report alcohol data, provide empirical data useful for optimal design of daily process sampling, and provide an initial demonstration of the use of transdermal alcohol assessment to characterize drinking dynamics associated with risk for alcohol dependence.

Singer: SEM modeling with singular moment matrices part II: ML-estimation of sampled stochastic differential equations **Singer-2012**

Hermann Singer. "SEM modeling with singular moment matrices part II: ML-estimation of sampled stochastic differential equations". In: *The Journal of Mathematical Sociology* 36.1 (Jan. 2012), pp. 22–43. ISSN: 1545-5874. DOI: 10.1080/0022250x.2010.532259.

Abstract: Linear stochastic differential equations are expressed as an exact discrete model (EDM) and estimated with structural equation models (SEMs) and the Kalman filter (KF) algorithm. The oversampling approach is introduced in order to formulate the EDM on a time grid which is finer than the sampling intervals. This leads to a simple computation of the nonlinear parameter functionals of the EDM. For small discretization intervals, the functionals can be linearized, and standard software permitting only linear parameter restrictions can be used. However, in this case the SEM approach must handle large matrices leading to degraded performance and possible numerical problems. The methods are compared using coupled linear random oscillators with time-varying parameters and irregular sampling times.

Sjoerds et al.: Behavioral and neuroimaging evidence for overreliance on habit learning in alcohol-dependent patients **Sjoerds-deWit-vandenBrink-etal-2013**

Zsuzsika Sjoerds et al. "Behavioral and neuroimaging evidence for overreliance on habit learning in alcohol-dependent patients". In: *Translational Psychiatry* 3.12 (Dec. 2013), e337–e337. ISSN: 2158-3188. DOI: 10.1038/tp.2013.107.

Abstract: Substance dependence is characterized by compulsive drug-taking despite negative consequences. Animal research suggests an underlying imbalance between goal-directed and habitual action control with chronic drug use. However, this imbalance, and its associated neurophysiological mechanisms, has not yet been experimentally investigated in human drug abusers. The aim of the present study therefore was to assess the balance between goal-directed and habit-based learning and its neural correlates in abstinent alcohol-dependent (AD) patients. A total of 31 AD patients

and 19 age, gender and education matched healthy controls (HC) underwent functional magnetic resonance imaging (fMRI) during completion of an instrumental learning task designed to study the balance between goal-directed and habit learning. Task performance and task-related blood oxygen level-dependent activations in the brain were compared between AD patients and healthy matched controls. Findings were additionally associated with duration and severity of alcohol dependence. The results of this study provide evidence for an overreliance on stimulus-response habit learning in AD compared with HC, which was accompanied by decreased engagement of brain areas implicated in goal-directed action (ventromedial prefrontal cortex and anterior putamen) and increased recruitment of brain areas implicated in habit learning (posterior putamen) in AD patients. In conclusion, this is the first human study to provide experimental evidence for a disturbed balance between goal-directed and habitual control by use of an instrumental learning task, and to directly implicate cortical dysfunction to overreliance on inflexible habits in AD patients.

Smith et al.: From ecological momentary assessment (EMA) to ecological momentary intervention (EMI): Past and future directions for ambulatory assessment and interventions in eating disorders **Smith-Juarascio-2019**

Kathryn E. Smith and Adrienne Juarascio. "From ecological momentary assessment (EMA) to ecological momentary intervention (EMI): Past and future directions for ambulatory assessment and interventions in eating disorders". In: *Current Psychiatry Reports* 21.7 (June 2019). DOI: 10.1007/s11920-019-1046-8.

Abstract: Purpose of Review: Ambulatory assessment methods, including ecological momentary assessment (EMA), have often been used in eating disorders (EDs) to assess the type, frequency, and temporal sequencing of ED symptoms occurring in naturalistic environments. Relatedly, growing research in EDs has explored the utility of ecological momentary interventions (EMIs) to target ED symptoms. The aims of the present review were to (1) synthesize recent literature pertaining to ambulatory assessment/EMA and EMI in EDs, and (2) identify relevant limitations and future directions in these domains. Recent Findings: With respect to ambulatory assessment and EMA,

there has been substantial growth in the expansion of constructs assessed with EMA, the exploration of state- vs. trait-level processes, integration of objective and passive assessment approaches, and consideration of methodological issues. The EMI literature in EDs also continues to grow, though most of the recent research focuses on mobile health (mHealth) technologies with relatively minimal EMI components that adapt to momentary contextual information. Summary: Despite these encouraging advances, there remain several promising areas of ambulatory assessment research and clinical applications in EDs going forward. These include integration of passive data collection, use of EMA in treatment evaluation and design, evaluation of dynamic system processes, inclusion of diverse samples, and development and evaluation of adaptive, tailored EMIs such as just-in-time adaptive interventions. While much remains to be learned in each of these domains, the continual growth in mobile technology has potential to facilitate and refine our understanding of the nature of ED psychopathology and ultimately improve intervention approaches.

Song et al.: Bayesian estimation of random coefficient dynamic factor models

Song-Ferrer-2012

Hairong Song and Emilio Ferrer. "Bayesian estimation of random coefficient dynamic factor models". In: *Multivariate Behavioral Research* 47.1 (Feb. 2012), pp. 26–60. ISSN: 1532-7906. DOI: 10.1080/00273171.2012.640593.

Abstract: Dynamic factor models (DFMs) have typically been applied to multivariate time series data collected from a single unit of study, such as a single individual or dyad. The goal of DFMs application is to capture dynamics of multivariate systems. When multiple units are available, however, DFMs are not suited to capture variations in dynamics across units. The aims of this study are (a) to propose a random coefficient DFM (RC-DFM) to statistically model variations of dynamics across multiple units using the Bayesian method, (b) to illustrate the use of the proposed procedure by applying RC-DFMs to affect data collected from multiple dyads in romantic relationships, and (c) to evaluate the performance of the RC-DFMs with Bayesian estimation through simulation analyses. The results from the simulation analyses show that the Bayesian estimation of RC-DFMs works well in recovering parameters including both fixed and random effects. A number of practical considerations are provided to guide future research on using Bayesian methods for estimating multivariate time series from multiple units.

Tan et al.: A time-varying effect model for intensive longitudinal data **Tan-Shiyko-Li-etal-2012**

Xianming Tan et al. “A time-varying effect model for intensive longitudinal data”. In: *Psychological Methods* 17.1 (2012), pp. 61–77. ISSN: 1082-989X. DOI: 10.1037/a0025814.

Abstract: Understanding temporal change in human behavior and psychological processes is a central issue in the behavioral sciences. With technological advances, intensive longitudinal data (ILD) are increasingly generated by studies of human behavior that repeatedly administer assessments over time. ILD offer unique opportunities to describe temporal behavioral changes in detail and identify related environmental and psychosocial antecedents and consequences. Traditional analytical approaches impose strong parametric assumptions about the nature of change in the relationship between time-varying covariates and outcomes of interest. This article introduces time-varying effect models (TVEMs) that explicitly model changes in the association between ILD covariates and ILD outcomes over time in a flexible manner. In this article, we describe unique research questions that the TVEM addresses, outline the model-estimation procedure, share a SAS macro for implementing the model, demonstrate model utility with a simulated example, and illustrate model applications in ILD collected as part of a smoking-cessation study to explore the relationship between smoking urges and self-efficacy during the course of the pre- and postcessation period.

A. B. Taylor et al.: Four applications of permutation methods to testing a single-mediator model **Taylor-MacKinnon-2012**

Aaron B. Taylor and David P. MacKinnon. “Four applications of permutation methods to testing

a single-mediator model". In: *Behavior Research Methods* 44.3 (Feb. 2012), pp. 806–844. DOI: 10.3758/s13428-011-0181-x.

Abstract: Four applications of permutation tests to the single-mediator model are described and evaluated in this study. Permutation tests work by rearranging data in many possible ways in order to estimate the sampling distribution for the test statistic. The four applications to mediation evaluated here are the permutation test of ab, the permutation joint significance test, and the noniterative and iterative permutation confidence intervals for ab. A Monte Carlo simulation study was used to compare these four tests with the four best available tests for mediation found in previous research: the joint significance test, the distribution of the product test, and the percentile and bias-corrected bootstrap tests. We compared the different methods on Type I error, power, and confidence interval coverage. The noniterative permutation confidence interval for ab was the best performer among the new methods. It successfully controlled Type I error, had power nearly as good as the most powerful existing methods, and had better coverage than any existing method. The iterative permutation confidence interval for ab had lower power than do some existing methods, but it performed better than any other method in terms of coverage. The permutation confidence interval methods are recommended when estimating a confidence interval is a primary concern. SPSS and SAS macros that estimate these confidence intervals are provided.

Tibshirani: Regression shrinkage and selection via the lasso: A retrospective

Tibshirani-2011

Robert Tibshirani. "Regression shrinkage and selection via the lasso: A retrospective". In: *Journal of the Royal Statistical Society Series B: Statistical Methodology* 73.3 (Apr. 2011), pp. 273–282. ISSN: 1467-9868. DOI: 10.1111/j.1467-9868.2011.00771.x.

Abstract: In the paper I give a brief review of the basic idea and some history and then discuss some developments since the original paper on regression shrinkage and selection via the lasso.

Tofighi et al.: Indirect effects in sequential mediation models: Evaluating methods for hypothesis testing and confidence interval formation Tofighi-Kelley-2019

Davood Tofighi and Ken Kelley. "Indirect effects in sequential mediation models: Evaluating methods for hypothesis testing and confidence interval formation". In: *Multivariate Behavioral Research* 55.2 (June 2019), pp. 188–210. DOI: 10.1080/00273171.2019.1618545.

Abstract: Complex mediation models, such as a two-mediator sequential model, have become more prevalent in the literature. To test an indirect effect in a two-mediator model, we conducted a large-scale Monte Carlo simulation study of the Type I error, statistical power, and confidence interval coverage rates of 10 frequentist and Bayesian confidence/credible intervals (CIs) for normally and nonnormally distributed data. The simulation included never-studied methods and conditions (e.g., Bayesian CI with flat and weakly informative prior methods, two model-based bootstrap methods, and two nonnormality conditions) as well as understudied methods (e.g., profile-likelihood, Monte Carlo with maximum likelihood standard error [MC-ML] and robust standard error [MC-Robust]). The popular BC bootstrap showed inflated Type I error rates and CI under-coverage. We recommend different methods depending on the purpose of the analysis. For testing the null hypothesis of no mediation, we recommend MC-ML, profile-likelihood, and two Bayesian methods. To report a CI, if data has a multivariate normal distribution, we recommend MC-ML, profile-likelihood, and the two Bayesian methods; otherwise, for multivariate nonnormal data we recommend the percentile bootstrap. We argue that the best method for testing hypotheses is not necessarily the best method for CI construction, which is consistent with the findings we present.

Tofighi et al.: Monte Carlo confidence intervals for complex functions of indirect effects Tofighi-MacKinnon-2015

Davood Tofighi and David P. MacKinnon. "Monte Carlo confidence intervals for complex functions of indirect effects". In: *Structural Equation Modeling: A Multidisciplinary Journal* 23.2 (Aug. 2015), pp. 194–205. DOI: 10.1080/10705511.2015.1057284.

Abstract: One challenge in mediation analysis is to generate a confidence interval (CI) with high coverage and power that maintains a nominal significance level for any well-defined function of indirect and direct effects in the general context of structural equation modeling (SEM). This study discusses a proposed Monte Carlo extension that finds the CIs for any well-defined function of the coefficients of SEM such as the product of k coefficients and the ratio of the contrasts of indirect effects, using the Monte Carlo method. Finally, we conduct a small-scale simulation study to compare CIs produced by the Monte Carlo, nonparametric bootstrap, and asymptotic-delta methods. Based on our simulation study, we recommend researchers use the Monte Carlo method to test a complex function of indirect effects.

Trull et al.: Ambulatory assessment **Trull-EbnerPriemer-2013**

Timothy J. Trull and Ulrich Ebner-Priemer. "Ambulatory assessment". In: *Annual Review of Clinical Psychology* 9.1 (Mar. 2013), pp. 151–176. ISSN: 1548-5951. DOI: 10.1146/annurev-clinpsy-050212-185510.

Abstract: Ambulatory assessment (AA) covers a wide range of assessment methods to study people in their natural environment, including self-report, observational, and biological/physiological/behavioral. AA methods minimize retrospective biases while gathering ecologically valid data from patients' everyday life in real time or near real time. Here, we report on the major characteristics of AA, and we provide examples of applications of AA in clinical psychology (a) to investigate mechanisms and dynamics of symptoms, (b) to predict the future recurrence or onset of symptoms, (c) to monitor treatment effects, (d) to predict treatment success, (e) to prevent relapse, and (f) as interventions. In addition, we present and discuss the most pressing and compelling future AA applications: technological developments (the smartphone), improved ecological validity of laboratory results by combined lab-field studies, and investigating gene-environment interactions. We conclude with a discussion of acceptability, compliance, privacy, and ethical issues.

Ugille et al.: Multilevel meta-analysis of single-subject experimental designs: A simulation study **Ugille-Moeyaert-Beretvas-etal-2012**

Maaike Ugille et al. "Multilevel meta-analysis of single-subject experimental designs: A simulation study". In: *Behavior Research Methods* 44.4 (May 2012), pp. 1244–1254. ISSN: 1554-3528. DOI: 10.3758/s13428-012-0213-1.

Abstract: One way to combine data from single-subject experimental design studies is by performing a multilevel meta-analysis, with unstandardized or standardized regression coefficients as the effect size metrics. This study evaluates the performance of this approach. The results indicate that a multilevel meta-analysis of unstandardized effect sizes results in good estimates of the effect. The multilevel meta-analysis of standardized effect sizes, on the other hand, is suitable only when the number of measurement occasions for each subject is 20 or more. The effect of the treatment on the intercept is estimated with enough power when the studies are homogeneous or when the number of studies is large; the power of the effect on the slope is estimated with enough power only when the number of studies and the number of measurement occasions are large.

Usami et al.: A unified framework of longitudinal models to examine reciprocal relations **Usami-Murayama-Hamaker-2019**

Satoshi Usami, Kou Murayama, and Ellen L. Hamaker. "A unified framework of longitudinal models to examine reciprocal relations". In: *Psychological Methods* 24.5 (Oct. 2019), pp. 637–657. ISSN: 1082-989X. DOI: 10.1037/met0000210.

Abstract: Inferring reciprocal effects or causality between variables is a central aim of behavioral and psychological research. To address reciprocal effects, a variety of longitudinal models that include cross-lagged relations have been proposed in different contexts and disciplines. However, the relations between these cross-lagged models have not been systematically discussed in the literature. This lack of insight makes it difficult for researchers to select an appropriate model when analyzing longitudinal data, and some researchers do not even think about alternative cross-lagged models. The present research provides a unified framework that clarifies the conceptual and mathematical similarities and differences between these models. The unified framework shows that existing longitudinal models can be effectively classified based on whether the model posits unique factors and/or dynamic residuals and what types of common factors are used to model changes. The latter is essential to understand how cross-lagged parameters are interpreted. We also present an example using empirical data to demonstrate that there is great risk of drawing different conclusions depending on the cross-lagged models used.

van Buuren et al.: mice: Multivariate Imputation by Chained Equations in R

vanBuuren-GroothuisOudshoorn-2011

Stef van Buuren and Karin Groothuis-Oudshoorn. "mice: Multivariate Imputation by Chained Equations in R". In: *Journal of Statistical Software* 45.3 (2011). DOI: 10.18637/jss.v045.i03.

Abstract: The R package mice imputes incomplete multivariate data by chained equations. The software mice 1.0 appeared in the year 2000 as an S-PLUS library, and in 2001 as an R package. mice 1.0 introduced predictor selection, passive imputation and automatic pooling. This article documents mice, which extends the functionality of mice 1.0 in several ways. In mice, the analysis of imputed data is made completely general, whereas the range of models under which pooling works is substantially extended. mice adds new functionality for imputing multilevel data, automatic predictor selection, data handling, post-processing imputed values, specialized pooling routines, model selection tools, and diagnostic graphs. Imputation of categorical data is improved in order to bypass problems caused by perfect prediction. Special attention is paid to transformations, sum scores, indices and interactions using passive imputation, and to the proper setup of the predictor matrix. mice can be downloaded from the Comprehensive R Archive Network. This article provides a hands-on, stepwise approach to solve applied incomplete data problems.

Van den Noortgate et al.: A multilevel meta-analysis of single-subject experimental design studies **VandenNoortgate-Onghena-2008**

Wim Van den Noortgate and Patrick Onghena. "A multilevel meta-analysis of single-subject experimental design studies". In: *Evidence-Based Communication Assessment and Intervention* 2.3 (Sept. 2008), pp. 142–151. ISSN: 1748-9547. DOI: 10.1080/17489530802505362.

Abstract: A multilevel approach is proposed to synthesize the results of single-subject experimental research by combining either raw data or effect-size measures. A multilevel meta-analysis of single-subject data allows us to make inferences not only about the effects for the participants that are investigated, but also about the generality of the effects. It also permits a systematic exploration of the possible moderating effect of study and participant characteristics. We also discuss the possibility of combining single-subject experimental research and group-comparison research in one single meta-analysis, and the difficulties involved, especially in making effect-size measures comparable for both kinds of studies.

van Erp et al.: Prior sensitivity analysis in default Bayesian structural equation modeling. **vanErp-Mulder-Oberski-2018**

Sara van Erp, Joris Mulder, and Daniel L. Oberski. "Prior sensitivity analysis in default Bayesian structural equation modeling." In: *Psychological Methods* 23.2 (June 2018), pp. 363–388. ISSN: 1082-989X. DOI: 10.1037/met0000162.

Abstract: Bayesian structural equation modeling (BSEM) has recently gained popularity because it enables researchers to fit complex models and solve some of the issues often encountered in classical maximum likelihood estimation, such as nonconvergence and inadmissible solutions. An important component of any Bayesian analysis is the prior distribution of the unknown model parameters. Often, researchers rely on default priors, which are constructed in an automatic fashion without requiring substantive prior information. However, the prior can have a serious influence on the estimation of the model parameters, which affects the mean squared error, bias, coverage rates, and

quantiles of the estimates. In this article, we investigate the performance of three different default priors: noninformative improper priors, vague proper priors, and empirical Bayes priors-with the latter being novel in the BSEM literature. Based on a simulation study, we find that these three default BSEM methods may perform very differently, especially with small samples. A careful prior sensitivity analysis is therefore needed when performing a default BSEM analysis. For this purpose, we provide a practical step-by-step guide for practitioners to conducting a prior sensitivity analysis in default BSEM. Our recommendations are illustrated using a well-known case study from the structural equation modeling literature, and all code for conducting the prior sensitivity analysis is available in the online supplemental materials.

Ver Hoef: Who invented the delta method? VerHoef-2012

Jay M. Ver Hoef. "Who invented the delta method?" In: *The American Statistician* 66.2 (May 2012), pp. 124–127. ISSN: 1537-2731. DOI: 10.1080/00031305.2012.687494.

Abstract: Many statisticians and other scientists use what is commonly called the "delta method." However, few people know who proposed it. The earliest article was found in an obscure journal, and the author is rarely cited for his contribution. This article briefly reviews three modern versions of the delta method and how they are used. Then, some history on the author and the journal of the first known article on the delta method is given. The original author's specific contribution is reproduced, along with a discussion on possible reasons that it has been overlooked.

Voelkle et al.: Continuous time modelling with individually varying time intervals for oscillating and non-oscillating processes Voelkle-Oud-2012

Manuel C. Voelkle and Johan H. L. Oud. "Continuous time modelling with individually varying time intervals for oscillating and non-oscillating processes". In: *British Journal of Mathematical and Statistical Psychology* 66.1 (Mar. 2012), pp. 103–126. DOI: 10.1111/j.2044-8317.2012.02043.x.

Abstract: When designing longitudinal studies, researchers often aim at equal intervals. In practice, however, this goal is hardly ever met, with different time intervals between assessment waves and different time intervals between individuals being more the rule than the exception. One of the reasons for the introduction of continuous time models by means of structural equation modelling has been to deal with irregularly spaced assessment waves (e.g., Oud & Delsing, 2010). In the present paper we extend the approach to individually varying time intervals for oscillating and non-oscillating processes. In addition, we show not only that equal intervals are unnecessary but also that it can be advantageous to use unequal sampling intervals, in particular when the sampling rate is low. Two examples are provided to support our arguments. In the first example we compare a continuous time model of a bivariate coupled process with varying time intervals to a standard discrete time model to illustrate the importance of accounting for the exact time intervals. In the second example the effect of different sampling intervals on estimating a damped linear oscillator is investigated by means of a Monte Carlo simulation. We conclude that it is important to account for individually varying time intervals, and encourage researchers to conceive of longitudinal studies with different time intervals within and between individuals as an opportunity rather than a problem.

Voelkle et al.: An SEM approach to continuous time modeling of panel data: Relating authoritarianism and anomia **Voelkle-Oud-Davidov-etal-2012**

Manuel C. Voelkle, Johan H. L. Oud, et al. "An SEM approach to continuous time modeling of panel data: Relating authoritarianism and anomia". In: *Psychological Methods* 17.2 (2012), pp. 176–192. DOI: 10.1037/a0027543.

Abstract: Panel studies, in which the same subjects are repeatedly observed at multiple time points, are among the most popular longitudinal designs in psychology. Meanwhile, there exists a wide range of different methods to analyze such data, with autoregressive and cross-lagged models being 2 of the most well known representatives. Unfortunately, in these models time is only considered implicitly, making it difficult to account for unequally spaced measurement occasions or to compare parameter estimates across studies that are based on different time intervals. Stochastic differential

equations offer a solution to this problem by relating the discrete time model to its underlying model in continuous time. It is the goal of the present article to introduce this approach to a broader psychological audience. A step-by-step review of the relationship between discrete and continuous time modeling is provided, and we demonstrate how continuous time parameters can be obtained via structural equation modeling. An empirical example on the relationship between authoritarianism and anomia is used to illustrate the approach.

Vuorre et al.: Within-subject mediation analysis for experimental data in cognitive psychology and neuroscience **Vuorre-Bolger-2017**

Matti Vuorre and Niall Bolger. "Within-subject mediation analysis for experimental data in cognitive psychology and neuroscience". In: *Behavior Research Methods* 50.5 (Dec. 2017), pp. 2125–2143. ISSN: 1554-3528. DOI: 10.3758/s13428-017-0980-9.

Abstract: Statistical mediation allows researchers to investigate potential causal effects of experimental manipulations through intervening variables. It is a powerful tool for assessing the presence and strength of postulated causal mechanisms. Although mediation is used in certain areas of psychology, it is rarely applied in cognitive psychology and neuroscience. One reason for the scarcity of applications is that these areas of psychology commonly employ within-subjects designs, and mediation models for within-subjects data are considerably more complicated than for between-subjects data. Here, we draw attention to the importance and ubiquity of mediational hypotheses in within-subjects designs, and we present a general and flexible software package for conducting Bayesian within-subjects mediation analyses in the R programming environment. We use experimental data from cognitive psychology to illustrate the benefits of within-subject mediation for theory testing and comparison.

K. Wang: Understanding power anomalies in mediation analysis — Wang-2018

Kai Wang. "Understanding power anomalies in mediation analysis". In: *Psychometrika* 83.2 (June 2018), pp. 387–406. ISSN: 1860-0980. DOI: 10.1007/s11336-017-9598-1.

Abstract: Previous studies have found some puzzling power anomalies related to testing the indirect effect of a mediator. The power for the indirect effect stagnates and even declines as the size of the indirect effect increases. Furthermore, the power for the indirect effect can be much higher than the power for the total effect in a model where there is no direct effect and therefore the indirect effect is of the same magnitude as the total effect. In the presence of direct effect, the power for the indirect effect is often much higher than the power for the direct effect even when these two effects are of the same magnitude. In this study, the limiting distributions of related statistics and their non-centralities are derived. Computer simulations are conducted to demonstrate their validity. These theoretical results are used to explain the observed anomalies.

Lijuan Wang et al.: Investigating reliabilities of intraindividual variability indicators — Wang-Grimm-2012

Lijuan Wang and Kevin J. Grimm. "Investigating reliabilities of intraindividual variability indicators". In: *Multivariate Behavioral Research* 47.5 (Sept. 2012), pp. 771–802. ISSN: 1532-7906. DOI: 10.1080/00273171.2012.715842.

Abstract: Reliabilities of the two most widely used intraindividual variability indicators, ISD 2 and ISD, are derived analytically. Both are functions of the sizes of the first and second moments of true intraindividual variability, the size of the measurement error variance, and the number of assessments within a burst. For comparison, the reliability of the intraindividual mean, $\bar{X}$, is also derived. Simulations are conducted to confirm accuracy of the derived functions. The influences of different factors on the reliabilities of these indicators are also studied. Results show that the reliabilities of ISD^2 and ISD are lower than the reliability of $\bar{X}$ in all studied conditions and the reliability of ISD

is lower than that of ISD^2 in most studied conditions. The derived functions are applied to calculate the reliabilities of intraindividual variability indicators of negative affect measured over 56 days and perceptual-motor performance measured over five assessments. Implications for measurement-burst study designs are discussed.

Lijuan Wang et al.: Investigating inter-individual differences in short-term intra-individual variability **Wang-Hamaker-Bergeman-2012**

Lijuan Wang, Ellen Hamaker, and C. S. Bergeman. "Investigating inter-individual differences in short-term intra-individual variability". In: *Psychological Methods* 17.4 (2012), pp. 567–581. ISSN: 1082-989X. DOI: 10.1037/a0029317.

Abstract: Intra-individual variability over a short period of time may contain important information about how individuals differ from each other. In this article we begin by discussing diverse indicators for quantifying intra-individual variability and indicate their advantages and disadvantages. Then we propose an alternative method that models inter-individual differences in intra-individual variability by separately considering both the amplitude of fluctuations and temporal dependency in the data. In the proposed model, temporal dependency and amplitude of fluctuations are both included as random effects. Parameter estimation is done with a multiple-step approach using maximum likelihood, or with a recommended 1-step approach using a Bayesian method. The similarities and differences between the proposed method and some existing methods are discussed and investigated using diary study data from older adults. The results from empirical data analysis revealed that temporal dependency and amplitude of fluctuations have different predictability of health outcomes and thus should be modeled and considered separately.

Lijuan Wang et al.: On disaggregating between-person and within-person effects with longitudinal data using multilevel models **Wang-Maxwell-2015**

Lijuan Wang and Scott E. Maxwell. "On disaggregating between-person and within-person effects

with longitudinal data using multilevel models". In: *Psychological Methods* 20.1 (2015), pp. 63–83. ISSN: 1082-989X. DOI: 10.1037/met0000030.

Abstract: This article extends current discussion of how to disaggregate between-person and within-person effects with longitudinal data using multilevel models. Our main focus is on the 2 issues of centering and detrending. Conceptual and analytical work demonstrates the similarities and differences among 3 centering approaches (no centering, grand-mean centering, and person-mean centering) and the relations and differences among various detrending approaches (no detrending, detrending X only, detrending Y only, and detrending both X and Y). Two real data analysis examples in psychology are provided to illustrate the differences in the results of using different centering and detrending methods for the disaggregation of between- and within-person effects. Simulation studies were conducted to further compare the various centering and detrending approaches under a wider span of conditions. Recommendations of how to perform centering, whether detrending is needed or not, and how to perform detrending if needed are made and discussed.

Wichers et al.: Critical slowing down as a personalized early warning signal for depression **Wichers-Groot-Psychosystems-2016**

Marieke Wichers et al. "Critical slowing down as a personalized early warning signal for depression". In: *Psychotherapy and Psychosomatics* 85.2 (2016), pp. 114–116. ISSN: 1423-0348. DOI: 10.1159/000441458.

W. Wu et al.: A new procedure to test mediation with missing data through nonparametric bootstrapping and multiple imputation **Wu-Jia-2013**

Wei Wu and Fan Jia. "A new procedure to test mediation with missing data through nonparametric bootstrapping and multiple imputation". In: *Multivariate Behavioral Research* 48.5 (Sept. 2013), pp. 663–691. DOI: 10.1080/00273171.2013.816235.

Abstract: This article proposes a new procedure to test mediation with the presence of missing data by combining nonparametric bootstrapping with multiple imputation (MI). This procedure performs MI first and then bootstrapping for each imputed data set. The proposed procedure is more computationally efficient than the procedure that performs bootstrapping first and then MI for each bootstrap sample. The validity of the procedure is evaluated using a simulation study under different sample size, missing data mechanism, missing data proportion, and shape of distribution conditions. The result suggests that the proposed procedure performs comparably to the procedure that combines bootstrapping with full information maximum likelihood under most conditions. However, caution needs to be taken when using this procedure to handle missing not-at-random or nonnormal data.

Yu et al.: Psychometric validation of the Internet Gaming Disorder-20 Test among Chinese middle school and university students **Yu-Pesigan-Zhang-etal-2019**

Shu M. Yu, Ivan Jacob Agaloos Pesigan, et al. "Psychometric validation of the Internet Gaming Disorder-20 Test among Chinese middle school and university students". In: *Journal of Behavioral Addictions* 8.2 (May 2019), pp. 295–305. ISSN: 2063-5303. DOI: 10.1556/2006.8.2019.18.

Abstract: Background and aims: Internet gaming disorder (IGD) was proposed in Diagnostic and Statistical Manual of Mental Disorders of American Psychiatric Association as an area warranting more research attention. High prevalence of excessive Internet game use and related addictions has been reported in China, especially among youth; however, there is a lack of psychometrically and theoretically sound instruments for assessing IGD in the Chinese language. Methods: This study aimed to examine the psychometric properties of a Chinese version of the Internet Gaming Disorder Test (IGD-20 Test) among Chinese middle school ($n = 569$; $M_{\text{age}} = 13.34$; 46.2% females) and university students ($n = 523$; $M_{\text{age}} = 20.12$; 48.4% females) samples in Beijing, China. All participants voluntarily completed an anonymous questionnaire. Results: Confirmatory factor analysis results showed that the Chinese version of the IGD-20 Test had five factors (i.e., salience-tolerance, mood modification, withdrawal, conflict, and relapse). Measurement invariance was confirmed across the two samples. The test score was positively associated with the modified Young's Internet Addiction Test for gaming addiction. Concurrent validation was further demonstrated by the IGD-20 Test's positive correlation with weekly gameplay and depression symptoms. The latent profile analysis showed four different gamer classes (i.e., regular gamers, low-risk engaged gamers, high-risk engaged gamers, and probable disordered gamers), with the estimated prevalence of 2.1% of the last group. Discussion and conclusion: The IGD-20 Test was applicable to Chinese youth and its Chinese version generally demonstrated good psychometric properties.

Yu et al.: Cognitive and psychosocial health risk factors of social networking addiction

Yu-Wu-Pesigan-2015

Shu M. Yu, Anise Man Sze Wu, and Ivan Jacob Agaloos Pesigan. "Cognitive and psychosocial health risk factors of social networking addiction". In: *International Journal of Mental Health and Addiction* 14.4 (Dec. 2015), pp. 550–564. ISSN: 1557-1882. DOI: 10.1007/s11469-015-9612-8.

Abstract: The aim of this study was to examine the effects of the two cognitive factors proposed by social cognitive theory to be highly influential on behavior (i.e., outcome expectancies and self-efficacy), in addition to optimism and loneliness, on social networking addiction among university students. In the study, 395 Chinese students (145 males, $M_{\text{age}} = 19.05$, age range = 17-27 years) voluntarily completed an online, anonymous questionnaire regarding their Internet use. Almost all of the participants (99%) were found to be using online social networking sites, and findings showed that social networking addiction was strongly correlated with Internet addiction. As hypothesized, more negative outcome expectancies and lower self-efficacy with regard to reducing Internet use were associated with higher social networking addictive tendencies. The results of the path analysis showed that low optimism was an indirect risk factor of social networking addiction through outcome expectancies and self-efficacy, while loneliness was a direct risk factor. The findings provide practical implications to preventive intervention for social networking addiction among youth.

Yuan et al.: Biases and standard errors of standardized regression coefficients

Yuan-Chan-2011

Ke-Hai Yuan and Wai Chan. "Biases and standard errors of standardized regression coefficients". In: *Psychometrika* 76.4 (Aug. 2011), pp. 670–690. DOI: 10.1007/s11336-011-9224-6.

Abstract: The paper obtains consistent standard errors (SE) and biases of order O(1/n) for the sample standardized regression coefficients with both random and given predictors. Analytical results indicate that the formulas for SEs given in popular text books are consistent only when the population value of the regression coefficient is zero. The sample standardized regression coefficients are also biased in general, although it should not be a concern in practice when the sample size is not too small. Monte Carlo results imply that, for both standardized and unstandardized sample regression coefficients, SE estimates based on asymptotics tend to under-predict the empirical ones at smaller sample sizes.

Yzerbyt et al.: New recommendations for testing indirect effects in mediational models: The need to report and test component paths **Yzerbyt-Muller-Batailler-etal-2018**

Vincent Yzerbyt et al. "New recommendations for testing indirect effects in mediational models: The need to report and test component paths". In: *Journal of Personality and Social Psychology* 115.6 (Dec. 2018), pp. 929–943. DOI: 10.1037/pspa0000132.

Abstract: In light of current concerns with replicability and reporting false-positive effects in psychology, we examine Type I errors and power associated with 2 distinct approaches for the assessment of mediation, namely the component approach (testing individual parameter estimates in the model) and the index approach (testing a single mediational index). We conduct simulations that examine both approaches and show that the most commonly used tests under the index approach risk inflated Type I errors compared with the joint-significance test inspired by the component approach. We argue that the tendency to report only a single mediational index is worrisome for

this reason and also because it is often accompanied by a failure to critically examine the individual causal paths underlying the mediational model. We recommend testing individual components of the indirect effect to argue for the presence of an indirect effect and then using other recommended procedures to calculate the size of that effect. Beyond simple mediation, we show that our conclusions also apply in cases of within-participant mediation and moderated mediation. We also provide a new R-package that allows for an easy implementation of our recommendations.

Zamboanga et al.: A "player's" game? Masculinity and drinking games participation among White and Asian American male college freshmen

Zamboanga-Iwamoto-Pesigan-etal-2015

Byron L. Zamboanga, Derek K. Iwamoto, et al. "A "player's" game? Masculinity and drinking games participation among White and Asian American male college freshmen". In: *Psychology of Men & Masculinity* 16.4 (Oct. 2015), pp. 468–473. ISSN: 1524-9220. DOI: 10.1037/a0039101.

Abstract: Research with college students indicates that conformity to distinct masculine norms is associated with heavy drinking and alcohol-related problems. Drinking games (DGs) involve heavy alcohol consumption, and although such games have been characterized as a male-dominated activity, studies have not examined how gender-relevant factors such as conformity to masculine norms are associated with DGs participation. Moreover, the extent to which these associations vary as a function of race/ethnicity warrants further exploration because the exact role that these factors play in increasing college students' risk for DGs participation is unclear. Thus, the primary aim of this study was to examine the associations between distinct masculine norms and frequency of DGs participation (while controlling for typical alcohol consumption) in a sample of White ($n = 328$) and Asian American ($n = 136$) college men ($M_{\text{age}} = 18.11$ years, SD $= 0.35$). A secondary aim was to test the degree to which such relationships are similar between these groups. Male college freshmen from a public university completed self-report questionnaires. Results indicated that increased levels of conformity to the masculine norms of being a playboy and heterosexual presentation were significantly associated with more frequent DGs participation for White but

not Asian American college men. Implications for intervention and future research directions are discussed.

Zamboanga et al.: Frequency of drinking games participation and alcohol-related problems in a multiethnic sample of college students: Do gender and ethnicity matter?

Zamboanga-Pesigan-Tomaso-etal-2015

Byron L. Zamboanga, Ivan Jacob Agaloos Pesigan, et al. "Frequency of drinking games participation and alcohol-related problems in a multiethnic sample of college students: Do gender and ethnicity matter?" In: *Addictive Behaviors* 41 (Feb. 2015), pp. 112–116. ISSN: 0306-4603. DOI: 10.1016/j.addbeh.2014.10.002.

Abstract: Introduction: A drinking game (DG) is a high-risk, social drinking activity that consists of certain rules (i.e., when to drink and how much to consume) designed to promote inebriation and that requires each player to perform a cognitive and/or motor task (Zamboanga et al., 2013). Research suggests that non-White or female students who play DGs are at an increased risk of experiencing alcohol-related problems. Thus, this study examined whether the associations between DG participation and alcohol-related problems were similar for men and women and across ethnic groups. Method: College students ($N = 7409$; 73% women; 64% White, 8% Black, 14% Hispanic, 14% Asian) from 30 U.S. colleges/universities completed self-report questionnaires. Results: Controlling for age, site, Greek membership (i.e., membership in a fraternity or sorority), and typical alcohol consumption, results indicated that the association between DG participation and alcohol-related problems was stronger for men compared to women. With respect to ethnicity, the association between these variables was stronger among Black women than Black men. Conclusions: Findings from this large-scale study highlight the need to closely investigate how gender and ethnicity moderate the associations between DG participation and alcohol-related problems. College intervention efforts designed to address high-risk drinking behaviors such as DG participation might consider paying close attention to ethnic minority populations, perhaps particularly Black women.

G. Zhang: Testing process factor analysis models using the parametric bootstrap

Zhang-2018

Guangjian Zhang. "Testing process factor analysis models using the parametric bootstrap". In: *Multivariate Behavioral Research* 53.2 (Jan. 2018), pp. 219–230. ISSN: 1532-7906. DOI: 10.1080/00273171.2017.1415123.

Abstract: Process factor analysis (PFA) is a latent variable model for intensive longitudinal data. It combines P-technique factor analysis and time series analysis. The goodness-of-fit test in PFA is currently unavailable. In the paper, we propose a parametric bootstrap method for assessing model fit in PFA. We illustrate the test with an empirical data set in which 22 participants rated their effects everyday over a period of 90 days. We also explore Type I error and power of the parametric bootstrap test with simulated data.

G. Zhang et al.: Bootstrap standard error estimates in dynamic factor analysis

Zhang-Browne-2010

Guangjian Zhang and Michael W. Browne. "Bootstrap standard error estimates in dynamic factor analysis". In: *Multivariate Behavioral Research* 45.3 (May 2010), pp. 453–482. ISSN: 1532-7906. DOI: 10.1080/00273171.2010.483375.

Abstract: Dynamic factor analysis summarizes changes in scores on a battery of manifest variables over repeated measurements in terms of a time series in a substantially smaller number of latent factors. Algebraic formulae for standard errors of parameter estimates are more difficult to obtain than in the usual intersubject factor analysis because of the interdependence of successive observations. Bootstrap methods can fill this need, however. The standard bootstrap of individual timepoints is not appropriate because it destroys their order in time and consequently gives incorrect standard error estimates. Two bootstrap procedures that are appropriate for dynamic factor analysis are described. The moving block bootstrap breaks down the original time series into blocks and

draws samples of blocks instead of individual timepoints. A parametric bootstrap is essentially a Monte Carlo study in which the population parameters are taken to be estimates obtained from the available sample. These bootstrap procedures are demonstrated using 103 days of affective mood self-ratings from a pregnant woman, 90 days of personality self-ratings from a psychology freshman, and a simulation study.

M. X. Zhang et al.: Effects of social and outcome expectancies on hazardous drinking among Chinese university students: The mediating role of drinking motivations

Zhang-Ku-Wu-et-al-2020

Meng Xuan Zhang, Lisbeth Ku, et al. "Effects of social and outcome expectancies on hazardous drinking among Chinese university students: The mediating role of drinking motivations". In: *Substance Use & Misuse* 55.1 (2020), pp. 156–166. ISSN: 1532-2491. DOI: 10.1080/10826084.2019.1658784.

Abstract: Background and Objectives: Based on the theory of reasoned action, the present study investigated the relative effects of drinking outcome expectancies and parental norms, as well as the mediating effect of drinking motivations, on hazardous drinking in Chinese university students. Method: A sample of Chinese university students in Hong Kong and Macao ($N = 973$, $M = 19.82$, $SD = 1.57$, 48.9% males), who reported drinking in the past 3 months, voluntarily completed an anonymous questionnaire. Path analysis was used to test the effects of the variables on hazardous drinking. Results: All the psychosocial variables showed positive correlations with hazardous drinking. In the path model, controlling for sex, parental norms had both direct and indirect effects on hazardous drinking through social and enhancement motivations. Courage had the strongest indirect effect on drinking behavior through social, enhancement, and coping motivations, whereas the relationship between tension reduction and hazardous drinking was mediated by enhancement and coping motivations. Sociality and sexuality only had indirect effect through social and coping motivations respectively. Negative outcome expectancies had no direct nor indirect effects on hazardous drinking. Conclusions: Perceived approval from parents and positive alcohol outcome expectancies

may enhance individuals' tendency to engage in hazardous drinking by increasing their motivation to drink to be social, for enjoyment, and to cope with problems. Parents should explicitly show their disapproval of their children's drinking, and education efforts should focus on decreasing positive outcome expectancies and associated motivations for drinking among Chinese university students.

M. X. Zhang et al.: Psychometric properties of a Chinese version of the Brief Young Adult Alcohol Consequences Questionnaire (B-YAACQ)

Zhang-Pesigan-Kahler-etal-2019

Meng Xuan Zhang, Ivan Jacob Agaloos Pesigan, et al. "Psychometric properties of a Chinese version of the Brief Young Adult Alcohol Consequences Questionnaire (B-YAACQ)". In: *Addictive Behaviors* 90 (Mar. 2019), pp. 389–394. ISSN: 0306-4603. DOI: 10.1016/j.addbeh.2018.11.045.

Abstract: Aim: This study evaluated the psychometric properties of the Chinese version of the Brief Young Adult Alcohol Consequences Questionnaire (B-YAACQ). Method: In this study, 1616 Chinese university students (male = 58.66%; $M_{\text{age}} = 19.88$) reporting past-year drinking experience voluntarily completed an anonymous questionnaire. Rasch analysis, reliability analysis, and linear modeling were performed to examine the psychometric properties of the Chinese version of B-YAACQ. Results: Results of Rasch analysis and reliability analysis supported the assumption of uni-dimensionality, local independence, and internal consistency of the 24-item B-YAACQ in our Chinese sample. However, six items had marginal outfit statistics and/or potential gender bias; therefore, a model with 18 items was also tested after removing these items. The 18-item model showed excellent fit to the uni-dimensional model with no gender bias. However, the Pearson correlation between the 24-item and 18-item versions was $r = 0.98$, suggesting highly similar measurement. Both versions demonstrated concurrent validity through positive association with the Alcohol Use Disorder Identification Test (AUDIT) subscales, even after controlling for the effects of age and gender. Conclusion: This study is the first to validate a measurement tool for negative drinking consequences for university students in China. Despite some limitations, the original 24-item B-YAACQ was shown to have satisfactory psychometric properties when applied to Chinese

university students. We recommend the shorter 18-item version without significant gender bias for testing gender differences.

Z. Zhang et al.: Methods for mediation analysis with missing data Zhang-Wang-2012

Zhiyong Zhang and Lijuan Wang. "Methods for mediation analysis with missing data". In: *Psychometrika* 78.1 (Dec. 2012), pp. 154–184. DOI: 10.1007/s11336-012-9301-5.

Abstract: Despite wide applications of both mediation models and missing data techniques, formal discussion of mediation analysis with missing data is still rare. We introduce and compare four approaches to dealing with missing data in mediation analysis including listwise deletion, pairwise deletion, multiple imputation (MI), and a two-stage maximum likelihood (TS-ML) method. An R package bmem is developed to implement the four methods for mediation analysis with missing data in the structural equation modeling framework, and two real examples are used to illustrate the application of the four methods. The four methods are evaluated and compared under MCAR, MAR, and MNAR missing data mechanisms through simulation studies. Both MI and TS-ML perform well for MCAR and MAR data regardless of the inclusion of auxiliary variables and for AV-MNAR data with auxiliary variables. Although listwise deletion and pairwise deletion have low power and large parameter estimation bias in many studied conditions, they may provide useful information for exploring missing mechanisms.

Zyphur et al.: From data to causes I: Building a general cross-lagged panel model (GCLM) Zyphur-Allison-Tay-etal-2019

Michael J. Zyphur, Paul D. Allison, et al. "From data to causes I: Building a general cross-lagged panel model (GCLM)". In: *Organizational Research Methods* 23.4 (May 2019), pp. 651–687. ISSN: 1552-7425. DOI: 10.1177/1094428119847278.

Abstract: This is the first paper in a series of two that synthesizes, compares, and extends methods for causal inference with longitudinal panel data in a structural equation modeling (SEM) framework. Starting with a cross-lagged approach, this paper builds a general cross-lagged panel model (GCLM) with parameters to account for stable factors while increasing the range of dynamic processes that can be modeled. We illustrate the GCLM by examining the relationship between national income and subjective well-being (SWB), showing how to examine hypotheses about short-run (via Granger-Sims tests) versus long-run effects (via impulse responses). When controlling for stable factors, we find no short-run or long-run effects among these variables, showing national SWB to be relatively stable, whereas income is less so. Our second paper addresses the differences between the GCLM and other methods. Online Supplementary Materials offer an Excel file automating GCLM input for Mplus (with an example also for Lavaan in R) and analyses using additional data sets and all program input/output. We also offer an introductory GCLM presentation at https://youtu.be/tHnnaRNPbXs. We conclude with a discussion of issues surrounding causal inference.

Zyphur et al.: From data to causes II: Comparing approaches to panel data analysis

Zyphur-Voelkle-Tay-etal-2019

Michael J. Zyphur, Manuel C. Voelkle, et al. "From data to causes II: Comparing approaches to panel data analysis". In: *Organizational Research Methods* 23.4 (May 2019), pp. 688–716. ISSN: 1552-7425. DOI: 10.1177/1094428119847280.

Abstract: This article compares a general cross-lagged model (GCLM) to other panel data methods based on their coherence with a causal logic and pragmatic concerns regarding modeled dynamics and hypothesis testing. We examine three "static" models that do not incorporate temporal dynamics: random- and fixed-effects models that estimate contemporaneous relationships; and latent curve models. We then describe "dynamic" models that incorporate temporal dynamics in the form of lagged effects: cross-lagged models estimated in a structural equation model (SEM) or multilevel model (MLM) framework; Arellano-Bond dynamic panel data methods; and autoregressive latent trajectory models. We describe the implications of overlooking temporal dynamics in static models and show how even popular cross-lagged models fail to control for stable factors over time. We also

show that Arellano-Bond and autoregressive latent trajectory models have various shortcomings. By contrasting these approaches, we clarify the benefits and drawbacks of common methods for modeling panel data, including the GCLM approach we propose. We conclude with a discussion of issues regarding causal inference, including difficulties in separating different types of time-invariant and time-varying effects over time.